

BASIC GPS

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For use in the courses

TBA4236 Theoretical geomatics

TBA4245 Geodesy

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About the compendium, the contents

This compendium is a supplement to and an extension of chapters 4 and 5 in the Norwegian textbook "Grunnleggende landmåling" (Skogseth and Norberg).

You will also find references to Hofmann-Wellenhof "GNSS" and Holsen "Geodesy part 1" and "Geodesy part 2".

These textbooks are used in the courses TBA4236 Theoretical geomatics and TBA4245 Geodesy at Geomatics NTNU.

This compendium covers:

Part A: The GPS observations, sources of errors, baselines

- Processing of a GNSS baseline, the results

Part B: Transferring the GNSS baseline to the map projection plane

- Transferring the GNSS baseline from the WGS84 datum (ΔX , ΔY , ΔZ) to decomposed observations in the map projection plane (distance, plane bearing and zenith angle).

Part C: GNSS and heights. Geoid undulation. Corrections, to and from the plane

- The transformation from measured ellipsoidal heights/zenith angles to orthometric heights/zenith angles. Key words: The deviation of the vertical, the ellipsoid-geoid separation (geoid height), chapters 12 and 13.
- Corrections on the measured value between the ellipsoid/geoid and the map projection plane (chapter 14)

Appendix A:

- Textbooks, standards, internet addresses etc.

Some figures and text from the slides from the lectures, are also used in this compendium.

See also the example exercises no. 1 and 5 on Blackboard, they are covering topics from this compendium.

Geomatics, NTNU, 15.10.2009

NB: This English version is a translation of the original Norwegian compendium. Some chapters are still in Norwegian, but the required reading in the courses are in English (except chapter 10.3).

The plan is to translate the rest later. Until English versions are made, I have copied the chapters with Norwegian text in this document, the chapters 1 to 6, 8.1-8.2, 10.3, and 14.3, 14.4.

Chapters 12 and 13 are now translated. Chapter 10.3 is being translated, not quite finished.

See also the handouts on Blackboard and the user manual for the baseline processing program LeicaGeoOffice covering these topics.

The English language in this document is not perfect, but I hope it is understandable, and for sure better than a Google translation.

The original compendium in Norwegian was written in January 2001, and some pages are updated in newer versions of the compendium.

Geomatics, NTNU, 9.10.2017, Terje Skogseth

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1. Introduction

This compendium is a supplement to and an extension of the GNSS chapters 4 and 5 in the Norwegian textbook "Grunnleggende landmåling" (Skogseth and Norberg).

It is recommended that you read and refresh your knowledge of the basics of the different satellite systems.

Some topics are described rather shortly. For more about these topics, see another textbook in this course: Hofmann-Wellenhof et al "GNSS", especially for the topics atmospheric noise, multipath (flerveisinterferens), cycle slip (fasebrudd), ambiguity of the phase observations, description of the satellite systems.

For coordinate and height systems and datums that are used in Norway, see textbooks from previous courses.

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The plan is to translate the rest later. Until English versions are made, we have kept the chapters with Norwegian text in this document:

Chapters 1 to 6

Chapter 10.3

Chapters 14.3 and 14.4.

PART A: OBSERVATIONS, SOURCES OF ERRORS, BASELINES

2. Observations

Signalene vi mottar fra GPS-satellittene, er påmodulert en del informasjon som vi trenger for å kunne utføre beregninger av posisjonen for satellitter og mottakerstasjon. Det som blir målt, er *avstandene mellom mottakerantennen og satellittene* ved hvert måletidspunkt. Disse avstandene kan vi bestemme på to forskjellige måter:

- ved målinger på koden, som er modulert på bæreølgen(e)
- ved fasemålinger på bæreølgen(e)

Målenøyaktighet:

Målenøyaktigheten på en avstand mellom satellitt og en mottaker, er for kodemålinger i størrelsesorden flere meter. Det tilsvarende for fasemålinger er i cm/mm-området.

Skal vi nytte GNSS til landmåling, må vi derfor bruke fasemålinger. Ikke alle GPS-mottakere registrerer fasen på bæreølgen, idet denne måleteknikken gjør at mottakerne blir dyrere og mer kompliserte.

2.1 Code signal, code measurements

Idet satellitten sender ut sitt signal, blir en gitt kodeserie modulert på signalet. Forutsatt at klokka i mottakeren går synkront med klokka i satellitten, blir en tilsvarende signalkode generert i mottakeren på akkurat samme tidspunkt. Det er unike koder for hver satellitt, slik at mottakeren skal kunne skille satellittsignalene fra hverandre.

Det er to typer koder som moduleres på bæreølgene L1 og L2:

- *C/A-code*: Coarse/Acquisition-code. Er tilgjengelig for sivile brukere. Den har en frekvens på 1,023 MHz (som gir en bølgelengde på ca 300 m, og den har en repetisjonstid på 1 millisekund). C/A-koden er modulert på L1-frekvensen og ikke på L2.
- *P-code*: Precision-code. Gir større nøyaktighet og er reservert for spesielle brukere (militære i hovedsak). Den har en frekvens på 10,23 MHz (som gir en bølgelengde på ca 30 m, og den har en repetisjonstid på ca 266,4 dager). P-koden er modulert på både L1- og L2-frekvensene. P-koden kan krypteres til Y-kode ved å bruke "anti spoofing", A-S. Systemstøykilden "selective availability", SA, ble slått av 2. mai 2000. P-koden var ikke påvirket av SA-støyen, og ga en betydelig bedre nøyaktighet med SA-støy, enn målinger på C/A-koden.

Se kapittel 4 for mer om kodemåling.

2.2 Carrier waves, phase measurements

Fasemålinger er at mottakeren måler avstandsendinger mellom satellittene og antennen, ved å måle fasen på bæreølgen til satellittsignalet. Uten å gå nærmere inn på selve teknikken, kan vi si at mottakeren registrerer denne avstandsendingen i form av antallet hele bølgelengder pluss deler av en bølgelengde (jfr fasemålinger for elektroniske avstandsmålere). Bølgelengdene for de to bæreølgene L1 og L2 er kjente, dermed kan vi beregne avstandsendingene til satellittene.

Satellittklokkenes høye stabilitet over tid, gir også en tilsvarende høy kvalitet på frekvensene på de genererte signalene som sendes ut fra satellittene. Basisfrekvensen er på 10,23MHz, og fra denne basisfrekvensen blir de to fundamentale frekvensene L1 og L2 generert.

- L1: $10,23 \cdot 154 = 1575,42$ MHz, gir en bølgelengde i vakuum på 19,0 cm
- L2: $10,23 \cdot 120 = 1227,60$ MHz, gir en bølgelengde i vakuum på 24,4 cm

Vi skiller mellom en-frekvente (måler på L1) og to-frekvente mottakere, som måler på både L1 og L2. Se kapittel 5 for mer om fasesmåling.

2.3 Observation equations. Linear combinations of measurements

GNSS kan anvendes på flere måter og på ulike nøyaktighetsnivå. For å tilfredsstille de forskjellige nøyaktighetskrav, kan man danne bestemte lineære kombinasjoner av de fundamentale kodemålingene eller fasemålingene.

Vi har fire viktige GPS-observasjonsligninger:

1. P1: En ligning for kode-pseudoavstander på L1-frekvensen
2. P2: En ligning for kode-pseudoavstander på L2-frekvensen
3. L1: En ligning for fasemålinger av bæreboelgen på L1-frekvensen
4. L2: En ligning for fasemålinger av bæreboelgen på L2-frekvensen

GNSS-målinger er en form for avstandsmålinger som er beheftet med systematiske feil. Alle typer av ovennevnte observasjonsligninger, vil inneholde lineære ledd som tar hånd om virkningene av systematiske feil som mottakerklokkens stand (offset, avvik fra GPS-tid), satellittklokkens stand, ionosfærisk og troposfærisk forsinkelse, satellittbanefeil.

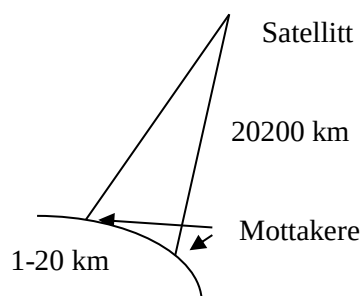
Med GNSS-data er både absolutt og relativ punktbestemmelse mulig. Absolutt punktbestemmelse begrenser seg til en nøyaktighet på noen få meter. Dette er som en følge av usikkerheten i satellittposisjon, klokkefeil og forplantningsforsinkelse i ionosfæren og troposfæren. Ved bruk av lange observasjoner (minst 12 - 24 timer) og spesialprogrammer (Statens kartverk bruker blant annet GIPSY), kan en absoluttbestemmelse på cm-nivå oppnås.

For de fleste geodetiske og geodynamiske formål vil det være nødvendig med relativ posisjonsbestemmelse for å oppnå tilstrekkelig nøyaktighet. Feilkildene som påvirker GPS-signalene, slik som satellittbanefeil, klokkefeil i satellitt- og mottakerklokke og forplantningsforsinkelse i atmosfæren, vil i stor grad ha samme virkning på signaler fra en og samme satellitt, men som går til mottakere i forskjellige stasjoner. Formålet med differansemetoder er å trekke fordeler av dette, for å øke nøyaktigheten ved relativ posisjonsbestemmelse. Bruk av fasemåling for relativ posisjonsbestemmelse består i å danne differansen mellom målinger. På den måten vil effekten av de forskjellige feil som er felles for de målinger man danner differansen mellom, bli eliminert eller sterkt redusert.

- Anvendes lineære kombinasjoner av observasjonene, får man redusert og til dels eliminert virkningen av systematiske feil som:
 - virkningen av klokkefeil, tvetydighet i antall hele bølgelengder (perioder, sykler, N).
- Ved å danne differanser mellom observasjoner, reduseres også virkningen av:
 - satellittbanefeil og virkningen av forsinkelse som skyldes ionosfære og troposfære.

Denne reduksjonen er mer effektiv, jo kortere avstanden er mellom de to mottakerne som måler samtidig. Hvorfor?

Se figur 2.1, der avstanden mellom mottakerne er betydelig overdrevet (er ca 8000 km på figuren, mens en vanlig vektor er på 1 – 20 km)



Figur 2.1 Avstander mellom mottakere og satellitt

Ved korte avstander kan man regne med at signalene fra satellitten og ned til de to mottakerne blir forstyrret (forsinket) ca. like mye. Ved å ta differansen mellom observasjonene fra de to mottakerne, er det bare den normalt svært lille forskjellen i forsinkelser som blir igjen.

Fra de opprinnelige observasjonsligningene kan en danne:

- Singel-differanser
- Dobbelt-differanser
- Trippel-differanser

Mer om de i kapittel 8.

Andre mer spesielle kombinasjoner som kan brukes i beregningene av en basislinje, er:

- L3/P3: ionsfærefri. Benytter seg av at L1 og L2 har ulike forsinkelser gjennom ionosfæren (forsinkelsen er frekvensavhengig). Denne bør/må brukes ved beregning av lange vektorer (lengre enn ca 20 km).
- L4/P4
- L5: "wide lane", differansen mellom L1 og L2, gir en bølgelengde på L3 på 86,2 cm, noe som gjør det enklere å finne heltallene i forhold til de kortere L1 og L2
- L6/P6: "narrow lane", summen av L1 og L2, har en bølgelengde på 10,7 cm, ca. halve av bølgelengden for L1 og L2.
- L7: mer robust mot flerveisinterferens ("multipath")

For mer om disse kombinasjonene, vises det til andre lærebøker.

3. Definition of time. GPS time standard. GPS calendar

Vi har sett av innledningskapitlene at det er avstandene mellom satellittene og mottakeren(e) som bestemmes ved GNSS. Nøyaktigheten på denne avstandsbestemmelsen er helt avhengig av at klokken som brukes i satellittene og i mottakerne, er av meget høy kvalitet, dette gjelder spesielt satellittklokkene. I dette kapitlet skal det defineres de tidsstandardene som er aktuelle i forbindelse med GPS.

Tidsstandarter som stjerneid og soldid er nærmere beskrevet i litteratur som omhandler geodetisk astronomi, men disse tidsbegrepene benyttes også i GNSS-beregningene, da satellittene kan ses på som himmellegemer på samme måte som stjerner, planeter og måner.

3.1 Time standards

Det er mange tidsstandarter, eller mange måter å måle tid på, som er i bruk.

Vi kan skille mellom følgende hovedgrupper, som alle er knyttet opp mot ulike periodiske bevegelser eller prosesser:

1. Jordas vandring rundt sola
2. Atom oscillasjoner (svingninger)
3. Jordas egenrotasjon, rotasjonen med Jordas polakse som rotasjonsakse

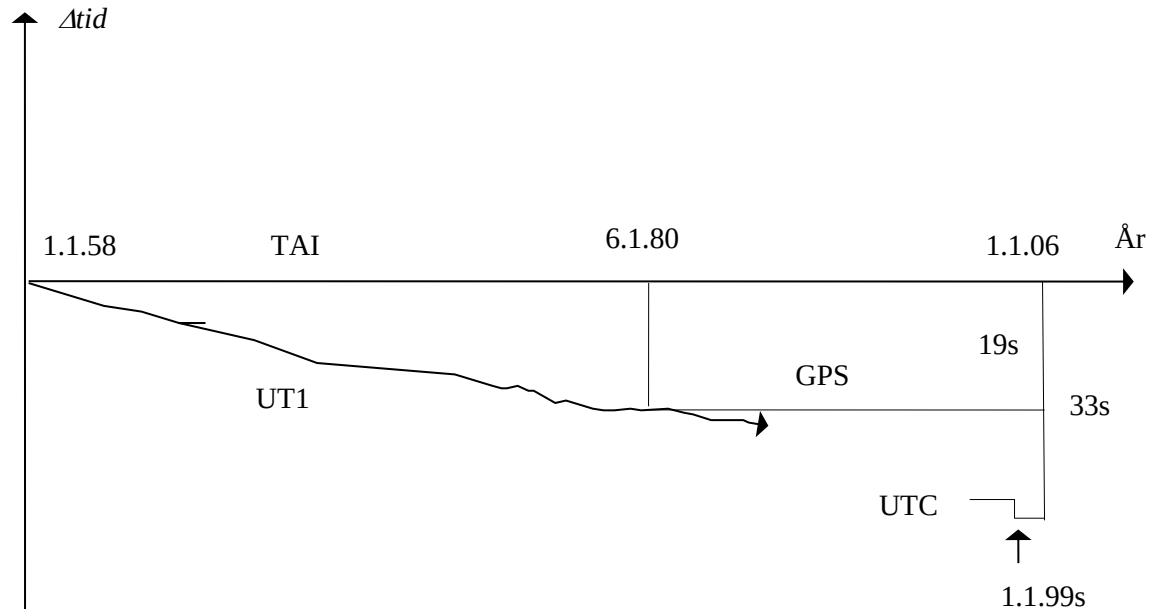
1. **Jordas vandring rundt sola:** Se annen litteratur for nærmere beskrivelse.

- *Barysentrisk tid:* Inersielt tidssystem (Newton)
- Dynamisk tid: Kvasi-inersielt, brukt også i forbindelse med satellitt baneberegninger rundt Jorda.

2. **Atom oscillasjoner** (svingninger): Se definisjonen av ett sekund i SI-systemet.

- International Atomic Time, TAI or IAT. Ett sekund i TAI er definert som ett SI-sekund ved middelvannstand. TAI er en jevn og stabil tidsskala som ikke er påvirket av Jordas uregelmessige rotasjon.
- *UTC:* Coordinated Universal Time. Tidssignalene er i UTC. UTC tidsskalaen er definert som å være et visst antall hele sekunder i differanse fra TAI (atomtida), slik at differansen mellom UTC og UT1 skal være mindre enn 0,9 sekund i absoluttverdi. UT1 er definert i forhold til Jordas egenrotasjon, se nedenfor. Blir differansen større, må det innføres et hopp på 1 sekund på UTC-tidsangivelsen (hoppet gjøres 31.12, eller også 30.6 hvis det er behov for flere hopp i løpet av et kalenderår). På datoen 1. januar 1958 viste TAI-klokkene og UTC-klokkene det samme. Siden 1. januar 1958, har UTC-tidsskalaen endret seg stegvis med ett sekund, og forskjellen per 1. januar 2006 var på 33 sekunder.

- **GPS-tid:** GPS-tidsskalaen var sammenfallende (lik) med UTC-tidsskalaen den 6. januar 1980. I GPS-tidsskalaen innføres det ikke noen stegvise sprang som det gjøres for UTC. GPS-tidssystemet har derfor et konstant forskjell på 19 sekunder sammenlignet med TAI.
- **GLONASS-tid:** Følger UTC, og har da også et sprang på 1 sekund med uregelmessige mellomrom.



Figur 3.1 Sammenhengen mellom tidsangivelser: UTC, stjernetid, IAT (eller TAI), UT1, GPS-tid.

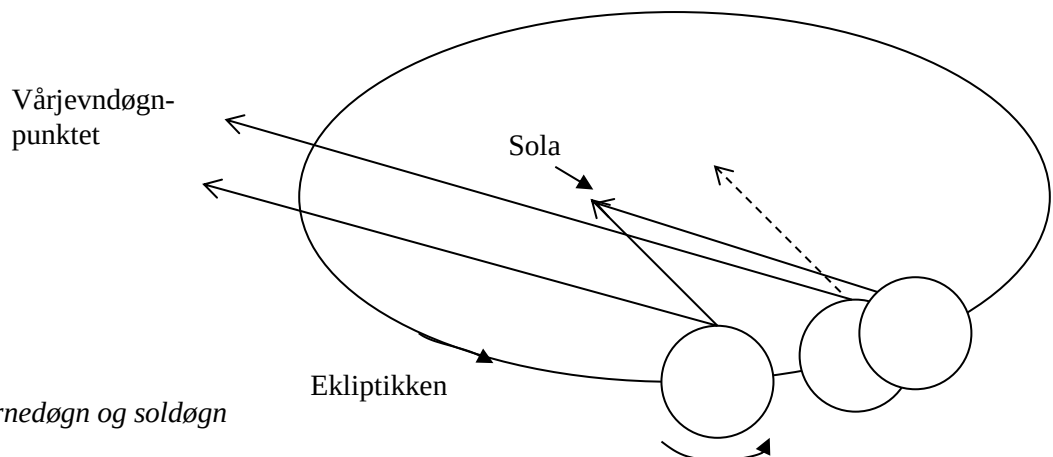
Per 1.1.2006 var det følgende sammenhenger, se også figur 3.1:

$$\begin{aligned} \text{TAI} &= \text{GPS} + 19 \text{ s} && \text{en konstant forskjell} \\ \text{TAI} &= \text{UTC} + n \cdot 1,000 \text{ s} && \text{der } n = 33 \text{ per 1.1.2006} \end{aligned}$$

Når TAI-klokka 1.1.2006 viste 12 00, viste GPS-klokka 11 59 41 og UTC-klokka 11 59 27.

Det er blitt en forskjell på 33 sekunder på 48 år mellom TAI og UTC, siden 1958, det tilsvarer ca 0,7 sekunder per år.

3. **Jordas egenrotasjon**, rotasjonen med Jordas polakse som rotasjonsakse. Definisjonen av et døgn (24 timer) er knyttet til denne rotasjonen. Det skilles mellom et stjernetøgn og et soldøgn, og det skilles mellom midlere og sann verdi på døgnetts varighet. Stjernetid er kun basert på Jordas egenrotasjon, ett stjernetøgn er definert i forhold til et punkt på stjernehimmelen (som i prinsippet ligger uendelig langt unna Jorda), mens soltiden også tar hensyn til Jordas gang rundt sola.

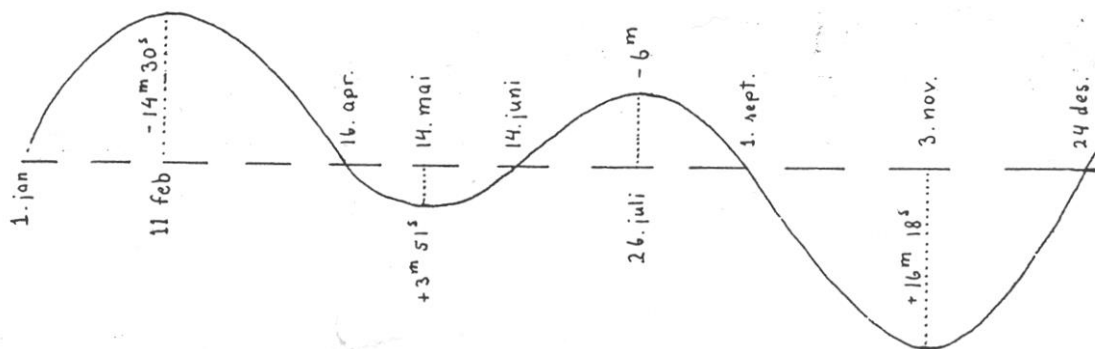


Figur 3.2 Stjernetøgn og soldøgn

Figur 3.2 viser en forenklet, skissemessig framstilling av forskjellen mellom sol- og stjernetid (det er ikke tatt hensyn til at stjernetid og soldøgnet starter ved hhv øvre og nedre kulminasjon, og størrelsene er noe overdrevet). Det framgår av figuren at Jorda må rotere litt lenger enn en runde, for å få sola i samme posisjon som ett soldøgnet tidligere, mens et stjernetid svarer en runde på jordrotasjonen. Se definisjonen av stjernetid, der det vises av Vårjevndøgnpunktet på himmelen brukes som referansepunkt

Begrepene er nærmere beskrevet i fagfeltet *geodetisk astronomi*. Her gis det en kort beskrivelse av de ulike definisjonene:

- **Soltid:** Soldøgnet er tiden mellom to påfølgende nedre kulminasjoner av solas sentrum. Nedre kulminasjon betyr at sola er i laveste posisjon på himmelkula, står i nord. Forskjellen mellom korteste og lengste sanne soldøgnet er på ca 51 sekunder.
- **Sann soltid:** Jordas vandring rundt sola skjer med ulik hastighet (jfr Keplers lover), noe som gir at også den sanne døgnlengden varierer i løpet av året. Sann soltid er derfor et upraktisk tidsmål å angi tiden på.
- **Midlere soltid:** For å få samme døgnlengde for alle døgn i løpet av et år, tenker man seg en midlere sol som går med konstant hastighet langs ekvator. Dermed har man definert et tidsmål som gir en konstant gange på klokken.
Soltiden vil på et gitt tidspunkt være forskjellig fra sted til sted med ulik geografisk lengde, da døgnets start er når sola står i nord (nedre kulminasjon) for et sted. Se om Greenwich midlere soltid, for definisjon av en referansetid, slik at man unngår det praktiske problemet med uendelig mange lokale tidssystemer.
- **Tidsjevning** = sann soltid – midlere soltid. Figur 3.3 viser at en variabel døgnlengde (se definisjonen av soltid ovenfor) kan gir en oppsummert forskjell på tidsjevningen på opptil ca pluss/minus 15 min i løpet av året.



Figur 3.3 (fra Berge). Tidsjevning

- **Greenwich midlere soltid:** Valgt referansetid, også kalt verdenstid, *UT* (universal time).
- **UT1 (Universal Time 1):** Er knyttet til Jordas rotasjonshastighet, vinkelhastighet. UT1 er derfor et tidsmål med ujevn gange, da Jordas rotasjonshastighet varierer over året (ulik massefordeling, Jorda roterer raskere om vinteren enn om sommeren) og fra år til år. Jordas rotasjonshastighet avtar fra år til år, slik at døgnets lengde øker med ca 0,002 sekund per døgn. Se UTC for et tidsmål med jevn gange.
- **Sonetid:** Av praktiske hensyn er Jorda delt inn i tidssoner, der man innen hver sone har samme tid. I prinsippet er bredden på hver tidssone 15 lengdegrader ($360 / 24$), da lengdeforskjell er tidsforskjell. Norge tilhører den Middeuropeiske tidssonen (CET: Central European Time), og vi har soloppgang før Greenwich, slik at vår klokkes sonetid viser 1 time mer enn UT:
$$\text{CET} = \text{UT} + 1 \text{ time}$$
- **Stjernetid:** Et stjernetid er tidsrommet mellom to påfølgende øvre kulminasjon av Vårjevndøgnpunktet. Vårjevndøgnpunktet står nå ca. i stjernetegnet Væren, og det er definert som skjæringspunktet mellom ekliptikken og ekvator på *himmelkula*, se beskrivelse i kompendiet om koordinatsystemer. Stjernetid er kun basert på Jordas egenrotasjon, 1 stjernetid er definert i

forhold til et punkt på stjernehimmelen (som i prinsippet ligger uendelig langt unna Jorda), mens soltiden også tar hensyn til Jordas gang rundt sola. Se figur 3.2.

- *Sann stjernetid*: En ujevn tidsskala, jfr. sann soltid.
- *Midlere stjernetid*: En jevn tidsskala, jfr. midlere soltid. Det er korrigert for nutasjon, som er månens innflytelse på stillingen i verdensrommet for Jordas rotasjonsakse.

Sammenhenger mellom stjernetid og soltid:

- 1 tropisk år: Tiden det tar for at Jorda har gått eksakt 1 runde rundt sola
- 1 tropisk år = 365,2422 middelsoldøgn = 366,2422 stjernedøgn
- 1 stjernedøgn (24 stjernetimer) = 23 h 56 min 04,1 sek midlere soltid
- 1 midlere soldøgn (24 soltimer) = 24 h 03 min 56,6 sek stjernetid

GPS-satellittene har en omløpstid på 11 h 58 min 02 sek midlere soltid = 12 h stjernetid, altså to runder per stjernedøgn.

Eksempel: Hva viser klokka vår når den sanne Sola står i sør i Trondheim den 18. november?

Problem: Beregn når den sanne Sola står i sør den 18. november for Lerkendalsbygget (=NTH), som har den geografiske lengden $L = 10^\circ 24' 38''$ Øst Greenwich. (Sann soltid vises på *solur*.)

Løsningsforslag:

Klokka vår (jfr. tidssignalet) viser tiden for den middeleuropeiske sonen (MT = CET). Vår sonetid er 1 time i forskjell fra Greenwich midlere soltid (UT: Universal Time):

$$\text{CET} = \text{UT} + 1 \text{ time.}$$

Den sanne Sola står i sør kl. 1200 sann soltid for *stedet* (NTH), ikke sonen.

Vi må derfor regne om kl. 1200 sann soltid til sann soltid vist i sonetid (UT eller CET), der lengdeforskjellen mellom stedets (NTHs) meridian og Greenwichmeridianen må være kjent.

Lengdeforskjellen omgjort til tidsenheter lik: $L = 10^\circ 24' 38'' = 0 \text{ h } 41 \text{ min } 39 \text{ sek}$

På grunn av jordrotasjonens retning (Sola står opp i øst), vil NTH (øst for Greenwich) få Sola i sør før Greenwich.

Dermed står den sanne Sola i sør på NTH kl. 11 h 18 min 21 sek sann soltid etter UT-klokka, eller kl. 12 h 18 min 21 sek sann soltid etter CET-klokka (sonetid).

For å finne ut når den *sanne Sola* står i sør i midlere soltid, må vi korrigere for tidsjevningen (se kurven på figur 3.3). Tidsjevningen for 18. november, er ca. +15 min.

$$\text{Midlere soltid} = \text{Sann soltid} - \text{Tidsjevning}$$

Den sanne Sola står i sør på NTH (Lerkendalsbygget) ca. kl. 12 h 03 min midlere soltid på CET-klokka (vår klokke) den 18.11.

3.2 GPS calendar

Juliansk dato brukes som datoangivelse i GPS-systemet, i stedet for å angi datoen som dag/måned/år.

En Juliansk dato angir antall soldager som er gått siden en bestemt startdato, 1. januar kl. 1200 i år 4713 før Kr.f. (År 4713 fKr er et historisk mye brukt startår for tidsregistreringer innen klassisk astronomi.)

I definisjonen av datumet WGS84, er det regnet antall Julianske dager med J2000,0 som referansetidspunkt (1. januar 2000, kl 1200).

Et Juliansk århundre har 36525 soldager (UT-dager, tiden UT brukes).

Dato	JD Juliansk dato	MJD (Modified Julian Date)	GPS-uke	Epoke
6.januar 1980, kl. 0000	2 444 244,5	44 244,0	0	Startdato for GPS
1. januar 2000, kl. 1200	2 451 545,0	51 544,5	1042 (søndagen)	J2000,0

Modifisert Juliansk dato (MJD) er "Juliansk dato" trukket fra 2400 000,5 dager, en mer praktisk datoangivelse. Legg merke til at ,5 gjør at ny dag regnes fra midnatt i MJD og ikke fra middag (kl 12) som i JD.

1 GPS-uke består av 7 Julianske dager, og GPS-uke brukes også som tidsenhet i GPS-systemet.

Midnatt mellom 21. og 22. august 1999 var datoen for start av GPS-uke nr. 1024 = 2^{10} , men denne "rollover"-en, der formatet for tidsangivelsen kunne bli sprengt, gikk uten problemer for de aller fleste GNSS-mottakerne (tilsvarende som overgangen til år 2000 stort sett gikk uten problemer).

3.3 Satellite clock, receiver clock

Det er avstanden mellom satellittene og mottakerne som bestemmes ved en GNSS-måling. For å få nøyaktige avstander, er signaltiden fra utsendelse til mottak, en kritisk faktor.

Satellittklokka

GPS-systemet gir en meget god nøyaktighet på absolutt eller relativ stedbestemmelse. Det er de høypresise atomklokkene i GPS-satellittene som er nøkkelen til denne nøyaktigheten. Klokkene i satellittene blir overvåket av GPS-kontrollsystemet, og de enkelte satellittklokkers avvik fra GPS-tidsstandarden blir beregnet og blir sendt ut som en del av signalene fra satellittene til mottakerne. Satellittklokkenes høye stabilitet over tid, gir også en tilsvarende høy kvalitet på frekvensene på de genererte signalene som sendes ut fra satellittene. Basisfrekvensen er på 10,23MHz, og fra denne basisfrekvensen blir de to fundamentale frekvensene L1 og L2 generert.

L1: $10,23 \cdot 154 = 1575,42$ MHz	gir en bølgelengde i vakuum på 19,0 cm
L2: $10,23 \cdot 120 = 1227,60$ MHz	gir en bølgelengde i vakuum på 24,4 cm

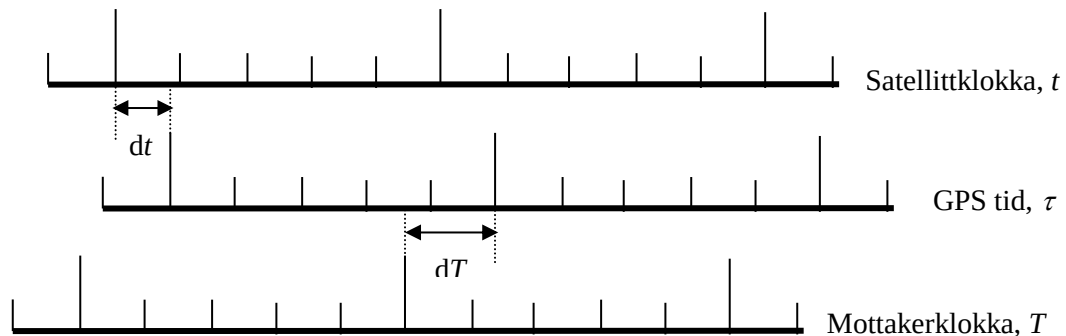
Mottakerklokka

Denne er ikke av samme kvalitet som satellittklokkene. Avviket fra GPS-tidsstandarden er ikke kjent med stor nok nøyaktighet, og dette avviket blir sett på som en ekstra ukjent størrelse, som må estimeres (eller fjernes) ved beregningene.

Tidsskalaer/klokker i GPS:

Vi har følgende tidsskalaer/klokker involvert i GPS:

- tidsskala t : utsendelse av et signal i satellitt-klokkas tidsskala
- tidsskala T : tid for mottakelse av dette signalet i mottaker-klokkas tidsskala
- tidsskala τ : tidsdifferansen mellom utsendt og mottatt signal må regnes i GPS-tid, som er tidsstandarden definert av GPS-systemet (se kapitlet om tidsstandarder).



Figur 3.4 Tidssignalene til satellittklokke, GPS-tidsstandarden og mottakerklokke

Av figur 3.4 ser vi at klokkekorreksjonene til GPS-tidsstandarden på satellitt- og mottaker-klokkene blir:

- For satellittklokka i satellitt a: $dt = \tau_a - t_a$ (3.1a)
- For mottakerklokka i mottaker b: $dT = \tau_b - T_b$ (3.1b)

De korrigerte avlesningene (GPS-tid) blir:

- For satellittklokka i satellitt a: $t_a + dt = \tau_a$ (3.2a)
- For mottakerklokka i mottaker b: $T_b + dT = \tau_b$ (3.2b)

Differansen ($\tau_b - \tau_a$) gir riktig tidsintervall i GPS-tidsstandarden, for den tiden det tar for signalet å gå fra satellitt til mottaker.

4. Code measurements

4.1 Definitions

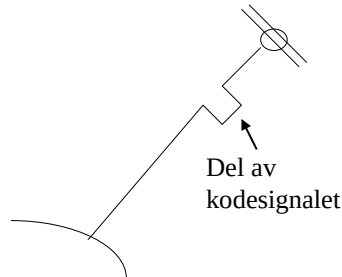
Se kapittel 2.1 for beskrivelse av ulike koder som blir modulert på satellittenes bæreølger:

- C/A-koden
- P-koden

Når en kodesekvens på satellittsignalet mottas av antennen, kan mottakeren måle hvor lang tid det har gått siden mottakeren genererte den samme sekvensen. Dette er den tiden signalet har brukt på sin vei fra satellitt til mottaker.

For å kunne lage (generere) en identisk kodesekvens i mottakeren til samme tid som den blir sendt ut av satellitten, er vi avhengige av at klokka i mottakeren går synkront med klokka i satellitten. Dette krever at vi har en atomklokke i mottakeren som er synkronisert til GPS-tid. En slik klokke er imidlertid så kostbar at mottakerne normalt leveres med mindre nøyaktige klokker. I stedet må mottakerklokkas avvik fra GPS-tid løses ut som en ukjent i beregningen av posisjonen for punktet vi ønsker å bestemme.

Siden den målte gangtiden for satellittsignalet inneholder mottakerklokkas avvik fra GPS-tid og altså ikke er en ekte avstandsmåling, blir avstanden mottakeren beregner av formelen, også kalt for *pseudodistance*.



Figur 4.1 Kodesignal

Den fundamentale kodemålingen er tidsdifferansen mellom tid for utsendelse av et signal (i satellitt-klokkas tidsskala, symbol t) og tid for mottakelse av dette signalet i en mottaker-klokkas tidsskala, symbol T . Denne tidsdifferansen må regnes i GPS-tid (symbol τ), som er tidsstandarden definert av GPS-systemet.

Se figur 3.1 for sammenhengen mellom τ , t og T .

Det er denne metoden som brukes for blant annet håndholdte GNSS-mottakere, de bruker C/A-koden. En ny posisjon kan bestemmes for f.eks. hvert sekund. En posisjonsbestemmelsen baserer seg bare på de avstandene som er målt *samtidig*.

Målt avstand mellom satellitten og mottakeren (forenklet formel, uten påførte feil, råavstand)

$$D = c \cdot \Delta\tau$$

c : lyshastigheten i vakuum

$\Delta\tau$: den tiden et utsendt kode-signal tar for å gå fra satellitt til mottaker. Tiden måles ved at koden fra satellitten inneholder informasjon om når den ble sendt ut, og ved at mottakerklokka viser når signalet kommer fram til mottakeren.

Legg merke til at avstanden til satellitten fås direkte (i sann tid) ved kodemåling. Dermed kan også en absolutt posisjon bestemmes i sann tid.

Se eksempel på feilbudsjett i tabellen i neste kapittel.

4.2 Error budget code measurements, 1 receiver, 1 satellite

Tabellen viser et feilbudsjett for de viktigste feilkildene for statistisk kodemålinger med 1 mottaker på C/A-koden. Tallverdiene er standardavvik for feiltypen (gjennomsnittsverdier), og viser hvor mye feil avstanden mellom en satellitt og mottakeren kan bli:

Feilkilde	Standardavvik i meter	Merknader ("normale" variasjoner)
Satellittklokka, stabilitet	2	1 - 3
Banefeil (eferemider)	4	2 - 7
SA, Selective Availability*	32	Kan variere mellom 0 og 100 i tallverdi
Ionosfære	5	2 - 15 (etter korreksjon)
Troposfære	1	0,5 - 2 (etter korreksjon)
Mottakerklokka		Løses ut som en ukjent
Støy på mottakeren	1	1 - 2
Flerveisinterferens (multipath)	1	0 - 4 (avhengig av reflekterende flater)

*: SA ble slått av (for godt?) 2. mai 2000

Systemets UERE (user equivalent range error) med og uten SA blir 33 m og 7 m på en målt avstand.

Beregning av UERE: $UERE = \sqrt{(2^2 + 4^2 + 32^2 + 5^2 + 1^2 + 1^2 + 1^2)} = 33 \text{ m}$

Med en PDOP på 2,5 (som tilsvarer en gjennomsnittlig god satellitt-geometri), fås en usikkerhet i posisjonen på mottakeren på 82,5 m og 17,5 m med og uten SA.

For mer om PDOP, se også side 162 i Grunnleggende landmåling, eller i en GNSS-lærebok.

4.3 Observation equation for a distance 1 satellite and 1 receiver

Vi beregner tidsforskjellen, $\Delta\tau$, mellom utsendt signal fra satellitt a og mottatt signal (kode) i mottaker b i GPS-tidsstandarden:

I vakuum (ingen forstyrrelser/feil) i GPS-tid: $\Delta\tau = \tau_b - \tau_a$

Målt tidsforskjell, avlesninger på mottaker- og satellittklokkene: $\Delta = T - t$

Målt tidsforskjell korrigert til GPS-tid, se ligning (3.1):

$$T - t = (\tau_b - dT) - (\tau_a - dt) = \Delta\tau + (dt - dT) \quad (4.1)$$

Vi har tidsforskjellen og multipliserer med lyshastigheten i vakuum, c , for å få avstanden:

$$P = c \cdot (T - t) = c \cdot \Delta\tau + c \cdot (dt - dT) \quad (4.2)$$

Det første leddet er signalets reisetid i vakuum: $\rho = c \cdot \Delta\tau$

Det andre leddet gir avvik i avstanden på grunn av satellittklokkas avvik fra GPS-tid ($c \cdot dt$)

Det tredje ledd gir avvik i avstanden på grunn av mottakerklokkas avvik fra GPS-tid ($c \cdot dT$)

Når vi også tar hensyn til at mottatt signal blir forsinket gjennom ionosfæren og troposfæren, og at koordinatene til satellitten ikke er nøyaktig kjent, får kodemålingen følgende form:

$$P = \rho + c \cdot (dt - dT) + d^{\text{ion}} + d^{\text{trop}} + d\rho \quad (4.3)$$

P : målt avstand, måleverdi (pseudoavstand)
 ρ : vektoren mellom mottakeren og satellitten

Ukjente størrelser i ligning (4.3):

- Mottakerens koordinater, disse ligger skjult i den sanne avstanden ρ , som er vektoren mellom mottaker og satellitt.
- Også dt og dT er ukjente, men satellittklokka avvik fra GPS-tid blir overvåket av kontrollsystemet, og et *antatt* avvik fra GPS-tidsskalaen sendes ut fra satellitten. Det kan derfor også være små restfeil på satellittklokka.
- De atmosfæriske forsinkelsene d^{ion} og d^{trop} kan en beregne med empirisk utviklede formler, eller modellere med parametre som inngår som ukjente i ligningen.
- Usikkerheten i banekoordinatene, $d\rho$, har man normalt ikke mulighet for å bestemme under målingene. Ved beregninger i ettertid (postprosessering) kan benytte presise banedata som er beregnet av målinger i et globalt sivilt overvåkningsnett (IGS). Disse banedataene kan lastes ned fra Internett, se litteraturlista bakerst.

5. Phase measurements

5.1 Introduction

Se kapittel 2.2 for mer om bæreølgene L1 og L2.

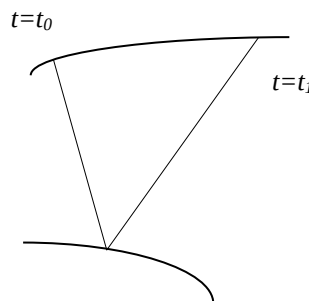
Fasemålinger er at mottakeren måler avstandsendinger mellom satellitter og antennen ved å måle fasen på bæreølgen til satellittsignalet.

Siden det som registreres, er *endring i avstand til satellitten fra det tidspunktet da vi begynte å måle*, er avstanden på starttidspunktet svært interessant. Hvis denne var kjent, ville fasemålingene våre kunne gi avstanden til satellitten på ethvert tidspunkt.

Avstanden på starttidspunktet kan deles opp i et helt antall bølgelengder pluss en del av en bølgelengde. Denne delen av en bølgelengde registrerer mottakeren på det tidspunktet da den begynner å måle.

Derfor er det bare *det hele antallet bølgelengder ved starttidspunktet som forblir ukjent*.

Ved å stå i ro i et visst tidsrom i punktene for basislinjen, kan man estimere (beregne) dette ukjente heltallet, og fasemålingen ved tidspunkt t_1 kan da ses på som en ren avstandsmåling til satellitten.



Figur 5.1 Fasemåling

Dette måleprinsippet, fasemålinger, brukes for geodetiske mottakere, da den gir cm-(mm)-nivå på bestemmelsen av en GNSS-vektor mellom to punkter. Måleprinsippet ligner på det som anvendes for målinger med en klassisk elektronisk avstandsmåler.

En forenklet formel for målt avstand ved tidspunktet $t = t_0$ (uten påførte feil, råavstand):

$$D = \Delta\lambda_{t_0} + N_{t_0} \cdot \lambda$$

- λ : Bølgelengden for bæreølgen, $\lambda_1=19,0$ cm på L1, eller $\lambda_2=24,4$ cm på L2.
 N : Antall hele bølgelengder når vi starter målingene ($t = t_0$). N er ukjent og må bestemmes ved basislinjeberegningen.
 ΔL : Deler av en bølgelengde, bestemmes ved fasemålinger.

Ved tidspunktet $t = t_1$, blir den forenklete formel for målt avstand (uten påførte feil, råavstand):

$$D = \Delta\lambda_{t_1} + N_{t_0} \cdot \lambda + \text{Int}(t_1 - t_0) \cdot \lambda$$

$\Delta\lambda_{t1}$: Deler av en bølgelengde, bestemmes ved fasemålinger.

$Int(t_1-t_0)$: Antall hele bølgelengder avstanden har endret seg siden tidspunktet t_0 .

De ukjente i disse forenklete observasjonsligningene er som vist N_{t0} og $\Delta\lambda_{t1}$, mens måleverdien $Int(t_1-t_0)$ registreres på mottakeren.

Legg merke til at avstanden til satellitten ikke fås direkte ved fasemåling, det må måles i et visst tidsrom for å bestemme heltallet N .

En forutsetning for fasemålinger er at mottakeren ikke mister signalet fra satellittene, det vil si at målingene er kontinuerlige. Hvis den mister kontakten med satellittsignalet, sier vi at det oppstår et *fasebrudd* (cycle slip). Idet mottakeren klarer å måle fasen på satellittsignalet igjen, vet den ikke hvor mange hele bølgelengder satellitten har nærmet eller fjernet seg i den tiden som har gått siden sist mottakeren hadde kontakt. Det lar seg gjøre å beregne fasebrudd i ettertid. Se også figur 5.4 i kapittel 5.3

5.2 Phase model, measured phase

Vi skal i det følgende se mer i detalj på hvordan en fullstendig observasjonsligning for en fasemåling kan utledes.

Mottakerne er i stand til å måle differansen (faseforskjellen) mellom bærebølgen som er generert av den interne oscillatoren i mottakeren, og bærebølgen som kommer fra satellittens oscillator.

Figur 5.2 viser bærebølgen før den er påmodulert koden (får modulert bærebølge).

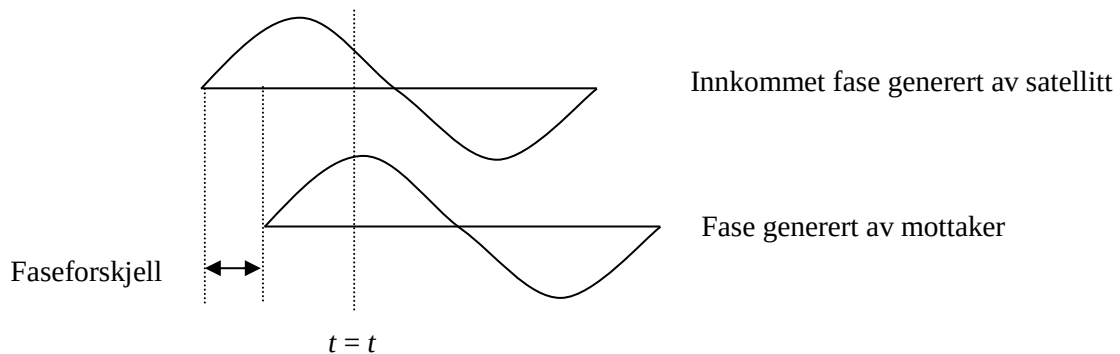


Figure 5.2 Phase difference at the point of time t

Denne differansen er "the carrier beat phase observable", fasemålingen, altså faseforskjellen mellom det innkommende Doppler-skiftet satellitt-bæresignalet og referansesignalet som er generert i mottakeren. Doppler-skiftet fås ved at det er en relativ bevegelse mellom satellitt og mottaker, noe som gir endring i mottatt frekvens i mottaker, se mer om Doppler i kapittel 7.1.1.

Disse to signalene blandes, slik at man får ut et signal som har en frekvens lik forskjellen mellom frekvensene for de to signalene. Denne frekvensen kalles *beatfrekvensen* (slagfrekvensen, svevningsfrekvensen).

Fasen til beatfrekvensen er lik forskjellen i fase for det innkomne signalet og det genererte signalet i mottakeren. Beatfrekvensen kan derfor anvendes videre i utviklingen av en observasjonsligning for fasemålinger, da den har informasjon om ønsket faseforskjell.

Faseforskjell uttrykt i perioder (cykler), se også figur 5.2:

$$\varphi = \varphi_i^k = \varphi^k(t) - \varphi_i(T) \quad (5.1)$$

$\varphi^k(t)$ er fasen til signalet som ble sendt ut fra satellitt k ved tidspunktet t (satellittklokke-tid), når signalet når mottakeren

$\varphi_i(T)$ er fasen på generert signal i mottaker ved mottakerklokke-tiden T

Ligningen kan omformes ved:

- (a) å bruke fase/frekvensforholdet
- (b) å relatere sendetid og mottakertid til GPS-tidsstandarden

- (a) Fase/frekvensforholdet som er gyldig for høystabile oscillatorer som satellitt-oscillatorene, over et kort tidsintervall, er generelt:

$$\varphi(t + \delta t) = \varphi(t) + f \cdot \delta t \text{ (målt i antall perioder (cyclers), } f \text{ er frekvensen)}$$

Med $\delta t = T - t$ (medgått tid fra utsendelse til mottak, målt med klokken i satellitt og mottaker), kan vi sette inn på begge sider av likhetstegnet:

$$\varphi^k(t + \delta t) = \varphi_i(T) = \varphi^k(t) + f \cdot (T - t)$$

som settes inn i (5.1), og vi får følgende uttrykk for fasedifferansen:

$$\varphi_i^k = \varphi^k(t) - \varphi_i(T) = \varphi^k(t) - \varphi^k(t) - f \cdot (T - t)$$

$$\varphi_i^k = -f \cdot (T - t) \quad (5.2)$$

- (b) Sendetid og mottakertid er relatert til hverandre. De systematiske feil som påføres et signal som sendes ut fra satellitten skal det også tas hensyn til, se også forrige kapittel om kodemåling for forklaring av symbolene:

Sendetidspunkt, i GPS-tid:	$\tau^k = t + dt$
Vektoravstand:	ρ
Signal-tid i vakuum:	ρ / c
Atmosfæriske forstyrrelser, i tid:	$(-d^{\text{ion}} + d^{\text{trop}}) / c$
Banefeil, i tid:	$d\rho / c$
Mottakertidspunkt, i GPS-tid:	$\tau_i = T + dT$

Som gir følgende sammenheng ved mottakertidspunktet, når det er tatt hensyn til forstyrrelser/feil på signalgangen mellom satellitt og mottaker

$$\tau_i = \tau^k + \Delta\tau = \tau^k + \rho / c + (-d^{\text{ion}} + d^{\text{trop}} + d\rho) / c$$

$$(T + dT) = (t + dt) + (\rho - d^{\text{ion}} + d^{\text{trop}} + d\rho) / c$$

Og med litt omstokking:

$$T - t = (dt - dT) + (\rho - d^{\text{ion}} + d^{\text{trop}} + d\rho) / c \quad (5.3)$$

Settes (5.3) inn i (5.2), fås følgende ligning for fasemodellen (målt fase):

$$\varphi_i^k = -f \cdot (T - t) = -\rho \cdot (f/c) - f \cdot (dt - dT) - (-d^{\text{ion}} + d^{\text{trop}} + d\rho) \cdot (f/c) \quad (5.4)$$

Sammenhengen mellom bølgelengde, lyshastighet i vakuum og frekvens, der λ er bæreølgen for frekvensene L1 eller L2:

$$\lambda = c / f$$

Innføres i (5.4), og gir en annen versjon av ligning (5.4):

$$\varphi_{\text{målt}} = \varphi_i^k = -\rho / \lambda - (c / \lambda) \cdot (dt - dT) - (-d^{\text{ion}} + d^{\text{trop}} + d\rho) / \lambda \quad (5.5)$$

φ_i^k er målt faseforskjell, for de andre symbolene og for den fullstendige observasjonsligningen for fasemåling (ligningen for total faseforskjell), se ligning (5.11) i neste kapittel.

5.3 Observation equation, the total phase difference

Fasemålingen ved et bestemt tidspunkt er basert på fasesynkronisering mellom mottakerklokka og det innkommende signalet. Men man kjenner ikke hvilken periode (cykel) som representerer den nøyaktige synkroniseringen.

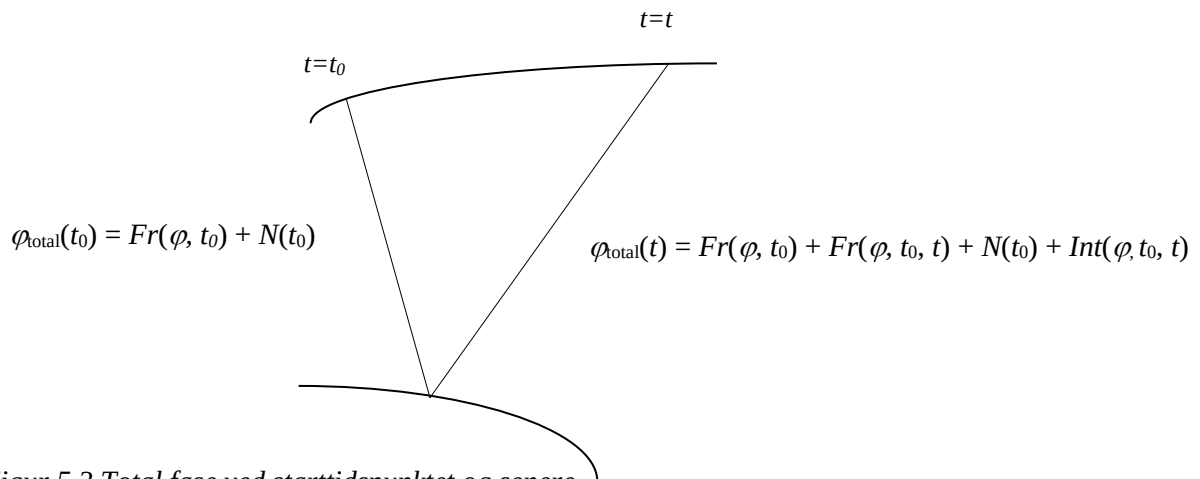
Den totale fasen, φ_{total} , består derfor av følgende deler (se også figur 5.3):

$$\text{Ved starttidspunktet } t = t_0: \varphi_{\text{total}}(t_0) = Fr(\varphi, t_0) + N(t_0) \quad (5.6)$$

$$\text{Ved tidspunktet } t = t: \varphi_{\text{total}}(t) = Fr(\varphi, t_0) + Fr(\varphi, t_0, t) + Int(\varphi, t_0, t) + N(t_0) \quad (5.7)$$

Forklaring av de ulike leddene i ligning (5.7):

$Fr(\varphi, t_0)$:	En fraksjonell fasedel, del av en hel bølgelengde ved tidspunktet t_0
$Fr(\varphi, t_0, t)$:	Endring i fraksjonell fasedel mellom starttidspunktet t_0 og tidspunktet t
$Int(\varphi, t_0, t)$:	Et helt antall perioder (cykler), regnet fra starttidspunktet t_0 og talt opp fram til tidspunktet t
$N(t_0)$:	Et ukjent heltall av perioder ved starttidspunktet t_0



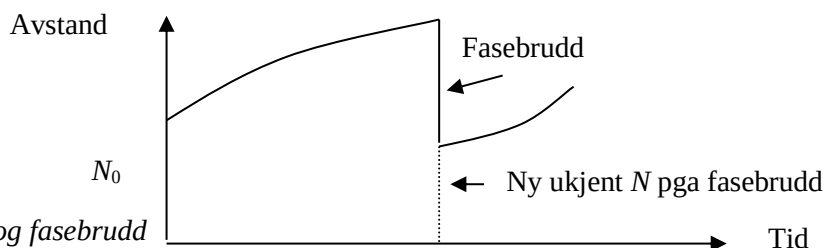
Figur 5.3 Total fase ved starttidspunktet og senere

Den totale fasen, φ_{total} , kan først beregnes når $N(t_0)$ er beregnet. Heltallet N kalles for *den heltallsukjente* ("cycle ambiguity").

Mottakeren starter målingene ved tidspunktet t_0 . Ved fasemålingen blir endringer i $Fr(\varphi)$ og $Int(\varphi, t_0, t)$ registrert (gir avstandsendringer fra starttidspunktet). For å få den reelle avstanden mellom mottaker og satellitt, må den ukjente N ved *starttidspunktet* bestemmes.

Er det ingen brudd i mottaket av signalet fra satellitten, er det 1 ukjent N per satellitt/mottaker-par og per bæreølge (L1, L2). Hvis det for eksempel måles mot 7 satellitter med en to-frekvent mottaker, er det til sammen 14 ukjente heltall som skal bestemmes.

Hvis mottakeren får et brudd i fasemålingene (mister signalet fra satellitten), vil man få flere heltallsukjente N -er per satellitt, da mottakeren starter tellingene på nytt på et vilkårlig heltall etter bruddet (etter hvert brudd). Programvaren som beregner vektorene, vil kunne reparere de fleste av disse fasebruddene i ettertid, eventuelt vil man mot noen satellitter bli stående igjen med flere N -er i beregningene.



Figur 5.4 Avstandsending og fasebrudd

Målt fase er sammensatt av målt faseforskjell ved starttidspunktet t_0 og avstandsendringen mellom tidspunktene t_0 og t , se under ligning (5.7) for forklaring av leddene i ligningen:

$$\varphi_{\text{målt}} = Fr(\varphi, t_0) + Fr(\varphi, t_0, t) + Int(\varphi, t_0, t) \quad (5.8)$$

Sammenligner (5.7) og (5.8) og får:

$$\varphi_{\text{total}} = \varphi_{\text{målt}} + N(t_0) \quad (5.9)$$

En fasemåling i en mottaker mot en satellitt på et bestemt tidspunkt, kan skrives som (se ligning (5.5) og (5.9)):

$$\varphi_{\text{total}} = -\rho / \lambda - (c / \lambda) \cdot (dt - dT) - (-d^{\text{ion}} + d^{\text{trop}} + d\rho) \cdot \lambda + N(t_0) \quad (5.10)$$

I ligning (5.10) er benevnningen perioder (cykler). Ved å multiplisere (5.10) med bæreboelgen λ (L1 eller L2) fås benevnningen meter.

Definisjon av sammenheng og fortegn: $\Phi_{\text{total}} = -\lambda \cdot \varphi$

som gir observasjonsligningen for fasemåling (i enheten meter):

$$\Phi_{\text{total}} = \rho + c \cdot (dt - dT) + \lambda \cdot N(t_0) - d^{\text{ion}} + d^{\text{trop}} + d\rho \quad (5.11)$$

- Φ_{total} : Total fase, måleverdi, i meter
 ρ : Vektoravstanden satellitt-mottaker, inneholder koordinatene til mottaker og satellitt
 c : Lyshastighet i vakuum
 $dt - dT$: Klokkefeil på satellitt- og mottakerklokke
 λ : Bølgelengden for bæreboelgen (L1 eller L2)
 N : Den heltallsukjente ved starttidspunktet t_0
 d^{ion} : Ionosfærisk forsinkelse
 d^{trop} : Troposfærisk forsinkelse
 $d\rho$: Feil på satellittens banedata

Ligning (5.11) er direkte sammenlignbar med ligningen for pseudoavstand (5.5), med unntak av:

- Leddet med N , det ukjente heltallet ved starttidspunktet
- Fortegnet på leddet for ionosfæreforsinkelse. Endringen av fortegnet skyldes at signalforsinkelsen gjennom ionosfæren er forskjellig for bæreboelgen (fasemåling) og for gruppehastigheten (kodemåling). Forsinkelsen gjennom ionosfæren er også frekvensavhengig, noe som vil gi ulik forsinkelse for bæreboelgene L1 og L2 for en GNSS-mottaker (kan utnyttes for å beregne/modellere forsinkelsen)
- Måleverdien Φ_{total} kan ikke ses på som avstanden mellom satellitt og mottaker, slik som måleverdien (pseudoavstanden) ved kodemålingen. Dette kan også sees av høyresiden i ligning (5.11), der ρ er vektoravstanden og $\lambda \cdot N$ er ca. avstanden ved starttidspunktet.

6. Sources of errors and factors of uncertainties of GNSS

Det er mange feilkilder og usikkerhetsfaktorer som påvirker nøyaktigheten på satellittbestemte punkter og vektorer. Vi skal her se litt nærmere på noen av disse, og spesielt vurdere hvilken innvirkning de vil ha, når vi nytter metoden med relativ posisjonsbestemmelse i landmåling. For mer om spesielt temaene flerveisinterferens, antennefeil, atmosfæren og A-S, se andre lærebøker.

- Feil på banedataene for satellittene

- Feil på koordinatene og høyden på referansepunktet ved basisberegninger
- Klokkefeil for satellittene
- Klokkefeil i mottakeren
- Flerveisinterferens (multipath) i mottakerstasjonen
- Antennefeil
- Operatørfeil
- Atmosfæriske forhold, spesielt i ionosfæren og troposfæren
- SA ("Selective availability")
- A-S ("anti-spoofing")

6.1 Error and uncertainties of the satellite orbits (ephemerides error)

Satellittene går i svakt elliptiske baner rundt Jorda med en hastighet på ca. 4 km/s. Satellittene er utsatt for mange forstyrrende krefter som påvirker baneforløpet (tiltrekningskraft fra Jorda, sola og månen, solstrålingstrykk, Jordas magnetfelt osv.). Disse forstyrrende kreftene gjør det vanskelig å beregne og predikere satellittbanene helt eksakt. De får faktisk så stor innvirkning at banekursen av og til må korrigeres ved hjelp av satellittenes jetmotorer. Siden informasjonen om satellittbanene som ligger i satellittmeldingen som sendes ut med GPS-signalet, er beregnet på grunnlag av målinger som ofte er gjort flere timer i forveien, vil denne baneinformasjonen inneholde feil.

På bestemmelsen av enkeltpunkter kan banefeil på 25 m i verste fall gi tilsvarende feil i punktets koordinater. Ved relativ punktbestemmelse vil imidlertid virkningen av den samme feilen bli redusert til 1 ppm av vektorlengden.

6.1.1 Orbits, from the GPS system and on internet

Virkningene av banefeil er størst når de utsendte banedata brukes, de er predikterte banedata og er av dårligere kvalitet, enn banedata beregnet i ettertid.

IGS ("International GNSS Service for Geodynamics") er en internasjonal organisasjon som blant annet leverer produkter som:

- Nøyaktige banedata (eferemider), beregnet i ettertid
- Nøyaktig informasjon om satellittenes klokker
- Ionosfæriske modeller

IGS har flere innsamlings- og regnesentre. Ett av de er CODE, som er "Centre for Orbit Determination in Europe", hvor Universitetet i Bern er en sentral aktør, se internettadresse i litteraturlista.

De nøyaktige banedata fra IGS, er beregnet ut i fra ca. 200 målestasjoner som samler inn GNSS-måledata og annen måledata kontinuerlig.

I tabellen nedenfor, skilles IGS mellom banedata som er spådd ("predicted", IGP), beregnet raskt i ettertid ("rapid", IGR) og de endelige og de beste verdier ("final", IGS). En ny banedata-beregning prøves ut (november 2000), den ligger mellom IGP og IGR i nøyaktighet, og kalles "ultra rapid", IGU.

Med bruk av mer presise banedata enn de som sendes ut fra satellittene, reduseres også feilene på posisjon og vektor tilsvarende.

Eksempel på nøyaktighet på ulike versjoner av banedata, som er tilgjengelig (februar 2001):

Orbits	Accuracy orbits	Accuracy sat. clock	When available	Where
Emitted "broadcast"	3 m		I sann tid	Utsendt via GPS
IGS Predicted, IGP	0,5 m	150 ns *	I sann tid	IGS datasentre
IGS Ultra rapid, IGU	0,25 m ?	0,5 ns	I sann tid	IGS datasentre
IGS Rapid, IGR	0,10 m	0,3 ns	After 24 hours	IGS datasentre
IGS Final, IGS	0,07 m		After 2 weeks	IGS datasentre

*: 1 ns = 1 nanosekund = $1 \cdot 10^{-9}$ sekund. (Hvor mye utgjør det i signallengde?)

Legg merke til at IGS-banedata i sann tid er vesentlig bedre enn de utsendte banedataene.

En internettadresse for presise banedata fra IGS, er: <http://igsb.jpl.nasa.gov>. Se litteraturlista bak i kompendier for flere adresser på internett.

6.1.2 Error on a measured baseline caused by erroneous orbits

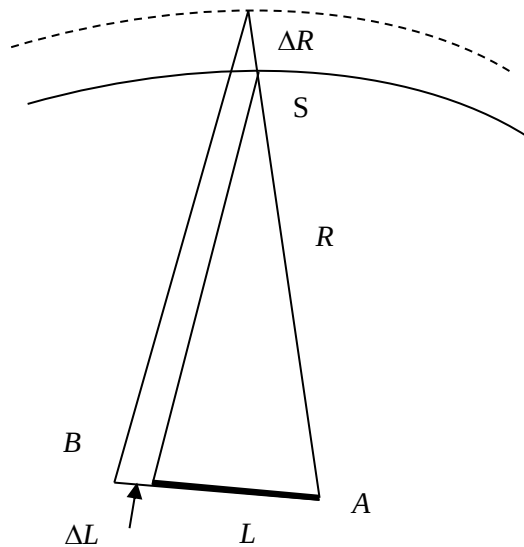
På figuren skal vektoren mellom A og B bestemmes. Punkt A er referansepunktet med gode koordinater og punkt B er det nye punktet, eller punktet med den mobile mottakeren, "roveren".

Satellitten S sender ut sine banekoordinater gjennom eferemidefilen, "broadcast orbit".

Vi kan dele banefeilen i tre komponenter:

- Radiell feil: Banefeil i retning av baneradien av satellittbanen
- Tangensiell feil: Banefeil langs satellittbanen
- Banefeil på tvers av satellittbanen

Radiell feil:



Av figur 6.1 kan følgende formel utledes:

$$\frac{L}{R} = \frac{L + \Delta L}{R + \Delta R}$$

$$\text{som gir: } \frac{\Delta L}{L} = \frac{\Delta R}{R} \quad (6.1)$$

Vi ser av formelen at en radiell feil i satellittbanedataene virker som en målestokksfeil.

Avstanden A-S beregnes av de kjente koordinatene for referansepunktet A og satellitten. Denne kjente avstanden mellom referansepunkt A og satellitten, får en feil på ΔR . Dermed må trekanten (B-A-S) på figur 6.1 forstørres, for å stemme med oppgitt, men feil banedata.

Figur 6.1 Radiell feil i banedata. NB: Figuren er fortegnet: Avstanden A-B er på ca 10000 km på figuren.

ΔR langs linja A-S er ikke helt identisk med den radielle banefeilen, da sentrum for satellittens bane ligger i jordsentrum og punkt A befinner seg på jordoverflata.

Eksempel, virkningen av 1 radiell satellittbanefeil:

Radiell feil for en satellitt, "broadcast orbit":	$\Delta R = 5 \text{ m}$
Gjennomsnittlig avstand fra mottaker til satellitten:	$R = \text{ca } 25000 \text{ km}$
Vektorlengde mellom de to mottakerne:	$L = 10 \text{ km}$

$$\text{Formel (6.1) gir en feil på vektoren på } 10 \text{ km på: } \Delta L = 5000 \cdot (10 / 25000) = 2 \text{ mm}$$

Av eksemplet ses at en radiell banefeil på 5 m, gir en feil på 0,2 ppm på vektoren.

Legg merke til at eksemplet ovenfor viser hvordan en feil på 1 satellitt forplantet seg til en feil på vektoren (målestokksfeil). Det måles mot flere satellitter samtidig. Satellittene har normalt ikke like store banefeil, og det er summen av virkningene av de ulike satellitters banefeil på vektoren mellom to mottakere som er interessant

Ved å gjøre et studie av varians/kovariansmatrisa (feilforplantning) til ulike satellittkonstellasjoner, kan følgende sammenheng utvikles (Zielenski 1988):

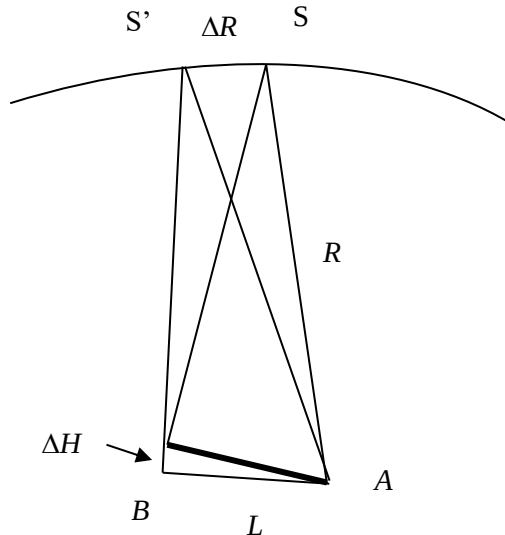
$$\frac{|\Delta R|}{10 \cdot |R|} \leq \left| \frac{\Delta L}{L} \right| \leq \frac{|\Delta R|}{4 \cdot |R|} \quad (6.2)$$

Merknad 2001: Det er logisk at vi får et lavere estimat på den totale virkningen av alle banefeil på vektoren A-B, sammenlignet med formel (6.1), som viser virkningen av 1 banefeil. Årsak: Banefeilene er ulike i størrelse og fortegn, de virker til dels mot hverandre (som tilfeldige feil), noe som gir de lavere grensene i formel (6.2). Men har ikke sjekket denne formelen mot originalen, før denne utgaven av kompendiet.

Banefeil langs satellittbanen:

Av figuren kan følgende formel utledes: $\Delta H / L = \Delta R / R$

$$\text{som gir: } \Delta H = \Delta R \cdot (L / R) \quad (6.3)$$



Vi ser av formelen at en banefeil langs satellittbanen, gir en høydefeil i punkt B, vektorens vertikalvinkel blir feil.

Avstanden A-S beregnes av de kjente koordinatene for referansepunktet A og satellitten. Denne kjente avstanden mellom referansepunkt A og satellitten, må dreies om A til S', som er den feilaktig oppgitte posisjonen langs satellittbanen, ΔR , (samtidig som den får en liten korleksjon i lengde). Dermed blir også vektoren A-B dreid med samme vinkel som A-S, for å stemme med oppgitt, men feil banedata.

Figur 6.2 Banefeil langs satellittbane. NB: Figur 6.2 er fortegnet: Avstanden A-B er på ca 10000 km.

Eksempel, virkningen av 1 banefeil langs satellittbanen:

Feil for en satellitt, "broadcast orbit":	$\Delta R = 5 \text{ m}$
Gjennomsnittlig avstand fra mottaker til satellitten:	$R = \text{ca } 25000 \text{ km}$
Vektorlengde mellom de to mottakerne:	$L = 10 \text{ km}$

Formel (6.3) gir en feil på vektoren på 10 km på: $\Delta H = 5000 \cdot (10 / 25000) = 2 \text{ mm}$

Banefeil på tvers av satellittbanen

Denne feiltypen minimal innflytelse på avstanden A-S ved normale størrelser på denne feilen. Årsaken er også den lange avstanden til satellitten, som gjør at en liten feil ikke endrer avstanden og retningen til satellitten.

6.2 Baseline processing: Errors on coordinates of the reference point

6.2.1 Error in the height of the reference point

Denne feiltypen virker på samme måte som radiell feil på satellittens banedata, se figur 6.1 og formel 6.1. Forskjellen på figuren er at ΔR nå er ved punkt A, og ikke ved satellitt S.

Avstanden A-S beregnes av de kjente koordinatene for referansepunktet A og satellitten S. Denne kjente avstanden mellom referansepunkt A og satellitten, får en feil på grunn av feilen ΔH i punkt A. Dermed må trekanten (B-A-S) på figuren forstørres/forminskes for å stemme med oppgitt, men feil høyde på A. Feilen forplanter seg til basislengden A-B slik:

$$\Delta L = -\Delta H \cdot (L / R) \quad (6.4)$$

Vi ser av formelen, at en feil på høyden på referansepunktet ved basisberegninger, virker som en målestokksfeil. Minusfortegnet forklares ved at for lav høydeverdi gir for lang avstand, L.

Eksempel, virkningen av en feil på høyden i referansepunktet:

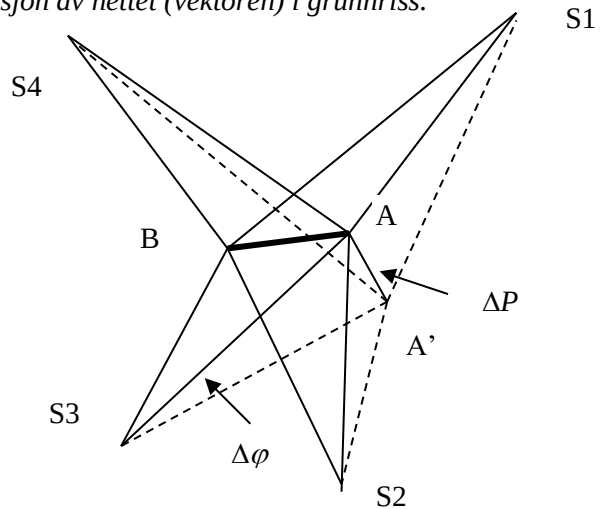
Feil på høyden, ortometrisk høyde i stedet for ellipsoidisk:	$\Delta H = 40 \text{ m}$
Gjennomsnittlig avstand fra mottaker til satellitten:	$R = \text{ca } 25000 \text{ km}$
Vektorlengde mellom de to mottakerne:	$L = 10 \text{ km}$

Gir en feil på vektoren på 10 km på: $\Delta L = 10 \cdot (0,040 / 25000) \cdot 10^6 = 16 \text{ mm} = 1,6 \text{ ppm}$

6.2.2 Error in the coordinates of the reference point

Denne feiltypen gir en deformasjon av nettet (vektoren) i grunnriss.

Figur 6.3 Deformasjon pga feil koordinater i referansepunktet A



Vi ser av figur 6.3, at en feil i koordinatene til referansepunktet A, kan gi en *dreining* og en *målestokksfeil* på vektoren mellom A og B. Størrelsen på denne feilforplantningen er avhengig av satellittkonstellasjon og vektorens lengde og retning.

Avstanden og retningen A-Si beregnes av de kjente koordinatene for referansepunktet A og satellitten Si. Denne kjente avstanden mellom referansepunkt A og satellitten, blir deformert til A'-Si, for å stemme med oppgitte, men feil koordinater på referansepunktet. Dette gir en dreining og en målestokkdeformasjon på vektormålingene.

NB: Med flere satellitter å ta hensyn til, blir den totale deformasjonen mer kompleks å beregne.

Eksempel på feil koordinater på referansepunktet:

Vi ser på satellitt S3 på figuren, der vektoren ΔP står ca. vinkelrett på retningen til satellitt S3.	
Feil på koordinatene, absoluttverdien fra mottakeren er brukt:	$\Delta P = 15 \text{ m}$
Gjennomsnittlig avstand fra mottaker til satellitten, i grunnriss:	$R = \text{ca. } 15000 \text{ km}$
Vektorlengde mellom de to mottakerne:	$L = 10 \text{ km}$

Vi får en dreining av retningen A-S3 på: $\Delta\phi = \Delta P / R = 0,00006 \text{ gon}$
 Som gir et tverravnvik på 10 mm på avstanden A-B på 10 km, som tilsvarer 1 ppm.

Eksemplet indikerer at vi må ha svært gode absoluttverdier på posisjonen på referansepunktet, bedre enn 10 meter, for at vi ikke skal få feil på vektoren på grunn av feile utgangskoordinater.

6.3 Satellite clock error

GPS-satellittene flere meget presise klokker. Problemet ligger i at klokkene ikke er fullkomment synkronisert til GPS-tid. Kontrollstasjonene beregner et avvik fra sann GPS-tid som brukeren mottar i satellittmeldingen ved målingene. Ved relative målinger, vil den resterende feilen bli eliminert.

6.4 Receiver clock error

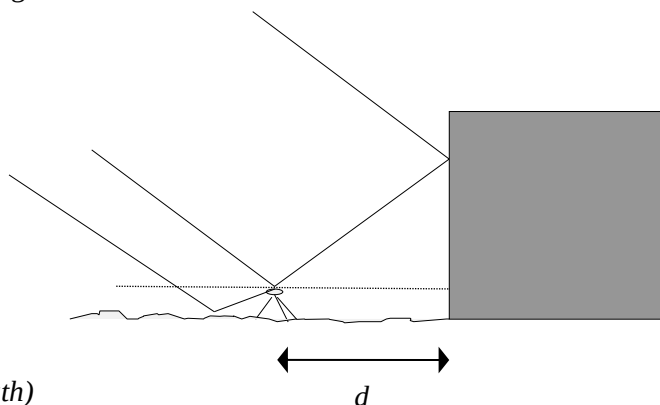
Mottakerens klokke ikke så nøyaktig som klokka i satellittene. Selv om vi beregner mottakerklokkes avvik fra GPS-tid, er denne beregningen ikke eksakt, og vi sitter igjen med restfeil som *ikke blir eliminert ved vanlig relativ måling*. Derfor betraktes mottakerens klokkefeil som en ukjent i beregningene.

I enkeltpunktbestemmelse med kodemålingene er den fjerde ukjente i tillegg til mottakerens koordinater, det må måles til minst 4 satellitter for å bestemme posisjonen (4 ukjente, minst 4 observasjoner trengs).

6.5 Multipath in the receiver station (flerveisinterferens)

En feilkilde som krever spesiell omtanke fra operatøren, er *flerveisinterferens* (multipath, MP).

Flerveisinterferens er at antennen mottar et satellittsignal som er reflektert fra en eller annen gjenstand i nærheten av antennen, istedenfor det signalet som kommer direkte fra satellitten.



Figur 6.4 Flerveisinterferens (multipath)

Typiske reflektorer kan være biler, trær, bakke eller bygninger, og refleksjonen kan være sterk eller svak (avhengig av den reflekterende flaten). I slike tilfeller er det ikke den riktige avstanden mellom antennen og satellitten som blir målt, fasemålingene på det direkte signalet blir forstyrret av det signalet som går via den reflekterende flaten og til antennen. Ved planlegging av målinger, bør operatøren være oppmerksom på mulige refleksjonskilder i nærheten av det punktet som skal måles inn.

MP gir feil i beregningene av avstandene til satellitter, og dermed også feil i beregningene av antennens posisjon.

Flerveisinterferens (multipath) er i dag en av de største feilkildene ved nøyaktige, relative/differensielle statiske og kinematiske GNSS-målinger. Den influerer på fase- og kodemålinger og gir et lavere signal/støyforhold.

Vi skiller mellom:

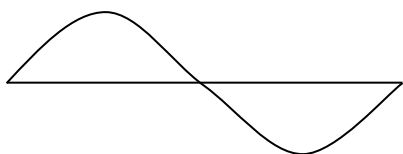
- indirekte signaler fra bakken, som ligger lavere enn antennen
- indirekte signaler fra objekter som høyere enn antennen (bygninger mm)

Fra bakken:

Bakken reflekterer signal godt, det er liten demping, spesielt vannflate. Det er alltid multipath fra bakken.

Antennen kan ha en beskyttende tallerken (ground plate) som demper/hindrer det indirekte signalet fra bakken, og man kan benytte spesielle antenner, choke-ring-antenner som er spesialkonstruert for å hindre flerveisinterferens nedenfra. Se figur 6.5.

Selv med tallerkener eller spesialantenner må det regnes med restfeil, da noe av det indirekte signalet når fram til antenna.



Flerveisinterferens på satellittsignalet som går via bakken, har en syklisk karakter, feilen varierer i løpet av tiden. En tommelfingerregel er at perioden på flerveisinterferens, der d er avstanden til den reflekterende flaten (se figur 6.4), er:

$$P = 40 \text{ min} / d \quad (6.5)$$

Perioden avhenger av antennehøyden, med en antennehøyde på 2 m, er perioden på 20 minutter. Dermed kan vi ved sanntidsmålinger med svært korte måletider, risikere å måle mens det er en maksimalverdi på denne feilkilden.

Betydningen av MP's periodiske oppførsel for GNSS-målinger:

Av formel (6.5) fås at perioden er på $P = 40$ min når $d = 1$ m, og $P = 4$ min når $d = 10$ m.

Den maksimale feilen på avstanden til en satellitt med flerveisinterferens (MP), kan være ca. 4-5 cm på fasemålinger. Man bør måle slik at man minst dekker hele perioden for MP. Ved korttidsstatiske målinger (måletid på 5-10 min) og ved RTK (måletid 10 sekunder), kan man risikere å måle i bare deler av perioden til MP og dermed få en systematisk feil på målingene. Dette gjelder spesielt hvis det er gode reflekterende flater svært nær antenna.

Over antenna:

Ved oppstilling nær bygninger, trær mm, nærmere enn ca 10 m, kan man få feil på flere cm på fasemålingen, men ikke typisk syklisk variasjon på feilen her, da feilen er avhengig av hvordan signalet blir reflektert.

Dette indirekte signalet har såpass lengre vei å gå (se figur 6.4) sammenlignet med bakkerefleksjonen, at det indirekte signalet lettere kan separeres fra det direkte, noe som utnyttes i beregningene for eliminering av MP.

Man kan separere og fjerne MP, de indirekte signalene, men kan derved få mer støy (fjerner noe av det direkte også), spesielt hvis den indirekte signallengden er på flere meter.

Choke ring antenne

Figur 6.5 Choke ring antenne. Hentet fra <http://www.airdata.ca/chokante.html>

Denne antenntypen er spesiallaget for å redusere flerveisinterferens betydelig, spesielt ved lave høydevinkler. Det spesielle med choke ring antennene er de konsentriske ringene rundt selve antennen. Denne typen antenner anbefales for permanente basestasjoner og for særlig nøyaktige målinger.

6.5.1 Summary, multipath:

Spesielt ved korte måletider kan flerveisinterferens være en av de største feilkildene ved GNSS-målingene, både fase- og kode-målinger. Denne feilkilden er lokal for en mottaker og den blir ikke eliminert ved relative målinger.

Ved sanntidsmålinger med måletider ned mot ca. 10 sekunder (1-2 epoker), må man derfor være spesielt oppmerksom på om flerveisinterferens kan ødelegge målingene. Målingene kan ha blitt utført da den tidsavhengige feilen var stor.

Gjenværende MP kan reduseres ved:

- Bruk av nyere antenner/software
- Antenneplassering, beskyttelse. Antenna kan skrus på bolten på bakken (jfr. Stamnettpunktene)
- Antennas følsomhet for lave signaler
- Mottakerens evne til å fjerne MP
- Måle i minst 1 periode, snittet av målingene blir bra.

For en basestasjon kan vi velge plassering, men for de punktene vi skal måle inn, må vi vurdere om hvordan effektene av MP kan reduseres.

6.6 Errors on the antenna, electronic noise

Ved nøyaktige målinger må en også passe på at antennene er dreid slik at nordpilen på antennene er mot nord. Dermed får en eliminert eventuell forskyvning i grunnriss av *fasesenteret* for antenna (nullpunktet i antenna), *forutsatt at en bruker samme type antenne for alle mottakerne*.

I vertikalretning kan fasesenteret variere, avhengig av frekvens (L1, L2) og senitvinkel på innkommet signal. Ved svært nøyaktige målinger, må derfor de antennene som er i bruk, kalibreres, slik at det er mulig å korrigere for variasjonen på fasesenteret. For vanlig bruk, er denne feilkilden i vertikalretning innenfor målenøyaktigheten.

Ren elektronisk støy og svakheter i *mottakerens registreringsenhet* gir målefeil som det ofte er vanskelig å eliminere. Samtidig kan slike svakheter føre til at mottakeren har lett for å miste kontakt med satellittsignalet, og vi får brudd i fasemålingen (cycle slip).

6.7 Errors done by the surveyor

Vær spesielt oppmerksom på:

- *Kontroll av utstyr* som trefoten (det optiske loddet og libellen).
- *Nøyaktighet ved oppstilling* av stativ, og ved bruk av optisk lodd og libelle. Husk at vektoren måles med en nøyaktighet på ca. 5 mm for en vektor på 1 – 5 km, da må kvaliteten på oppstillingen være ”på millimeteren”.
I de ca. 930 punktene i Stamnettet til Statens kartverk, kan antennen skrus direkte ned på bolten, uten bruk av stativ. Dermed elimineres ev feil i oppstillingen av antenna, samtidig som flerveisinterferens fra bakken også elimineres.
- *Måling av antennehøyde*. Helst bør det stilles opp minst to ganger i et punkt, med uavhengige målinger av antennehøyde. En feil i antennehøyden i et punkt der det bare er stilt opp en gang, vil normalt ikke bli oppdaget i beregningene.

6.8 Atmospheric problems, especially in ionosphere and troposphere

På sin vei gjennom atmosfæren blir radiosignaler fra satellittene avbøyd, slik at signalveien beskriver en krum linje istedenfor en rett. Samtidig blir også hastigheten til signalene forandret. Begge disse fenomenene gir feil i avstandene vi måler til satellittene. Gjennom de øvre lagene av atmosfæren, *ionosfæren*, virker denne avbøyningen ulikt på forskjellige signalfrekvenser.

Troposfæren og ionosfæren kan modelleres, eller man kan legge inn beregnede modeller for atmosfæriske korreksjoner fra IGS for beregninger *i ettertid*. Se bakerst for internettadresser.

Fortegn på korreksjoner

I observasjonsligningene for kodemåling (4.3) og fasemåling (5.11), ser vi at fortegnet på leddet for ionosfære- og troposfære-forsinkelse/feil er ulike. Endringen av fortegnet skyldes at signalforsinkelsen gjennom ionosfæren er forskjellig for bæreølgen (fasemåling) og for gruppehastigheten (kode-måling). Bæreølgen for fasemålingene får høyere hastighet, mens gruppehastigheten for kodemålingene blir forsinket i ionosfæren. I tallstørrelse er korreksjonene like store.
Dette medfører at fasemålingene gir for lang avstand og kodemålingene for kort, sammenlignet med den geometriske avstanden mellom satellitt og mottaker.

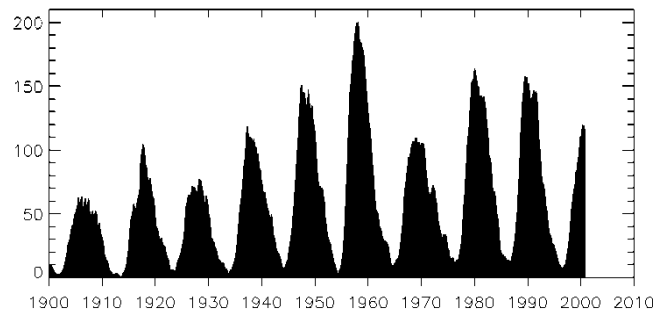
Ved relative målinger vil atmosfærisk forstyrrelser i stor grad bli eliminert, men denne feiltypen er normalt hoved årsaken hvis beregningene ikke klarer å løse ut de ukjente heltallene (N) ved fasemåling.

6.8.1 Ionosphere

Ionosfæren er et skall av elektroner og elektrisk ladede atomer rundt Jorda. Mellom ca 50 –1000 km over jordoverflaten.

Dette ioniserte mediet influerer på GNSS-signalene.

Aktiviteten i ionosfæren har en syklus på ca. 11 år. Som det framgår av figur 6.6, passerte vi i 2003 toppen av den ionosfærisk aktiviteten.



Figur 6.6 Solflekk-sykluser 1900-2001 (fra <http://sidc.oma.be/>)

Forsinkelsen gjennom ionosfæren er frekvensavhengig, noe som vil gi ulik forsinkelse for bæreølgene L1 og L2 for en GPS-mottaker (kan utnyttes for å beregne/modellere forsinkelsen). Ved å ha mottakere som måler på begge GPS-signalene, L1 og L2, kan vi derfor til en viss grad beregne målefeilen ut fra en kjent matematisk modell. Vi oppnår derfor bedre målenøyaktighet med tofrekvente GPS-mottakere.

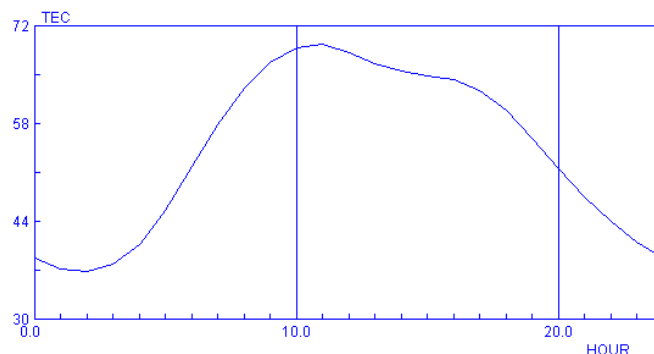
Et problem ved mye støy, er at koden på L2 er ikke tilgjengelig for sivile mottakere på grunn av A-S (anti-spoofing, se dette). De ulike GPS-mottakerne har ulike metoder for å følge L2-signalene uten kjennskap til denne koden. Disse metodene kan være mindre robuste ved høy aktivitet i ionosfæren.

Virkningen av ionosfærisk støy varierer med år, årstider og tid på døgnet. Den er veldig vanskelig å forutsi. Har vi enfrekvente mottakere (som bare måler på L1), har vi ingen mulighet til å estimere målefeilen. Her i nordområdene har ionosfærisk støy skapt ekstra store problemer for GPS-brukere.

I modellering av ionosfærisk aktivitet, brukes måleenheten TEC, totalt electron count, som er antall elektroner for et "rør" med tverrsnitt 1 m^2 tenkt lagt langs signalbanen. TEC er et mål for elektrontettheten.

TEC er foruten solaktiviteten, også avhengig av:

- Breddegrad
- Varierer med døgnet: Laveste verdier om natta, da vi ligger i skyggen av solvindene
- Varierer med høydevinkelen for mottaker til satellitt. Står satellitten lavt, har signalet en lengre vei gjennom ionosfæren



Figur 6.7 TEC for stedet $67^\circ\text{N } 14^\circ\text{Ø}$, den 7.mai 2001. En TEC-enhet (TECU) er $1 \cdot 10^{16} \text{ el/m}^2$

Figur 6.7 viser TEC beregnet ved bruk av modellen IRI (The International Reference Ionosphere), som ligger på adressen: http://omniweb.gsfc.nasa.gov/vitmo/iri_vitmo.html Vi ser av figuren at det som ventet er best å måle om natten....

Se også en global modell på JPL-siden: <http://iono.jpl.nasa.gov/>

Eksempel på beregning av forsinkelser gjennom ionosfæren:"Normal"verdi på TEC: $2 \cdot 10^{16}$ el/m²Svært høy verdi på TEC: $100 \cdot 10^{16}$ el/m²

Fasemålinger:

Bølgelengde	λ	Forsinkelser		1% endring	
		Normal TEC	Høy TEC	Normal TEC	Høy TEC
L1	0,190 m	-0,325 m	-16,237 m	0,003 m	0,162 m
L2	0,242 m	-0,535 m	-26,742 m	0,005 m	0,267 m

Problemer for fasemålingene: Endringene ved høy TEC utgjør ca 1 bølgelengde, kan få feil på avstanden til satellitt på 1 hel bølgelengde. Kan tolkes som fasebrudd (cycleslip), men skyldes ionosfæren.

Pseudorangemålinger (kodemålinger): Samme tallverdier men motsatt fortegn!

6.8.2 Troposphere

I de lavere lagene av atmosfæren, *troposfæren*, hjelper det ikke å måle på to frekvenser. Avstandsfeilen vil være den samme på både L1 og L2.

Troposfæren deles i to:

- Tørr del, ytterst: Har 90% av forstyrrelsen på GPS-signalene. Er lettest å modellere
- Våt del, nærest Jorda: Har 10% av forstyrrelsen. Er mest vanskelig å modellere

Til å estimere avstandsfeilen på grunn av troposfæren, kan man benytte empiriske matematiske modeller, der værdata i målestasjonen (temperatur, trykk og relativ fuktighet) settes inn i modellen. Hvis man ikke måler værdata i stasjonen under observasjonsperioden, nytter man standardverdier i modellen. Måler man over kortere avstander, har erfaring vist at bruk av standardverdier gir best resultat.

Modeller kan beregne gode verdier på forsinkelsene, som typisk er på 1 cm på tørr del og 5 cm på våt del.

6.8.3 Summary, the atmosphere

Ved relative målinger vil atmosfærisk forstyrrelser i stor grad bli eliminert, men denne feiltypen er normalt hoved årsaken hvis beregningene ikke klarer å løse ut de ukjente heltallene (N) ved fasemåling.

- Bruk kortere vektorer ved støyproblemer
- Måle ved antatt lav ionosfærisk aktivitet
- Benytte spesielle kombinasjoner i beregningene av en basislinje: L3/P3: ionsfærefri. Benytter seg av at L1 og L2 har ulike forsinkelser gjennom ionosfæren (forsinkelsen er frekvensavhengig).

6.9 SA ("Selective availability")

Da GPS-systemet ble planlagt av det amerikanske forsvaret, regnet man med å oppnå en dårligere nøyaktighet enn den man i praksis fikk da GPS gikk inn i sin testperiode. Av forsvarsmessige hensyn ble det derfor bestemt å degradere nøyaktigheten i systemet for vanlige sivile navigasjonsbrukere. Dette blir kalt "selective availability", SA. Den absolutte posisjonen kan endre seg med 100 m i løpet av noen minutter på grunn av SA. Eierne (USA) fjernet SA permanent 2. mai 2000.

Det russiske GLONASS-systemet har ikke en tilsvarende pålagt støy på sitt system.

Degraderingen av nøyaktighet ser ikke ut til å ha påvirket nøyaktigheten ved landmåling i særlig stor grad.

Se også tabellen i kapittel 4.2.

6.10 A-S ("anti-spoofing")

Eier av GPS-systemet kan kryptere den presise P-koden til en Y-kode som ikke er tilgjengelig for vanlige brukere. P-koden anvendes også ved fasemålinger, og en kryptering vil kunne skape problemer for geodetiske GNSS-målinger.

7. Single-, double- and triple differences

In chapter 2.3 various linear combinations of the original observations are described, and the chapter also gives an overview of the background for using differences between the original observations between the satellites and the antennas/receivers.

By using linear combinations of the observations, or by forming differences between the observations, we can reduce and even eliminate the effect of systematic errors, as for instance the effect of clock errors and ambiguity of the number of cycles (N), satellite orbit errors and the effect of delays that are related to the ionosphere and troposphere. The shorter the distance between the two receivers measuring at the same time is, the more effective this reduction will be. See figure 1.1.

The original observation equations of a distance between a satellite and a receiver, is derived in chapters 4 (code) and 5 (phase), the equations (4.3) and (5.11):

$$\text{Code: } P = \rho + c \cdot (dt - dT) + d^{\text{ion}} + d^{\text{trop}} + d\rho \quad (7.1)$$

$$\text{Phase: } \Phi_{\text{total}} = \rho + c \cdot (dt - dT) + \lambda \cdot N(t_0) - d^{\text{ion}} + d^{\text{trop}} + d\rho \quad (7.2)$$

The terms in equations (7.1) and (7.2):

- P : Code: measured distance (pseudorange)
- Φ_{total} : Phase: the measured total phase, in meters
- ρ : Vector distance satellite-receiver, containing the coordinates of the receiver and the satellite
- c : Speed of light in vacuum
- $dt - dT$: Clock errors on the satellite- and the receiver clocks
- λ : Wave length of the carrier waves (L1, L2, L5)
- N : Phase: The unknown integer when the logging starts, time: t_0
- d^{ion} : Ionospheric delay (notice the opposite signs for code and phase)
- d^{trop} : Tropospheric delay
- $d\rho$: Error on the orbit of the satellite

The equations (7.1) and (7.2) are also called the non-difference observation equations.

From the original observation equations, which is a distance measurement between a satellite and a receiver, we can derive:

- Single difference
- Double difference
- Triple difference

The difference between GPS observations can be formed between receivers, between satellites, between point in time, or a combination of these.

These symbols will be used for the most common differences (difference operations):

- Δ : The difference between two receivers (check the form of the triangle)
- ∇ : The difference between two satellites (check the form of the triangle)
- δ : The difference between two points in time

See an example in chapter 7.4, for the connection between the various differences.

7.1 Single differences

The starting point is the original observation equations. From these equations the differences between two and two observations are formed. There are several ways of forming the differences, dependent on which sources of errors we want to eliminate or reduce:

- The difference between two points in time, δ , where the ambiguity, N , is eliminated
- The difference between two receivers, Δ , where the satellite clock error is eliminated
- The difference between two satellites, ∇ , where the receiver clock error is eliminated

7.1.1 Single difference between two points in time (Doppler effect)

One satellite and one receiver, two points in time.

As a consequence of the satellite moving relative to the receiver, the frequency of the signal which is sent from the satellite, will vary continuously. This is the Doppler effect. Another example of the Doppler effect, is the sound signal at a railway crossing: The signal has a higher frequency when the train passenger is coming closer to the crossing, and a lower frequency when the passenger is moving away from the signal.

The predecessor to GPS, Transit NNSS, used this principle for geo-referencing, taking advantage of the derived observation equation does not contain the term with the unknown integer, N , the ambiguity.

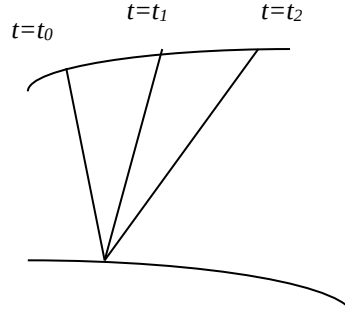


Figure 7.1 Single differences, time

The fundamental GPS-Doppler observation gives the change in phase between two points in time on the received signal. The equation of the phase difference between two points in time can be written as shown below, it are based on the original observation equations (7.1) and (7.2):

$$\delta \Phi_{\text{avstand}} = \Phi(t = t_2) - \Phi(t = t_1)$$

$$\delta \Phi_{\text{avstand}} = [\rho_2 + c \cdot (dt_2 - dT_2) + \lambda \cdot N(t_0) - d_2^{\text{ion}} + d_2^{\text{trop}} + d\rho_2] - \\ - [\rho_1 + c \cdot (dt_1 - dT_1) + \lambda \cdot N(t_0) - d_1^{\text{ion}} + d_1^{\text{trop}} + d\rho_1]$$

$$\delta \Phi_{\text{avstand}} = \delta \rho + c \cdot (\delta dt - \delta dT) - \delta d^{\text{ion}} + \delta d^{\text{trop}} + \delta d\rho \quad (7.3)$$

By forming this single difference, we have a derived observation without the unknown integer, N .

The Doppler observations can be used in several ways, Doppler counts means counting cycles:

1. "Intermittently Integrated Doppler (IID)"

The Doppler counts is carried out in a short time lag. The counter starts on zero again after a short break and we get a new observation. Thus we get independent observations, but we do not utilize the observation capacity as we are not using the signals in the breaks.

2. "Consecutive Doppler Counts (CDC)"

The Doppler counts is done in short intervals, and the counter is starting from zero at the end of the previous interval, but without losing any cycles. These observations can be interpreted as a change on the distance in the integration intervals.

3. "Continuously Integrated Doppler (CID)"

The Doppler counts is carried out continuous for the whole passage of a satellite. The observations can be interpreted as differences in distances from the start of the logging of data ($t = t_0$). As the distance to the satellite at the start is unknown, the observations can be considered as the difference between two distance observations which both are corrupted with the common systematic errors.

7.1.2 Single differences between satellites

One receiver, two satellites, same time

A receiver can measure towards two satellites at the same time, $t = t_1$.

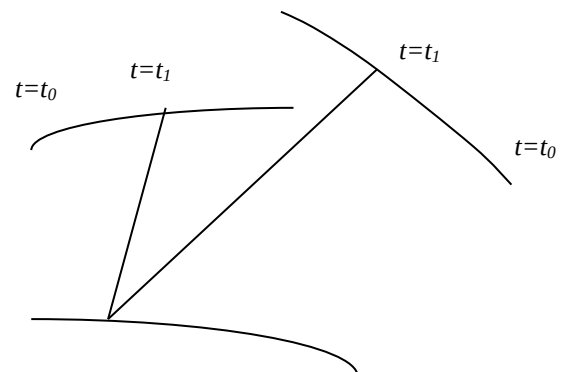


Figure 7.3 Single difference, satellites and receiver, same point of time

We can form the difference between the *code observations* on the P- or C/A-code towards the two satellites. Based on the original observation equation (7.1), we obtain:

$$\begin{aligned}\nabla P &= P(\text{satellite S2}) - P(\text{satellite S1}) \\ \nabla P &= [\rho_{S2} + c \cdot (dt_{S2} - dT) + d_{S2}^{\text{ion}} + d_{S2}^{\text{trop}} + d\rho_{S2}] - [\rho_{S1} + c \cdot (dt_{S1} - dT) + d_{S1}^{\text{ion}} + d_{S1}^{\text{trop}} + d\rho_{S1}] \\ \nabla P &= \nabla \rho + c \cdot (dt_{S2} - dt_{S1}) + \nabla d^{\text{ion}} + \nabla d^{\text{trop}} + \nabla d\rho \\ \nabla P &= \nabla \rho + c \cdot dt + \nabla d^{\text{ion}} + \nabla d^{\text{trop}} + \nabla d\rho\end{aligned}\quad (7.4)$$

In a similar way we obtain the difference equation for the *phase observations*, based on equation (7.2):

$$\begin{aligned}\nabla \Phi &= \Phi(\text{satellite S2}) - \Phi(\text{satellite S1}) \\ \nabla \Phi_{\text{Distance}} &= \nabla \rho - c \cdot \nabla dt + \lambda \cdot \nabla N - \nabla d^{\text{ion}} + \nabla d^{\text{trop}} + \nabla d\rho\end{aligned}\quad (7.5)$$

In equations (7.4) and (7.5) are:

- ∇dt : difference in clock error for the two satellites
- ∇d : difference in ionospheric and tropospheric delays
- ∇N : difference in the number of complete cycles (integers) at the start of logging

In (7.5), the single differences between satellites, *the term containing the receiver clock error is eliminated*.

7.1.3 Single differences between receivers

Two receivers, one satellites, same time

Two receivers are logging data from the same satellite at the same point of time, $t = t_1$.

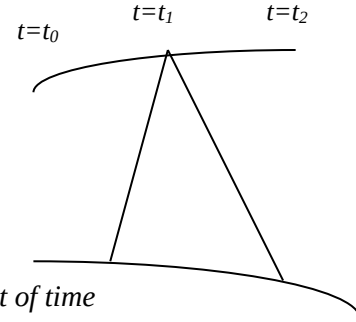


Figure 7.2 Single difference, satellite and receivers, same point of time

We can form the difference between the *code observations* on the P- or C/A-code stored in the two receivers towards the satellite. Based on the original observation equation (7.1), we obtain:

$$\begin{aligned}\Delta P &= P(\text{receiver M2}) - P(\text{receiver M1}) \\ \Delta P &= [\rho_{M2} + c \cdot (dt - dT_{M2}) + d_{M2}^{\text{ion}} + d_{M2}^{\text{trop}} + d\rho] - [\rho_{M1} + c \cdot (dt - dT_{M1}) + d_{M1}^{\text{ion}} + d_{M1}^{\text{trop}} + d\rho] \\ \Delta P &= \Delta \rho + c \cdot (-dT_{M2} + dT_{M1}) + \Delta d^{\text{ion}} + \Delta d^{\text{trop}} \\ \Delta P &= \Delta \rho - c \cdot \Delta dT + \Delta d^{\text{ion}} + \Delta d^{\text{trop}}\end{aligned}\quad (7.6)$$

In a similar way we obtain the difference equation for the *phase observations*, based on equation (7.2):

$$\begin{aligned}\Delta \Phi &= \Phi(\text{receiver M2}) - \Phi(\text{receiver M1}) \\ \Delta \Phi_{\text{Distance}} &= \Delta \rho - c \cdot \Delta dT + \lambda \cdot \Delta N - \Delta d^{\text{ion}} + \Delta d^{\text{trop}}\end{aligned}\quad (7.7)$$

In equations (7.6) and (7.7) are:

- ΔdT : difference in clock error for the two receivers
- Δd : difference in ionospheric and tropospheric delays
- ΔN : difference in the number of complete cycles (integers) at the start of logging

The equations (7.6) and (7.7), the single differences between receivers, *eliminates or reduces effects of errors on the observations*:

- Satellite clock error is eliminated
- Orbit error and atmospheric delays are strongly reduced, as the distance between the two receivers (the baseline) is short compared to the distance from the receivers to the satellite (which is ca. 20000-25000 km).

7.2 Double differences

We start with the single difference equations. If we are using two receivers and two satellites and using data for a period of time, there are three different possibilities to form difference between single difference equations:

- $\Delta\nabla = \Delta^1 - \Delta^2$: The difference between two receivers and two satellites, same point of time. This double difference is used for a standard baseline processing, where one of the satellites is normally chosen as a reference satellite for all the double differences.
- $\Delta\delta = \Delta^1_{t_1} - \Delta^1_{t_2}$: The difference between two receivers, one satellites and two points of time. This double difference eliminates the unknown integer, N , and the satellite clock error.
- $\nabla\delta$: The difference between one receivers, two satellites and two points of time. Not discussed here.

7.2.1 Double differences with two receivers, two satellites, same time

We are using two receivers, two satellites, same point of time.

We derive the double difference equations from the single difference equations in the chapters 7.1.2 and 7.1.3. A double difference can be formed from the single differences between the receivers, and this double difference involves both receivers and satellites.

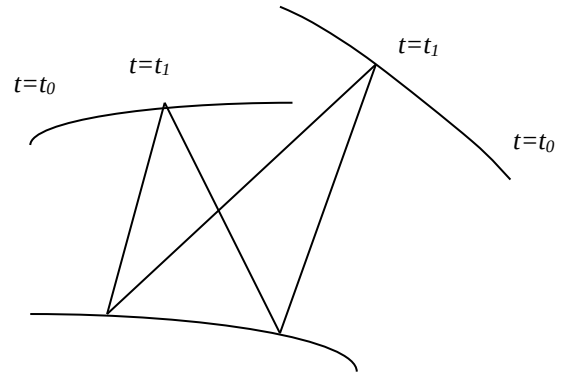


Figure 7.4 Double differences between receivers and satellites, same point of time, t_1

The double differences in this chapter can be derived in two ways:

1. We form the difference between two single differences that are between two receivers and one satellite, but with reference to two different satellites, see figure 7.4: $\Delta\nabla = \Delta^1 + \Delta^2$
2. We form the difference between two single differences that are between one receiver and two satellites, but with reference to two different satellites, see figure 7.4: $\nabla\Delta = \nabla_1 + \nabla_2$

The two ways of deriving the double difference will give the same equations, the sequence (what to do first and last) has no impact on the final result: $\nabla\Delta = \Delta\nabla$

Code measurements: Starting with the single difference in equation (7.6), we obtain the double difference equation by using method no. 1 above:

$$\begin{aligned}\nabla\Delta P &= [\Delta\rho - c \cdot \Delta dT + \Delta d^{\text{ion}} + \Delta d^{\text{trop}}]_{M1,M2,S2} - [\Delta\rho - c \cdot \Delta dT + \Delta d^{\text{ion}} + \Delta d^{\text{trop}}]_{M1,M2,S1} \\ \nabla\Delta P &= \nabla\Delta\rho + \nabla\Delta d^{\text{ion}} + \nabla\Delta d^{\text{trop}}\end{aligned}\quad (7.8)$$

Phase measurements: The double difference equation is derived in a similar way as the code, starting with equation (7.7):

$$\nabla\Delta\Phi = \nabla\Delta\rho + \lambda \cdot \nabla\Delta N - \nabla\Delta d^{\text{ion}} + \nabla\Delta d^{\text{trop}}\quad (7.9)$$

The double differences in equations (7.8) and (7.9) eliminate the clock errors of both the satellites and the receivers:

- The difference between the clock errors in the two receivers. This source of error is found in the single differences between receivers. Δ .
- The difference between the clock errors in the two satellites. This source of error is found in the single differences between receivers. ∇ .

Baseline processing, reference satellite:

Double differences are normally used in the baseline processing. During the observation period for a GNSS baseline minimum 4 satellites up to more than 12 satellites are observed at the same time. The double difference equations are normally computed by using a *reference satellite* as a common satellite in the differences, see an example of a baseline processing in chapter 9.

By using a reference satellite, the double differences are derived from the single differences involving two receivers and two satellites at a point of time, t . One of the satellites in each double difference between two satellites, is the reference satellite.

These double differences will be correlated, because the observation (single difference) to the reference satellite is used in all the double differences that involves the reference satellite. See the example exercise in chapter 7.5 and more stuff in chapter 9.4.1.

7.2.2 Double differences with two receivers, one satellites, two points of time

We are using two receivers, one satellite, two point of time.

A double difference equation will in this case show the changes of the single difference between one satellite and two receivers from one point of time to the next point of time, from t_1 to t_2 on figure 7.5.

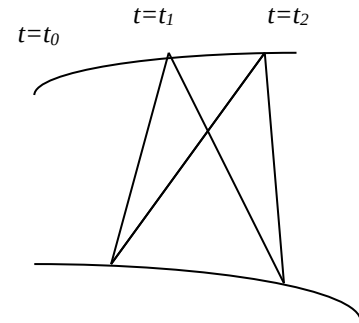


Figure 7.5 Double difference, receivers and time

Starting with the single difference between two receivers and one satellite in equation (7.6) and (7.7) in chapter 7.1.3, we obtain the double difference equations:

Code measurements:

$$\delta\Delta P = [\Delta\rho - c \cdot \Delta dT + \Delta d^{\text{ion}} + \Delta d^{\text{trop}}]_{t=t_2} - [\Delta\rho - c \cdot \Delta dT + \Delta d^{\text{ion}} + \Delta d^{\text{trop}}]_{t=t_1}$$

$$\delta\Delta P = \delta\Delta\rho - c \cdot \delta\Delta dT + \delta\Delta d^{\text{ion}} + \delta\Delta d^{\text{trop}} \quad (7.10)$$

Phase measurements:

$$\delta\Delta\Phi = [\Delta\rho - c \cdot \Delta dT + \lambda \cdot \Delta N - \Delta d^{\text{ion}} + \Delta d^{\text{trop}}]_{t=t_2} - [\Delta\rho - c \cdot \Delta dT + \lambda \cdot \Delta N - \Delta d^{\text{ion}} + \Delta d^{\text{trop}}]_{t=t_1}$$

$$\delta\Delta\Phi = \delta\Delta\rho - c \cdot \delta\Delta dT - \delta\Delta d^{\text{ion}} + \delta\Delta d^{\text{trop}} \quad (7.11)$$

The unknown integer (unknown no. of cycles), N , at the start of logging data, is eliminated in equation (7.11). Therefore this double difference can be used for detecting cycle slip (temporary loss of contact with a satellite).

7.3 Triple differences between receivers and satellites

A triple difference is derived by taking the difference between two double differences referred to two separate satellites and to two different, corresponding points of time.

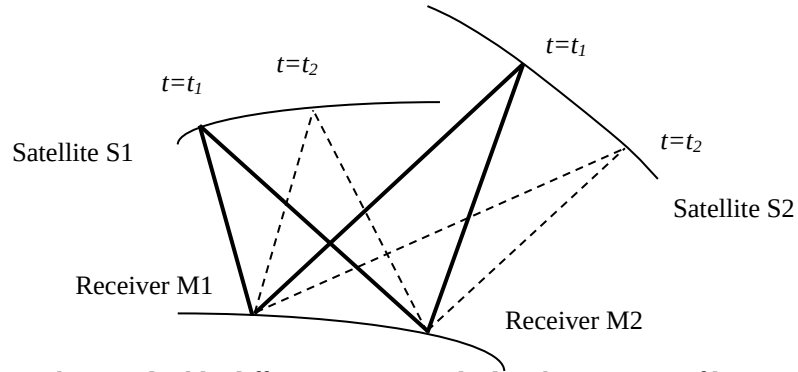


Figure 7.6 Triple difference. The two double differences are marked with two types of lines

To derive the triple differences we can start with the double differences either in chapter 7.2.1 or chapter 7.2.2, both ways will give the same equations of the triple difference: $\delta \nabla \Delta = \nabla \delta \Delta$

Code measurements:

The triple difference equation for the code, derived from the double differences in equation (7.8):

$$\delta \nabla \Delta P = \nabla \Delta P(t = t_2) - \nabla \Delta P(t = t_1)$$

$$\delta \nabla \Delta P = \delta \nabla \Delta \rho + \delta \nabla \Delta d^{\text{ion}} + \delta \nabla \Delta d^{\text{trop}} \quad (7.12)$$

Phase measurements:

The triple difference equation for the phase, derived from the double differences in equation (7.9):

$$\delta \nabla \Delta \Phi = \nabla \Delta \Phi(t = t_2) - \nabla \Delta \Phi(t = t_1)$$

$$\delta \nabla \Delta \Phi = \delta \nabla \Delta \rho - \delta \nabla \Delta d^{\text{ion}} + \delta \nabla \Delta d^{\text{trop}} \quad (7.13)$$

Sources of error that are eliminated in the double difference equations, are not present in the triple difference equations. In addition, the unknown integer at the start of logging data, N , is eliminated in the triple difference equation (7.13). *This is a very important property/characteristic of the triple differences:*

- By using the triple differences we almost automatically obtain a detection of a cycle slip (temporary loss of contact with a satellite). See an example in the next chapter.
- The baseline or the position can be calculated without calculating the unknown integers N for each satellite (the number of cycles at the start of logging data). The clock errors of the satellites and the receivers are also eliminated.

The triple differences can therefore be used to calculate a preliminary position for use in the final calculations. It is the double differences that are used for processing baselines that are shorter than ca. 15-20 km. See also chapter 8 about this topic.

7.4 Use of double differences and triple differences

The original non-differenced observations are influenced by systematic errors on the clocks in the receivers and the satellite. For the phase observations the unknown integers, N , also must be calculated. All these unknowns are eliminated by using the triple differences in equation (7.13). On the other hand, the number of observations will be reduced considerably.

In the double- and triple differences are one or both the clock errors eliminated from the mathematical model.

If we want to examine how the clocks are performing, both compared to the GPS time signal and the stability of the clocks over time: In this case we use original non-differential data to examine the both the receiver- and the satellite clock, or use single differences to examine either the receiver clock or the satellite clock.

Example of forming differences and detection of cycle slips

See figure 7.6 for names on the satellites (S1 and S2) and receivers (M1 and M2).

The original observation equation for phase observations between a satellite and a receiver are the equations (7.2) and/or (5.11):

$$\Phi = \Phi_{\text{total}} = \rho + c \cdot (dt - dT) + \lambda \cdot N(t_0) - d^{\text{ion}} + d^{\text{trop}} + d\rho$$

This example will show:

- cycle slips can be detected by forming and using triple differences
- the total number of observations will decrease by taking differences between observations

We assume a cycle slip with size F take place between the epochs (points of time for storing data) ($i-1$) and i , and that the cycle slip was on the L1 signal from satellite S2 to the receiver M1. The table shows the stored observations for the frequency L1 (completely similar for L2).

	Original observations on L1				Single diff. between 1 receiver and 2 satellites		Double .diff.	Triple diff
Epoch (time)	M1-S1	M2-S1	M1-S2	M2-S2	∇ , M1	∇ , M2	same time: $\nabla\Delta$	$\delta\nabla\Delta$
$i-2$	Φ_{M1S1}	Φ_{M2S1}	Φ_{M1S2}	Φ_{M2S2}	$\nabla\Phi_{M1S1S2}$	$\nabla\Phi_{M2S1S2}$	$\nabla\Delta\Phi$	
								$\delta\nabla\Delta\Phi$
$i-1$	Φ_{M1S1}	Φ_{M2S1}	Φ_{M1S2}	Φ_{M2S2}	$\nabla\Phi_{M1S1S2}$	$\nabla\Phi_{M2S1S2}$	$\nabla\Delta\Phi$	
								$\delta\nabla\Delta\Phi - F$
i	Φ_{M1S1}	Φ_{M2S1}	$\Phi_{M1S2}+F$	Φ_{M2S2}	$\nabla\Phi_{M1S1S2} + F$	$\nabla\Phi_{M2S1S2}$	$\nabla\Delta\Phi - F$	
								$\delta\nabla\Delta\Phi$
$i+1$	Φ_{M1S1}	Φ_{M2S1}	$\Phi_{M1S2}+F$	Φ_{M2S2}	$\nabla\Phi_{M1S1S2} + F$	$\nabla\Phi_{M2S1S2}$	$\nabla\Delta\Phi - F$	
								$\delta\nabla\Delta\Phi$
$i+2$	Φ_{M1S1}	Φ_{M2S1}	$\Phi_{M1S2}+F$	Φ_{M2S2}	$\nabla\Phi_{M1S1S2} + F$	$\nabla\Phi_{M2S1S2}$	$\nabla\Delta\Phi - F$	

Table 7.1 Forming differences. Cycle slip, F

Then number of stored observations:

- During the 5 epochs (registrations of observations) $4 \times 5 = 20$ original observation equations for the frequency L1 are stored.
- Forming the single differences by taking the difference between two original observations at the same point of time, and we get 10 single differences.
- Same procedure for the double differences from the single differences, see the table. The number of double differences are 5, again halving the number of observations.
- Finally we end up with 4 triple differences, they are derived by taking the difference between two double differences.

The number of observations in the processing of the baseline are considerable reduced, but the advantages by using double and triple differences are that we eliminate or reduce the influence of error sources connected to the observations.

Cycle slip:

In the column to the right in table 7.1 we see that the assumed cycle slip, F , is found in only one of the triple differences. Therefore a jump in the numerical value of the triple difference can be explained with a cycle slip. The size of the cycle slip can then be calculated and thus be eliminated from the observation equations of the triple, double and the single differences (we know the value of F in table 7.1).

7.5. Correlation between difference observations. Variance/covariance

Two types of correlations:

- Physical correlations
- Mathematical correlations
- **The physical correlations** is far more difficult to handle compared to the mathematical correlations. Inhomogeneity in the atmosphere is one example of a source of physical correlations. It is very difficult to obtain good values of the physical correlations.
- **The mathematical correlations** are theoretically easier to handle, but normally leads to a lot more calculations if the mathematical correlations is taken into consideration 100 %. In this compendium we look closer into the following two types of mathematical correlations:

(1) Baseline conversion to the map projection plane

In chapter 11.4 it is shown how the variance/covariance propagation law is derived and used. From accuracy measures of the original observations (the coordinate differences ΔX , ΔY and ΔZ of a measured baseline in the geocentric 3D Cartesian coordinate system) to the derived accuracy measures of the GNSS baseline components in the map projection plane (bearing, distance and height difference or zenith angle).

(2) Difference observations (single, double, triple differences)

The various difference equations can by the process of deriving the differences be mathematical correlated.

Regardless of the form of the difference equations (single, double, triple) we can regard the difference equation to be a result of doing a difference operation on the original non-differenced observations.

Difference operations:

- Δ : The difference between two receivers (check the form of the triangle)
- ∇ : The difference between two satellites (check the form of the triangle)
- δ : The difference between two points in time

The difference operations which is used, can be represented by a matrix operator, where the elements in the matrix consist of element with numerical values of +1, -1 and 0, see below.

Example:

Here we use the example in table 7.1 in the previous chapter. The original non-differenced observations are for the point of time $i-2$:

	Original observations on L1			
Epoch (time)	M1-S1	M2-S1	M1-S2	M2-S2
$i-2$	Φ_{M1-S1}	Φ_{M2-S1}	Φ_{M1-S2}	Φ_{M2-S2}

Table 7.2

The single differences are a linear combination of the observations in table 7.2:

$$\begin{aligned}\nabla \Phi_{M1-S1-S2} &= -\Phi_{M1-S1} + 0 & + \Phi_{M1-S2} &- 0 \\ \nabla \Phi_{M2-S1-S2} &= + 0 & - \Phi_{M2-S1} &+ 0 & + \Phi_{M2-S2}\end{aligned}\tag{7.14}$$

Mathematical connection between the single differences in equations (7.14) by using matrices, point of time $i-2$:

$$\mathbf{E}_{\nabla} = \mathbf{A}_{\nabla} \cdot \mathbf{F} = \begin{pmatrix} \nabla \Phi_{M1-S1-S2}^{i-2} \\ \nabla \Phi_{M2-S1-S2}^{i-2} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \Phi_{M1-S1}^{i-2} \\ \Phi_{M2-S1}^{i-2} \\ \Phi_{M1-S2}^{i-2} \\ \Phi_{M2-S2}^{i-2} \end{pmatrix}\tag{7.15}$$

The variance-covariance matrix of the original non-differenced observations can be set to the unity matrix: $\mathbf{C}_{FF} = \mathbf{I}$

Explanation: We assume that the original observations are of equal accuracy and that they are not correlated (they are independent). Equal accuracy can be discussed, but for this purpose small variations of the accuracy is not important for the result (proving correlations). Not correlated original observations can be discussed, as the satellites that are observed are the same for both the receivers, this is discussed in another course where analysing time series is a topic.

From the variance-covariance propagation law we directly obtain the variance-covariance matrix of the single differences, as the coefficient matrix $\mathbf{A}$ is known from the mathematical connection between the original observations and the single differences (equation 7.15).

Calculating the variance-covariance matrix of the single differences, $\mathbf{C}_{EE}$ by using the variance-covariance propagation law:

$$\mathbf{C}_{EE} = \mathbf{A}_V \cdot \mathbf{C}_{FF} \cdot \mathbf{A}_V^T = \mathbf{A}_V \cdot \mathbf{A}_V^T = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \quad (7.16)$$

With uncorrelated original observation equations, the single difference equations will also be uncorrelated.

The double differences in table 7.1 will be uncorrelated, as none of the single differences is used in more than *one* double difference.

Baseline processing with a reference satellite:

The double differences that are used in the baseline processing, will normally be calculated with a reference satellite as a common satellite for forming all the double differences, see the example covering baseline processing in chapter 9.4.1.

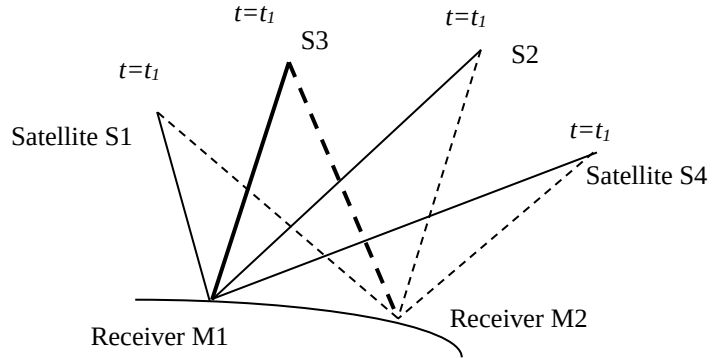


Figure 7.7 Common reference satellite for all the single differences, satellite S3

Using a third satellite, S3, as a reference satellite for the satellites S1 and S2 in figure 7.6 and 7.7, the double differences (between 2 receivers, 2 satellites (one of them is S3), same point of time) are formed by the single differences between 1 satellite, 2 receivers and the same point of time (t), $\nabla\Delta = \Delta^i - \Delta^3$:

$$\begin{aligned} \nabla\Delta\phi_{S3-S1-M1-M2} &= \Delta\phi_{S1-M1-M2} & + 0 & & - \Delta\phi_{S3-M1-M2} \\ \nabla\Delta\phi_{S3-S2-M1-M2} &= + 0 & + \Delta\phi_{S2-M1-M2} & & - \Delta\phi_{S3-M1-M2} \end{aligned}$$

Using matrices, obtaining the mathematical connection between the single differences and the double differences above:

$$\mathbf{D}_{\nabla\Delta} = \mathbf{B} \cdot \mathbf{E}_\Delta = \begin{pmatrix} \nabla\Delta\phi_{S3-S1-M1-M2}^t \\ \nabla\Delta\phi_{S3-S2-M1-M2}^t \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} \Delta\phi_{S1-M1-M2}^t \\ \Delta\phi_{S2-M1-M2}^t \\ \Delta\phi_{S3-M1-M2}^t \end{pmatrix}$$

The variance-covariance matrix of the double differences above, is calculated by using the variance-covariance propagation law:

9	C1	L1	D1	P2	L2	D2	# OF SATELLITES
							COMMENT
G 4	60	60	60	58	58	58	PRN / # OF OBS
G 9	60	60	60	60	60	60	PRN / # OF OBS
G12	60	60	60	60	60	60	PRN / # OF OBS
G14	60	60	60	60	60	60	PRN / # OF OBS
G17	60	60	60	60	60	60	PRN / # OF OBS
G26	60	60	60	60	60	60	PRN / # OF OBS
G27	60	60	60	60	60	60	PRN / # OF OBS
G30	60	60	60	60	60	60	PRN / # OF OBS
G32	60	60	60	60	60	60	PRN / # OF OBS
							END OF HEADER
9 10 7 9 10 20.0000000 0 9G27G26G17G30G 4G32G 9G12G14							
22273194.300 117046417.89058 -2505.559 22273192.678 91204992.63656 -1952.387							
21331408.782 112097303.71159 -1732.230 21331406.818 87348541.75657 -1349.793							

The observations are shown in the last three lines above.

The headline (first of the three lines) gives time and satellite information. Observations at the point of time 7.10.2009, at 9 h 10 min 20 seconds. At that point of time the receiver is logging data from 9 satellites, from satellite G27 to G14.

The last two lines are the logged observation values for the distance MOHOLT-G27.

The observation classes are C1, L1, D1, P2, L2 and D2.

Row 4-11, not shown: The observations towards the other 8 satellites.

7.6.2 Numerical example of single, double and triple differences. Cycle slip.

The first example is a numerical example of forming the single, double and triple differences from the original observations between 1 satellite and 1 receiver.

The second example is a numerical example of detecting a cycle slip by using the triple differences.

Example no. 1: Observations 7.10.2009, at 9 10 20-9 11 00

Original observations:

5 of the logged epochs in the points Moholt and ST46 towards the satellites G27 and G26, showing the observations C1, L1 and L2.

C1: C/A code pseudorange on L1 in meters

L1: phase observation on L1, in cycles

L2: phase observation on L2, in cycles

Moholt	G27	G27	G27
Point of time	C1	L1	L2
9 10 20	22273194,300	117046417,89058	91204992,63656
9 10 30	22277964,794	117071486,94148	91224526,95246
9 10 40	22282744,017	117096601,99048	91244097,10146
9 10 50	22287530,931	117121757,42248	91263698,72846
9 11 00	22292323,110	117146940,40248	91283321,82346
Moholt	G26 C1	G26 L1	G26 L2
9 10 20	21331408,782	112097303,71159	87348541,75657
9 10 30	21334711,203	112114658,26849	87362064,79447
9 10 40	21338029,157	112132094,76849	87375651,67247
9 10 50	21341361,659	112149607,55949	87389297,99447
9 11 00	21344706,266	112167183,77549	87402993,72947
ST46	G27 C1	G27 L1	G27 L2
9 10 20	22274666,759	117054162,02348	91211034,65146
9 10 30	22279440,703	117079249,08348	91230582,99946
9 10 40	22284221,678	117104373,02448	91250160,08946
9 10 50	22289008,971	117129530,04348	91269762,94846

9 11 00	22293803,412	117154724,56848	91289395,04146
ST46	G26 C1	G26 L1	G26 L2
9 10 20	21332237,125	112101659,77749	87351933,72047
9 10 30	21335544,453	112119040,15249	87365476,87147
9 10 40	21338865,651	112136493,38449	87379076,79047
9 10 50	21342200,000	112154015,57049	87392730,42647
9 11 00	21345548,294	112171611,13049	87406441,25847

Table 7.2 Original observations values

Single differences (SD)

From the original observations in table 7.2, the single differences are calculated, between 1 receiver and 2 satellites, same point of time:

Moholt	G27 - G26	G27 - G26	G27 - G26
Point of time	C1	L1	L2
9 10 20	941785,518	4949114,17899	3856450,87999
9 10 30	943253,591	4956828,67299	3862462,15799
9 10 40	944714,860	4964507,22199	3868445,42899
9 10 50	946169,272	4972149,86299	3874400,73399
9 11 00	947616,844	4979756,62699	3880328,09399
ST46	G27 - G26	G27 - G26	G27 - G26
Point of time	C1	L1	L2
9 10 20	942429,634	4952502,24599	3859100,93099
9 10 30	943896,250	4960208,93099	3865106,12799
9 10 40	945356,027	4967879,63999	3871083,29899
9 10 50	946808,971	4975514,47299	3877032,52199
9 11 00	948255,118	4983113,43799	3882953,78299

Table 7.3 Single differences

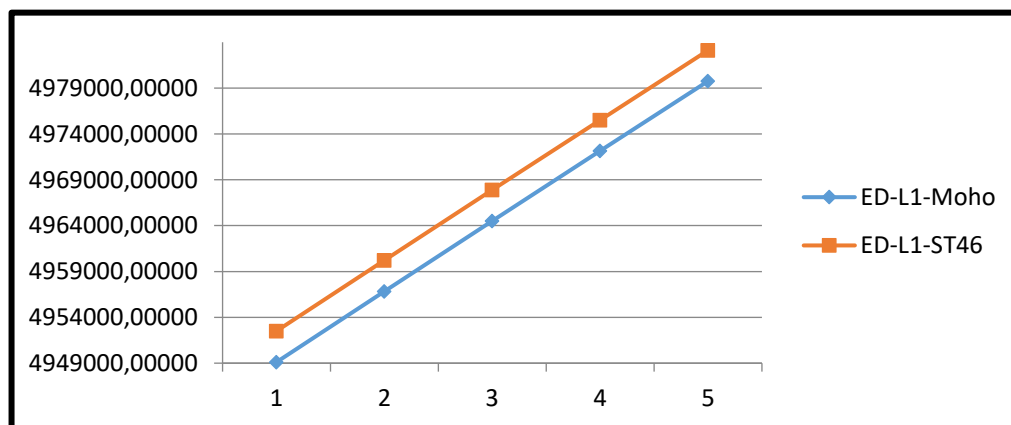


Figure 7.8 Single differences (ED = SD in the figure)

Double differences (DD)

From the single difference observations in table 7.3, the double differences are calculated, between 2 receivers and 2 satellites, same point of time:

	G27 - G26	G27 - G26, Moh - ST46	G27 - G26, Moh - ST46
Point of time	C1	L1	L2
9 10 20		-3388,06700	-2650,05100
9 10 30		-3380,25800	-2643,97000
9 10 40		-3372,41800	-2637,87000
9 10 50		-3364,61000	-2631,78800
9 11 00		-3356,81100	-2625,68900

Table 7.4 Double differences

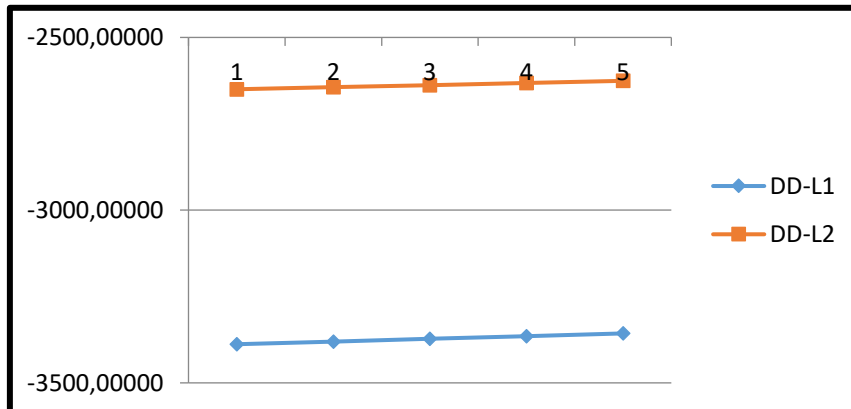


Figure 7.9 Double differences

Triple differences (TD)

From the double difference observations in table 7.4, the triple differences are calculated, between 2 receivers, 2 satellites, 2 points of time:

	G27 - G26	G27 - G26, Moh - ST46	G27 - G26, Moh - ST46
Points of times	C1	L1	L2
9 10 20, 9 10 30		-7,80900	-6,08100
9 10 30, 9 10 40		-7,84000	-6,10000
9 10 40, 9 10 50		-7,80800	-6,08200
9 10 50, 9 11 00		-7,79900	-6,09900

Table 7.5 Triple differences

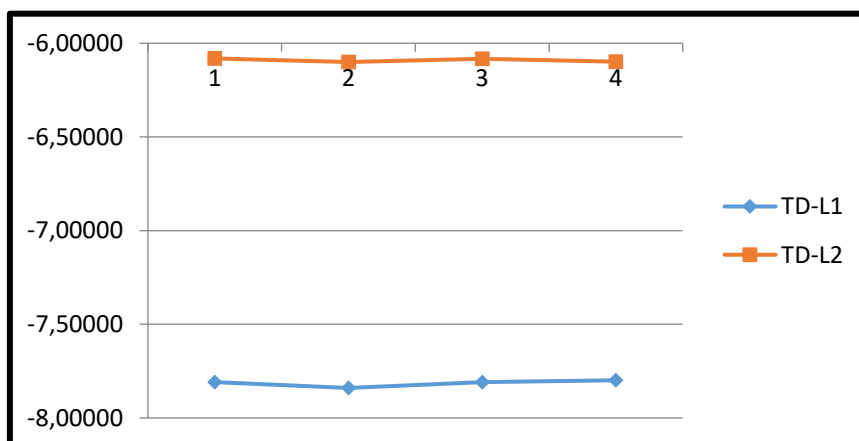


Figure 7.10 Triple differences

The number of observations in the example?

Two-frequency receivers, carrier waves L1 and L2. Observation time is 40 seconds (9 10 20, 30, 40, 50, 00), with 10 seconds between the epochs (storing data)

The number of original observations? The number of single differences?

The number of double differences? The number triple differences?

Answers:

Original observations: 5 epochs · 2 sat. · 2 receivers = 20 on L1, and also 20 obs. on L2.

Single differences: $20 / 2 = \underline{10}$ on L1, and 10 on L2

Double differences: $10 / 2 = \underline{5}$ on L1, and 5 on L2

Triple differences: 4 on L1, and 4 on L2

The number of unknowns and the degree of freedom in the example?

Two frequency receivers (L1 and L2), with 9 satellites involved, and using satellite G26 as a reference satellite. Observation time was 20 minutes (see textbooks for short time static processing). Double differences are normally used for the baseline processing.

How many unknowns in total, Conclusions?

The unknowns:

9 satellites involved: 9 unknown N -s on L1 and 9 unknown N -s on L2

The base line: 3 unknown coordinate differences between the two receivers

Sum: 21 unknowns

Original observations in 1 epoch on L1: 9 sat. · 2 receivers = 18, and also 18 on L2

Original observations in 1 minute: $18 \cdot 6$ epochs = 108, and also 108 on L2

Original observations in 20 minutes: $108 \cdot 20$ minutes = 2160, and also 2160 on L2

Single differences (SD) 2 receivers, 1 satellite, same point of time (Δ):

SD in 1 epoch: 9 sat · 1 epoch = 9 on L1, and 9 on L2

Double differences (DD) in 1 epoch (reference satellite G26):

1 epoch = 8 on L1, and 8 on L2

Double diff. in 1 minute: $8 \cdot 6 = 48$ on L1, and 48 on L2

Total no. of double differences, the measuring time was 20 minutes.

$48 \cdot 20$ minutes = 960 obs. on L1, and 960 obs. on L2.

The degree of freedom, $f = n - e$:

Original observations: (18 on L1 + 18 on L2) · 6 epochs · 20 minutes = 2320

Double differences, total no.: $(960 + 960) = 1920 = n$

Total no. of unknowns: $e = 9 + 9 + 3 = 21$

Degree of freedom, using double differences as observations: $f = 1920 - 21 = 1899$

Conclusion:

The degree of freedom is very large compared to surveys done with a total station where 50 % more observations than unknowns are Ok.

Even with a survey of 5 minutes, the degree of freedom is very large, $f = 460 - 21 = 439$.

This is also the reason to use a logging rate of 10 seconds or more, we do not “need” all the received signals to process a baseline. We need to observe a minimum period of 5-20 minutes (dependent on the baseline length) to have a geometry change that makes it possible to solve the unknown integers (the N -s). See also chapter 9.4.4 for a notice about the accuracy measures of a baseline, including multiplying the variance-covariance matrix of the baseline with a factor of 10^2 .

Numerical example no. 2. Triple differences for cycle slip detection

In table 1 in chapter 7.4 the theory for detecting cycle slip (“fasebrudd”) were shown. Here is a numerical example. The same example as above, but here satellite G04 also is involved.

Double differences, 2 receivers and 2 satellites, same point of time.

	G27 - G26, Moh- ST46	G27 - G26, Moh- ST46	G27 - G04, Moh- ST46	G27 - G04, Moh- ST46
	L1	L2	L1	L2
9 09 50	-3411,59500	-2668,38500	-6691,67398	-5219,94407
9 10 00	-3403,74400	-2662,26800	-6713,36897	-5236,91099
9 10 10	-3395,90800	-2656,16099	-6735,10598	-5253,82796
9 10 20	-3388,06700	-2650,05100	-6756,86387	-5258,90499
9 10 30	-3380,25800	-2643,97000	-6778,53997	-5275,62299

Table 7.6 Double differences

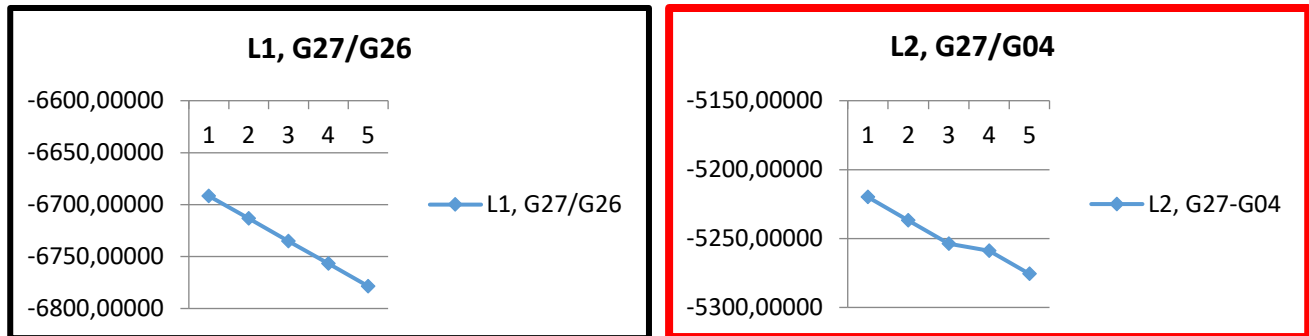


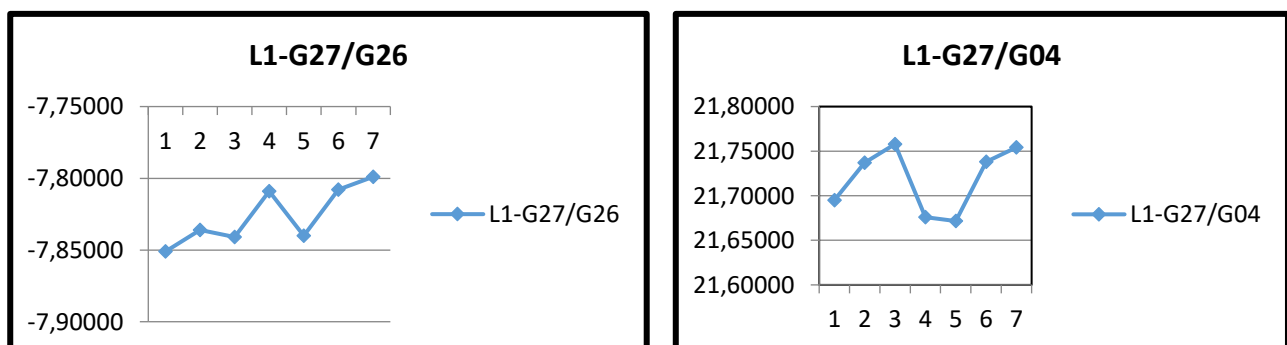
Figure 7.11 Double differences

Triple differences, 2 satellites, 2 receivers and 2 point of times

	G27 - G26, Moh- ST46	G27 - G26, Moh- ST46	G27 - G04, Moh- ST46	G27 - G04, Moh- ST46
	L1	L2	L1	L2
9 09 50, 9 10 00	-7,85100	-6,11700	21,69499	16,96692
9 10 00, 9 10 10	-7,83600	-6,10701	21,73701	16,91697
9 10 10, 9 10 20	-7,84100	-6,10999	21,75789	5,07703
9 10 20, 9 10 30	-7,80900	-6,08100	21,67610	16,71800

Table 7.7 Triple differences

Notice the jump in the values in the right hand column in table 7.7. From the neighboring triple difference values the assumed value should be ca. 16,84 and not 5,07703. There is a cycle slip on satellite G04 of 12 cycles on L2. We have to look more into the observations to check if the cycle slip is from satellite G04 to the receiver in Moholt or the receiver in point ST46.



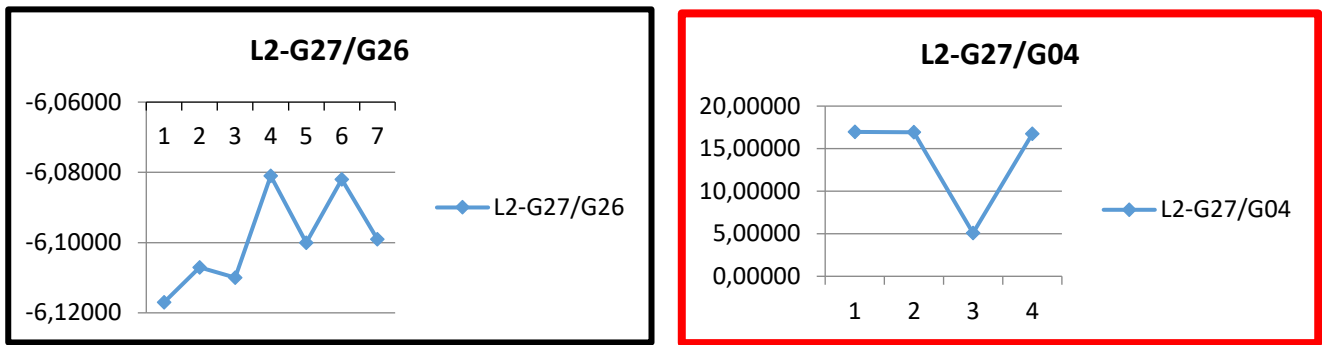


Figure 7.12 Triple differences. NB: Notice the different values of the steps of vertical scales

Results from the baseline processing by using Leica GeoOffice

Cycle Slip Statistics for baseline ST46 – MOHOLT, copied from the report:

Time	Satellite	Frequency	Slip value	Flags
10/07/2009 09:08:15	G04	L2	-1.00	flagged
10/07/2009 09:10:05	G04	L2	-12.00	flagged

The report confirms that the cycle slip found in table 7.7 has a slip value of -12 cycles.

The time tag is not exactly the same, this is probably due to the difference between the GPS-time standard compared to the GMT time standard on our watches.

8. Relative observations and baseline processing

2015, for use in the course TBA4236 Theoretical Geomatics:

Only a chapter that are part of the syllabus, are translated: 8.3

The other chapters, 8.1 and 8.2, are more relevant in the course TBA4231 Applied Geomatics.

8.1 Relative observations

Absoluttbestemmelse:

Bruk av måledata fra bare en mottaker, gir en absoluttbestemmelse. Ved kodemåling er usikkerheten i posisjonsbestemmelsen ca. 15-20 m (uten SA), se tabell i kapittel 4.2. Enkelpunktbestemmelse ved fasemåling brukes bare til innmåling av punkter til spesielle formål, som målinger i globale nett eller etablering av en ny basestasjon for sanntidsmålinger (RTK), da det her kreves svært lange observasjonstider og ofte spesielle beregningsprogrammer.

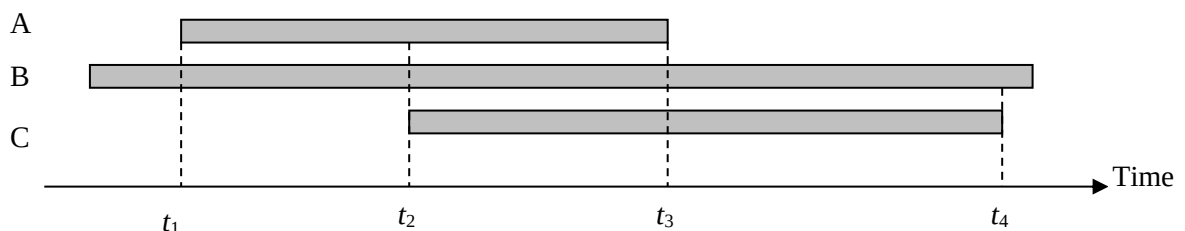


Figure 8.1 Observation time for the three receivers A, B and C

Relativbestemmelse, vektormåling:

Det måles med (minst) 2 mottakere samtidig. Det er bare den felles måletiden som kan utnyttes til beregning av vektoren mellom punktene, se figur 8.1.

Vektoren mellom mottakerne A og B kan bare beregnes av måledataene mellom tidspunktene t_1 og t_3 . Vektoren A-C kan bare beregnes av måledataene mellom tidspunktene t_2 og t_3 . Vektoren B-C kan bare beregnes av måledataene mellom tidspunktene t_2 og t_4 .

Independent baselines (non-trivial baselines):

I eksemplet i figur 8.1, registrerer alle 3 mottakerne måledata i tidsperioden mellom t_2 og t_4 . Vi kunne da ha beregnet alle 3 vektorene (A-B, A-C og B-C) ut i fra målingene i dette tidsrommet. Problemet er at vi bare får 2 uavhengige vektorer, den tredje vil kunne avledes av de to første, da det er de samme måledataene som brukes. Antall uavhengige vektorer vil være mindre enn antall mottakere som er i bruk i det aktuelle felles tidsrommet. Se mer om dette i "Grunnleggende landmåling" og standarden fra Kartverket om satellittbasert posisjonsbestemmelse.

Benyttes måledata fra ulike, ikke overlappende tidsrom, kan de beregnede vektorene ses på som uavhengige. I vårt eksempel kan vi få 3 uavhengige vektorer på flere måter, for eksempel slik:

- A-B: Tidsrommet $t_1 - t_2$ brukes
- A-C: Tidsrommet $t_2 - t_3$ brukes
- B-C: Tidsrommet $t_3 - t_4$ brukes

I tillegg kan enten vektoren A-B eller B-C også beregnes i tidsrommet $t_2 - t_3$

Står den ene mottakeren i et punkt med kjente koordinater og høyde, kan vi med den andre mottakeren bestemme koordinater og høyde for nye punkter, med de kjente verdiene som utgangspunkt.

Measuring methods, phase observations. Initializations

Stedfestingen, eller vektormålingen, kan vi gjøre i sann tid, eller ved beregninger i ettertid.

Under utviklingen av observasjonsligninga i kapittel 5, så vi at selve fasemålingen ga desimalene i antall bølgelengder ($\Delta\lambda$) for avstanden mellom satellitt og mottaker. De ukjente heltallene i starttidspunktet, N -ene (1 for hver satellitt), løses først ut ved målinger over tid, *initialiserings-perioden*.

Det er geometriendringen, det at satellittene endrer posisjon (asimut og høydevinkel endres), som er en forutsetning for at heltallene skal kunne bestemmes, slik at man får antall hele bølgelengder i starttidspunktet for vektorberegningen, $N \cdot \lambda$.

Ved fasemåling brukes det i hovedsak følgende målemetoder, der det også er ulikheter i hvordan initialiseringen utføres:

- (1) Klassisk statisk måling
- (2) Korttidsstatisk
- (3) Kinematisk måling

(1) Classic static:

Ca. 10 – 120 min observasjonstid, der nødvendig observasjonstid er mest avhengig av vektorens lengde og satellittgeometrien. For en vektor på opptil 10 km, er normalt ca. 60 min måletid tilstrekkelig.

Ved bruk av enfrekvente mottakere, må klassisk statisk måling benyttes, korttidsstatisk er ikke mulig.

(2) Short time static:

Mye kortere initialiseringstid, sammenlignet med klassisk statisk måling. Må ha tofrekvente mottakere, da både kode- og fase-målingene brukes for å få en rask bestemmelse av de ukjente heltallene. Måletiden er normalt ca. 2 – 10 min, der nødvendig observasjonstid er mest avhengig av vektorens lengde og satellittgeometrien.

For vektorer over ca. 10 – 15 km, må klassisk statisk måling benyttes, korttidsstatisk er ikke mulig.

(3) Kinematic:

Dette er målinger som utføres med en av mottakerne i bevegelse (mobil mottaker, eller "rover") og den andre fast i et kjent punkt ("referanse").

Betegnelsen kinematisk måling dekker også metoden med å stoppe opp underveis for å bestemme punkter (grensemerker, kummer, GIS-punkter generelt). Metoden kaller stopp-og-gå, "stop-and-go". De ukjente heltallene kan også bestemmes når roveren er i bevegelse (ingen statisk initialiseringsperiode), og vektoren mellom mottakerne bestemmes i tilnærmet sanntid. Metoden kalles på direkten ("on the fly")

Slike sanntidsmålinger kalles også RTK: "real time kinematic". Forutsetningen for RTK er at den mobile mottakeren får korreksjonssignaler fra referansestasjonen via radio, VHF, eller kommunikasjonssatellitt. Er dette ikke mulig, prosesseres posisjonene i ettertid.

Måletiden for å stedfeste et punkt (eller posisjon) ved kinematisk måling, er minimum 1 epoke (en registrering), altså ca. 10 sekunder ved stop-og-gå-metoden. Nøyaktigheten er lavere enn ved klassisk statistisk måling, ca. 1-2 cm for vektoren eller posisjonen.

For mer detaljerte retningslinjer for målingene, vises det til Standarden "Satellittbasert posisjonsbestemmelse", fra Kartverket

I faglitteraturen skilles det også mellom:

- (1) Når mottakeren som ikke måler i basestasjonen, er i ro (*relativ bestemmelse*).
Et eksempel er punktbestemmelse.
- (2) Er mottakeren i bevegelse under målingene, foretar vi en *differential* bestemmelse, f.eks. ved navigasjon

Kartverket har i sin nyeste versjon av satellitt-standard brukt differential measurements for begge typene ovenfor.

Vi bestemmer de nye punktene (eller navigerer) relativt til basepunktet vårt, også kalt referansepunktet: Med bruk av 2 mottakere bestemmes vektoren mellom de to mottakerne, koordinat-*differansene*: ΔX , ΔY , ΔZ . Vi bestemmer en basislinje eller en GNSS-vektor mellom to mottakere. Dette kan gjøres ved etterprosessering (beregninger i ettertid), eller i sanntid. Ved målinger i sanntid fås korrigerte koordinater i felten, slik at for eksempel grensepunkter kan settes ut direkte.

Figur 8.2 viser en enkel framstilling av prinsippet med relativ måling. Referanse-stasjonen kan ved hver måling sammenligne målt posisjon med gitt posisjon. Dermed kan man få en nøyaktig stedbestemmelse av den andre mottakeren også. Dette kan gjøres ved at de målte avstandene mellom satellittene og mottakeren i det ukjente punktet blir korrigerte ved hjelp av målingene i det kjente punktet. Korreksjonene kan påføres i sanntid (for eksempel RTK), eller ved beregninger i ettertid.

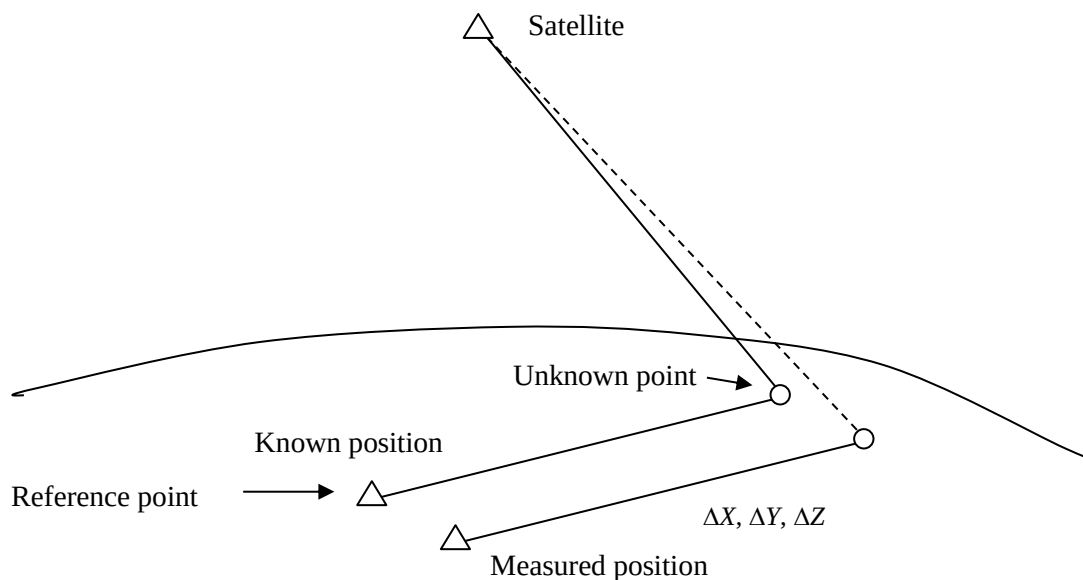


Figure 8.2 Absolute and relative measurements

8.2 Accuracy

Hvilken nøyaktighet oppnås ved å måle *samtidig* med 2 eller flere mottakere?

Standardavviket (gjennomsnittsnøyaktigheten) på vektoren (basislinja), eller på koordinatene (i grunnriss) til det nye punktet er ved klassiske statiske målinger:

Ca. 1-5 meter med C/A-kodemålere

Ca. 5 mm +1 mm/km med geodetiske mottakere (enfrekvente, L1)

Ca. 3-5 mm +1 mm/km med geodetiske mottakere (tofrekvente, L1 og L2)

Tallverdiene gjelder generelt, også når SA-støyen er på (...men SA ble slått av 2. mai 2000).

Hvorfor denne store forbedring fra absoluttbestemmelse med 1 mottaket til relativbestemmelsen med 2 mottakere som måler samtidig?

Ved beregningene er det *differansen* mellom målingene som er interessante.

Vi setter opp et feilbudsjett for de viktigste feilkildene for kodemålinger med 2 mottaker på C/A-koden. (Se også figur 1.1 som viser at avstanden mellom to mottakere, er svært liten sammenlignet med avstanden til satellitten.)

Tallverdiene er standardavvik for feiltypen (gjennomsnittsverdi), og viser hvor mye feil som blir påført målingene mellom en satellitt og mottakeren når det måles relativt (differensielt):

Error source	Standard deviation in meter	Merknader
Satellittklokka, stabilitet	0	
Banefeil	0	
SA, Selective Availability	0	
Ionosfære	0,5	Restfeil, ca samme strålegang gjennom atmosfæren
Troposfære	0	
Støy på mottakeren	1	
Mottakerklokka		Løses ut som en ukjent
Flerveisinterferens (multipath)	1	Avhengig av reflekterende flater

Table 8.1

Legg merke til at de største feilkildene med 1 mottaker, nulles ut: SA-støyen, satellittklokka og banefeil. Vi sitter igjen med forskjellen i størrelse for hvordan signalene forstyrres i atmosfæren, mens støy i mottaker og på grunn av flerveisinterferens (multipath) er de samme.

By using C/A code measurements:

Standardavviket (gjennomsnittlig nøyaktighet) på en posisjon bestemt med 1 mottaker i et ukjent punkt og med 1 mottaker som måler samtidig i et kjent punkt (basepunkt), er:

- Ca. 1-5 meter i grunnriss
- Ca. 1,5-7 meter i høyde

Med en PDOP på 2,5 (som tilsvarer en gjennomsnittlig god satellitt-geometri), fås en usikkerhet i posisjonen på mottakeren på ca. 2,5 – 12,5 m i grunnriss og 4 - 17 m i høyde.

For mer om PDOP, se også i «Grunnleggende landmåling», eller i en GNSS-lærebok.

Å måle relativt (eller differensielt) er helt avgjørende for å oppnå en nøyaktighet på ned mot 1 meter på posisjonsbestemmelsen med kodemålinger, eller på cm-nivå (eller bedre) med fasemålinger.

8.3 Methods for calculations, relative observations

Good coordinates and heights of the known reference points must be used in the processing of baselines (better than 10 meters), to avoid a deformation of the calculated baseline, see chapter 6.2.

Use independent baselines in the processing, see chapter 8.1.

One frequency solutions of the baseline can be used for baselines shorter than ca. 10-15 km, but the ionospheric delays (noise) will be better solved with a two frequency solution (L1 and L2 on GPS).

See also chapter 9 for more details about the baseline processing, the baseline processing example.

For more detailed guidelines for the processing, see also the standard "Satellittbasert posisjonsbestemmelse" from the Norwegian mapping authority, «Kartverket», and the user manuals of the actual baseline processing software (for instance Leica GeoOffice).

8.3.1 Double differences

As observation equations in the baseline processing normally double differences are used, double differences between two receivers and two satellites at the same point of time. A reference satellite is used as a common satellite when forming the double differences. See an example of a baseline processing in chapter 9, where a satellite that is moving high up on the sky is chosen as the reference satellite. We get the best signals from satellites moving high up on the sky (less noise) compared to satellites close to the horizon. See chapter 7.5 about the correlation between the double differences when a common reference satellite is used.

The main unknowns in a baseline processing are the coordinates of the unknown point and the unknown integers at the start of logging data, the N -s: 2 for each satellite if it is a two frequency receiver (L1 and L2), 1 for each satellite if it is a one frequency receiver (L1).

The other sources of error, like the errors on the receiver and satellite clocks, orbit errors, ionospheric and tropospheric delays, are eliminated or minimalized by using the double difference equations (see chapter 7.2 about the double differences).

The unknown integers, the N -s, are per definition integers (the number of complete cycles or periods).

Calculations with floating integer for the N -s ("float")

The baseline processing starts with a "float" solution, where the unknown N -s are calculated as decimal numeral together with the three coordinates of the unknown point.

The float solution gives normally the lowest value of the standard deviation of the observation with unit weight, as the float solution is a pure Least square solution ("Minste-kvadraters-løsning") *without any extra condition* (for instance that the N -s must be integers).

Calculations with integer conditions for the N -s ("fix")

From the float solution we obtain values on the unknown coordinates that normally are close to the final values. We are using the float solution to finally obtain a "fix" solution, which is a Least squares solution with a condition that all the integers, the N -s must be solved to be integers, the ambiguity resolution.

Based on the decimal numerals of the N -s from the float solution, several calculations are carried out, where different combinations of the N -s are used. The search criteria for choosing the integer values of the N -s in these calculations can vary. When choosing possible values of the N -s, the processing program search for combinations of integer values of N -s based on the preliminary value of the baseline and the preliminary values of the coordinates from the float solution.

The larger the search area is for finding the optimal solution of the integers, the N -s, the more combinations of the N -s must be tried. We can by statistical methods obtain an optimal reduction of the search area. The fix solution will normally give higher values of the standard deviation of the unit weight, as the fix solution is a Least squares method with a constrain.

8.3.2 Triple differences

For long baselines (more than ca. 20 km) the atmospheric noise / delays can be significant different for the incoming signals at the different receivers. Then it might be impossible to find an optimal combination of the N -s. In such cases can the triple difference give a better solution of the coordinates of the baseline, compared to a double difference estimation.

For shorter baselines can the triple differences be used to estimate a preliminary position, as the triple differences do not contain the term with the unknown integers, the N -s, and the terms with the clock errors of the satellites and receivers. See chapter 7.3 about how the N -s and the clock errors are eliminated from the triple differences.

As the condition of the N -s must be integers is not possible to include in a triple difference solution, the triple differences must not be used for processing of short baselines (shorter than ca. 20-30 km).

The triple differences is also used to repair cycle slips, as shown in chapter 7.4.

8.3.3 Example of float and fix solution of a baseline

The baseline is ca. 616 m and were measured in 1995 with an one frequency GPS receiver (from Trimble, measuring only on L1). There are 8 integers to be solved (8 biases"). See the table below for results and accuracy measures ("Sigma" = standard deviation).

In the *float solution* the integers, the *N*-s are solved as decimal numerals. Only the numbers behind the decimal point are shown. We see from the values that the 8 *N*-s differ very little from integer values. The largest deviation is for no. 6, of 0,058 with a standard deviation of 0,022. Based on these small deviations from an integer, we can anticipate that the calculations of the integers should go without problems. If the decimal numerals were around 0,4-0,6, we must anticipate to have problems with finding the optimal solution of the unknown integers, the *N*-s. In this case there are more than one combination of the integers that will give ca. the same values of the standard deviation of the unit weight from the least squares estimations. The observations are probably of low quality, the reasons could be too much noise on the received satellite signal, too long baseline or too short observation period.

	Float		Fix		Triple	
	Solution	Sigma	Solution	Sigma	Solution	Sigma
ΔX (m)	-532.102	.002	-532.107	.001	-532.090	.037
ΔY (m)	-153.293	.003	-153.292	.001	-153.283	.023
ΔZ (m)	270.364	.003	270.370	.002	270.368	.027
bias 1 (cycle)	.006	.015	.000			
bias 2 (cycle)	-.057	.011	.000			
bias 3 (cycle)	.013	.015	.000			
bias 4 (cycle)	.049	.019	.000			
bias 5 (cycle)	-.008	.014	.000			
bias 6 (cycle)	.058	.022	.000			
bias 7 (cycle)	-.007	.011	.000			
bias 8 (cycle)	.018	.012	.000			
Distance (m)	616.220		616.227		616.210	
RMS (cycles)	0.024		0.028		0.024	

Table 8.1 Float, fix and triple solutions of a baseline

The *fix solution*, as anticipated, was obtained without problems, as it is a short baseline and a long observation period.

The ratio F between $\sum(pvv)$ for the second best and the best combinations of integers (the *N*-s) is 26,36, with ca. 3,0 as a limit for this test.

$$F = \frac{\sum(pvv)_2}{\sum(pvv)_1} = \frac{5,27174}{0,19999} = 26,36$$

It was only the distance to satellite no. 6 (prn 9), which got a change of the integer value, *N*, compared to the closest integer values from the float solution. It was changed with 1, which is 19,0 cm on the frequency L1).

The fix solution has a slightly higher *rms* value than the float solution. (*rms*: "root-mean-square", is a kind of standard deviation.)

From the example in the table above, we see that the coordinate values of the triple solution differs with ca. 2 cm from the fix solution and that the standard deviations on the coordinates increases considerably. This is logical, as the condition of integer values of the *N*-s, is not part of the triple difference solution. Thus we do not get optimal values.

9. Example of a GNSS baseline processing

A printout from a baseline measured in March 2000 between two points in Trondheim. The baseline is measured with 2-frequency GPS receivers and the distance between the points is 2,2 km. The normal measuring time for such a short baseline is about ca 10 minutes using short-time-static method and ca. 45-60 minutes by classical static measurements. Here the measuring time is ca. 5 hours, because the measurements were used in an investigation of accuracy.

Leica 300 GPS receivers are used. Only GPS data was logged.

The contents of the documentation files covering the baseline processing from different receivers, different producers and different years are relatively alike.

The example shows:

- Assumptions and choices which are made before the calculations
- Overview showing which satellites it is observed against
- The criteria an estimated baseline must fulfil, to be good enough for further use
- The final results of the baseline processing, the coordinate differences and the accuracy measures

9.1 Various choices

9.1.1 About the project

Here you will find information about the program (including the version), coordinate system and time.

Processing software	: Leica SKI / Data processing version 2.3-1
Processing kernel	: PSI version 2.30
General header	: LEICA AG, CH-9435 Heerbrugg
Project name	: 21032000
Coordinate system	: WGS84
Time	: All results in local time (GPS + 2.00 hrs)

9.1.2 Parameters for the processing

Normally the baseline processing will be successful with the default values in the processing program. When we have problems, or wish to have a better accuracy of the baseline, we can improve the results by using other parameters or parameter values in the baseline processing.

Cut-off angle (deg)	: 15
Tropospheric model	: Hopfield
Ionospheric model	: No model
Solution type	: Standard
Ephemeris	: Broadcast
Data used	: Use Code and Phase
Phase Frequency	: Automatic
Code Frequency	: Automatic
Limit to resolve ambiguities (km)	: 26
a priori rms (mm)	: 15
Sampling rate for static (sec)	: Use all
Phase processing	: Automatic
Cycle slip detection	: Phase check & loss lock flag
Phase measurement rms (mm)	: 10
Update rate for kinematic (epoch)	: 5
Min. time to fix ambiguities. - L1 only (min):	9

Below some of the parameters in the table will be described. For a more complete descriptions, see also the user manuals.

- **Cut-off angle:** The value of the vertical angle indicate how high above the horizon a satellite must be, before starting to log data from the satellite. A common choice is now (2015) with both static and RTK measurements is 10° (15-20 years ago it was 15°). We wish to avoid using the satellite signals that have travelled longest through the problem areas *ionosphere* and *troposphere*, see figure 9.1. Today's better receivers and software makes it possible to use signals from low satellites. It is also possible to change the cut-off-angle to a higher value than 10° above the horizon, if there are problems with the ambiguity resolution, the integers N , see chapter 9.4.1.

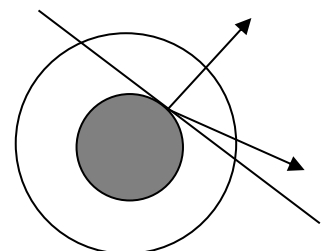


Figure 9.1 Directions to satellites. Earth (grey), the atmosphere, the horizon in the observation point

- *The troposphere and the ionosphere* can be modelled, or we can use estimated, real models of atmospheric corrections from IGS (the International GNSS service) for use in baseline processing.
- *Satellite orbits* (ephemeris): The coordinates of the satellites are received as a "broadcast" version, as a part of the satellite signal when measuring. These orbit data has the lowest accuracy, they are predicted. Estimated, real orbits can be imported to the processing program, see table in chapter 6.1.1.
- *The limit to fix the ambiguities*, the *N*-s, is normally set to ca. 20 km. For longer baselines the atmosphere is causing problems, and the atmospheric noise can make it impossible to obtain an optimal result of the unknown ambiguities.
- *A priori rms* (root mean square = standard deviation): Threshold value for phase measurements. The standard value, here 10 mm, normally should be used. For long baselines and long observation periods (more than ca. 1 hour, as here), The *rms* can be increased as especially the ionospheric noise is increasing with longer baselines. See also about statistical testing of the solution in chapter 9.4.1, where *rms* is used.

9.1.3 Choice of satellites

It is possible manually to choose satellites that shall not be used in the baseline processing, to disable satellites. This is an actual situation if we have problems with estimating the unknown integers, the *N*-s (ambiguities).

Satellites with many cycle slips, or satellites with data logged for a short period, are causing more trouble than contributing to an optimal result of the baseline processing, and can be the reason for not getting the baseline approved (not having the *ambiguities fixed*).

Manually disabled satellites : None

Some of the satellites can be "sick", either because of they are moving to another position relative to the other satellites, or that the satellite clock is not reliable any longer. Such satellites are also marked as not useable in the satellite message, which is received during the measurements.

9.2 Overview of the GNSS baseline(s). An example

In this baseline processing three baselines are processed. «Our» baseline is the second baseline, between the points *NTH-2* and *KU-46*. From the report file we can see that the control point *KU-46* is the reference point (starting point of the baseline) in the processing. We see from the table below that the ambiguities are fixed for all the baselines.

The frequencies L1 and L5 are used. L5 was in 2000 a "wide lane", a linear combination between L1 and L2 with a wave length of 0,862 m. This 4 times longer wave length than those of L1 and L2, was used to faster calculate the distances to the satellites.

STS indicates static observations (static survey). Other alternatives are kinematic survey, short-time static survey etc.

BL id	CH id	Rover	Reference	First common ep		Amb	Frq	Obs
1	1	NTH-2	hell	21.03.00	14:43:30	Y	1+5	STS
2	2	NTH-2	KU-46	21.03.00	12:54:30	Y	1+5	STS
3	3	NTH-2	storsand	21.03.00	13:31:30	Y	1+5	STS

9.2.1 Start values of the coordinates of the reference and the rover point

As shown in chapter 6.2.1 and 6.2.2, the position of the reference point must be good, better than 10 meters in absolute value. It is recommended that coordinates and heights of the known points are entered into the processing program before the data processing. Probably you have to use either Cartesian XYZ coordinates or geographic coordinates (latitude, longitude, ellipsoidal height). For the coordinate values of the rover (the other end point of the baseline), it is Ok to use the absolute position from the measurements in the point, the navigation solution shown below, or the estimated single point values. It is only the start point of the baseline, the reference point, which needs good absolute values.

Reference : Point id : KU-46					
X	2813149.4492	m	Y	518057.3274	m
Z	5681796.4309	m			
Lat	63 25 54.00750	N	Lon	10 26 3.96090	E
h 151.8450 m					
Rover : Point id : NTH-2					
X	2814787.6856	m	Y	516811.3629	m
Z	5681094.0626	m			
Lat	63 25 3.83044	N	Lon	10 24 14.25954	E
h 143.6955 m					

Point NTH-2 gets from the baseline processing in this example, almost the same good absolute coordinates as the reference point KU-46, see the accuracy of the baseline estimation. Therefore we can use the point NTH-2 as a reference point in another baseline processing, as long as we remember to use the new estimated coordinates of NTH-2.

For a network of baselines we do not need many points with a good accuracy at the start of the processing. We can perform the baseline processing in several stages, in such a way that we always have good absolute values in the starting point of the baseline (known or estimated coordinates).

9.2.2 Baseline processing, information

The total observation period was almost 5 hours, time: 21.03.2000: From 12:54:30 to 17:53:50. The table below shows that static survey (STS) was the observation method.

The time between each logging of data, the observation rate, was 10 seconds. For long baselines normally 15 seconds or more is used. For the short-time-static method 10 seconds is recommended, and for real time measurements 1 seconds or less is used. An *epoch* is one storage of data.

We can check the antenna height is correct entered, it normally consists of two components: Height reading (measured by the surveyor) and antenna offset (default value dependent of the antenna type).

Values of the eccentricity of the antenna can also be entered, normally these values are zero.

	Rover	Reference
Point id	NTH-2	KU-46
Sensor/Controller id	1898 /280300	1706 /280302
Operation mode	STS	STS
Observation rate (s)	10.0	10.0
Ht reading/Ant offset (m)	0.255 /0.139	0.037 /0.139
Eccentricity E/N/H (m)	0.000 /0.000 /0.000	0.000 /0.000 /0.000

9.3 Satellite information

9.3.1 The number of epochs

The table below shows how many (20) and which satellites that are involved during the survey and how many epochs that are registered for each satellite on the two frequencies L1 and L2 (phase) and on the codes. One *epoch* is one storage of data, see chapter 9.2.2 for more.

The total number of epochs in the survey are 1797. From satellite no. 17 it was registered data in a period of 1642 epochs $\times 10 = 16420$ seconds = $4^h 33^m 40^s$. Only the satellites 10, 24 and 30 have different numbers of registrations in the four columns. Different numbers indicates problems with the observations. There are a total of 13254 logged observations on L1 and 13251 on L2, gives 26504 logged observations on L1 and L2.

SV id	L1 phase	L2 phase	L1 code	L2 code
1	339	339	339	339
3	1233	1233	1233	1233
6	963	963	963	963
8	544	544	544	544
10	902	901	902	901
11	3	3	3	3
13	160	160	160	160
15	483	483	483	483
17	1642	1642	1642	1642
18	142	142	142	142
19	629	629	629	629
21	597	597	597	597

22	1556	1556	1556	1556
23	948	948	948	948
24	284	283	284	283
25	540	540	540	540
26	601	601	601	601
27	523	523	523	523
30	424	422	424	422
31	741	741	741	741

9.3.2 Satellite information. Azimuth and elevation (*høydevinkel*)

This table shows where the satellites are in the sky and the geometrical accuracy of the absolute position. The data from the satellites were registered with an interval of 10 seconds (epoch). In the table you will find a selection of the data, one line per hour.

The geometrical accuracy measures PDOP and GDOP and the elevation and azimuth of the satellites are found. Satellite no. 6 has at 12:54:30 an elevation of 74° and an azimuth of 161°.

PDOP was at the same point of time 4,0, which is larger than the mean value of ca. 2,5.. There were 5 satellites at 12:54:30, with a PDOP of 4,0; at 16:54:30 there were 11 satellites and a much lower PDOP value (1,8). Notice that the PDOP value decrease when the number of satellites increase.

A low DOP value indicates good geometry.

sat id :	6	25	30	1	22					
hh:mm:ss PDOP GDOP										
12:54:30	4.0	4.6	74/161	47/245	47/134	20/331	18/288			
sat id :	6	25	30	22	10	17	18			
hh:mm:ss PDOP GDOP										
13:54:30	2.2	2.5	59/100	27/220	20/138	44/284	28/ 62	41/167	15/343	
sat id :	6	3	22	10	17					
hh:mm:ss PDOP GDOP										
14:54:30	3.4	3.9	32/ 98	26/264	61/249	24/ 35	63/137			
sat id :	27	31	3	19	22	23	26			
17 21 8 15										
hh:mm:ss PDOP GDOP										
16:54:30	1.8	2.0	19/321	43/279	73/223	16/297	23/193	50/110	21/ 40	
31/ 71 33/162 20/329 24/234										
sat id :	11	31	3	23	21	15				
hh:mm:ss PDOP GDOP										
17:53:50	3.1	3.6	15/253	66/253	57/165	47/ 71	56/143	52/239		

9.4 Information about the calculations

The total number of satellites, and the standard deviation of the unit weight (*rms*, "root mean square"). With a total of 1797 epochs (almost 5 h observation time) and with two-frequency receivers and the a total of 26504 stored measurements on L1 and L2, the average number of GPS satellites per epoch are 7,38.

Total number of used measurements:	26504
Root mean square unit weight:	0.3957

9.4.1 Integer estimations (the N-s) ("ambiguity resolution")

In the processing in 2000, Leica used the method FARA ("Fast ambiguity resolution approach") for solve the ambiguity, the N-s. More about FARA and other methods to solve ambiguity in later courses.

In this baseline processing there are three statistical end tests that decide if we have obtained a good estimation of the baseline and that we have obtained significant values of the integers, the N-s:

Ambiguity tests:
test 1 - rms float not significantly bigger than rms a priori
test 2 - rms fix not significantly bigger than rms a priori
test 3 - rms fix is significantly smaller than rms fix 2

a priori	15.0 [mm]
rms float	3.2 [mm]
rms fix	3.5 [mm]
ratio	3.8615

1. For the "float" solution (the solution with decimal numerals and not integers, the N -s are decimal numerals): The standard deviation (*rms float*) must not be significant bigger than the a-priori standard deviation. Notice that the value 15 mm is used as the a priori value (= the assumed value), and not 10 mm which is the default value. For long baselines (and an observation period of more than ca. 1 hour, as here), can the a priori value be increased as the ionospheric noise increases.
Here the *rms float* is 3,2 mm and the a priori value is 15 mm => Ok
2. For the "fix" solution (the solution with the N -s as integers): The standard deviation (*rms fix*) must not be significant bigger than the a priori standard deviation.
Here the *rms fix* is 3,5 mm and the a priori value is 15 mm => Ok
3. For the "fix" solution with the lowest (and thus best) standard deviation: The standard deviation (*rms fix*) must be significant lower than the "fix" solution with the second lowest standard deviation (*rms fix 2*).
Here the ratio (*rms fix 2*) / (*rms fix*) = 3,8615 => Ok. From this ratio we calculate the value of (*rms fix 2*) to be 13,5 mm.

For this baseline processing all the three tests were Ok, thus the found fix solution can be used as the final and most optimal solution of the baseline.

The ratio between the second best (lowest) standard deviation and the standard deviation of the best fix solution was calculated to be 3,8615.

If this ratio had been 1,0 in test no. 3, there are two equivalent solutions of the unknown integers, the N -s, but with different values of the estimated coordinate differences of the baseline.

If test no. 3 fails, the baseline cannot be used. We can try a new estimation after changing some of the parameters of the baseline processing; for instance remove satellite(s), use another observation period etc., in worst case measure the baseline again.

A rule of thumb says that this ratio should be at least 2-3 to have two significant different fix solutions.

The table below shows which combinations of integers that are used in the two best fix solutions (those with the lowest *rms*). Double differences are used as observation equations in the baseline processing, and satellite no. 6 is used as a reference satellite for both the frequencies L1 and L2, at the start of the survey.

	Reference Satellite(s): 6/L1 6/L2										
set#	25/L1	25/L2	30/L1	30/L2	1/L1	1/L2	22/L1	22/L2	24/L1	24/L2	10/L1
	10/L2	17/L1	17/L2	18/L1	18/L2						
1	-11	12	-34	-35	13	120	-1	102	-37	22	16
	47	3	6	-5	14						
2	-11	12	-34	-35	13	120	-1	102	-37	22	16
	47	3	6	-4	15						

What is the reason for choosing satellite no. 6 as a reference satellite?

Look in the table showing azimuth and elevation in chapter 9.3.2. Satellite no. 6 is very high above the horizon (has an elevation of 74°). We get the best signals from this satellite (less noise, especially less atmospheric noise compared to satellites with a low elevation, as the signals from low satellites have a longer path through the atmosphere than satellites with a higher value of the elevation, see figure 9.1).

It is a large number of combinations of the unknown integers, the N -s, that are used for finding the best combination of the N -s).

The numerical values of the N -s in the table above is not of importance, it is the last two digits that are shown in these two best sets or combinations. Find the integers that are *different* when you compare the two rows. It is only the L1 and L2 values to satellite no. 18 that differs with 1 wave length, 5 and 4 on L1,

14 and 15 on L2. These two distances are changed with 1 wavelength each, when we compare set 1 with set 2, and the solution of the baseline using set 1 is significant better than the solution using set 2. 1 wavelength is ca. 19 cm on L1 and 24,4 cm on L2 in vacuum.

The table shows the solution of the unknown integers, the N -s, for 9 satellites at the start of the survey (8 satellites plus 1 reference satellite). When other satellites show up later in the survey, must the unknown integers for these new satellites be estimated.

Reference satellite

Especially for a long observation period, a change of reference satellite might be required. In our example, satellite no. 6 is the reference satellite at the start of the survey, but satellite no. 6 is not present during the whole survey. Therefore another satellite must be chosen as the new reference station when no. 6 cannot be used any longer. The last registration of data from satellite no. 6, was 15:34:50, and the baseline processing shows that at 14:29:10 satellite no. 22 was used as a new reference satellite. From 14:59:10 no. 17, and from 16:01:10 no. 3 were used as reference satellites. See the table in chapter 9.3.2, for azimuths and elevations of the satellites.

9.4.2 Cycle slips

Cycle slip is loss of signal from a satellite in a (short) period. The table below shows the cycle slips that were found in the processing of the baseline. The signal on L1 from satellite no. 30 was lost for ca. 20 seconds, and the slip value was 1 (a jump of 0.19 m). Only one cycle slip during a ca. 5 hours observation period, is almost an abnormal low figure.

If a certain satellite has had many cycle slips, and there is a problem with the processing of the baseline, we can remove a time slot with many cycle slips, eventually remove the satellite from the calculations.

Total no of cycle slips : 1						
Time	time from start (sec)	SV id	freq	slip value (cycle)	fraction (cycle)	
21.03.00 12:54:50	20	30	1	1.0	-0.014	

9.4.3 Final coordinates

The result of a baseline processing:

- The coordinates of the mobile point (the unknown point, "the rover") calculated with given coordinates of the reference point as a starting point.
- The coordinate differences ΔX , ΔY and ΔZ in the datum WGS84, Cartesian coordinates. It is these three original observations that later is transformed to the derived observations in the map projection plane: Bearing, distance, zenith angle or the height difference referred to the geoid. Thus the estimation of the coordinates and heights can be done in the map projection plane.
- The standard deviations (s) of the observations ΔX , ΔY and ΔZ . The values of the standard deviations calculated together with the baselines, are too low. See chapter 11.4.1 for more about the standard deviations and the correlations between the three components/observations of a baseline. It is recommended to multiply the derived standard deviations with a factor of ca. 10.
- Slope distance (or chord distance).

Rov:NTH-2 Ref:KU-46 Amb:Y Proc: L1+L2 phase 21.03.00 12:54:30											
Cartesian :											
X	2814807.7505	m	Y	516795.6314	m	Z	5681070.0818	m			
Δ X	1658.3013	m	Δ Y	-1261.6960	m	Δ Z	-726.3491	m			
sX	0.0001	m	sY	0.0001	m	sZ	0.0002	m			
Geodetic :											
Lat	63 25 2.99595	N	Lon	10 24 12.88320	E	h	129.8096	m			
Δ Lat	-51.01155		Δ Lon	-1 51.07770		Δ h	-22.0354	m			
sLat	0.0001	m	sLon	0.0001	m	sh	0.0002	m			
Distance :											
Slope	2206.6769	m									
sSlope	0.0001	m									

The geodetic coordinates are the *latitude* and the *longitude* (with the Greenwich meridian as the zero meridian) and *ellipsoidal height* in the datum WGS84.

Compare the estimated coordinates in the table above with the preliminary navigated coordinates of point *NTH-2*, see the table in chapter 9.2.1. The point is moved ca. 35 meters, indicating that the quality of the absolute position (only observations in *NTH-2* are used, and with the noise SA on, see chapter 6.9).

9.4.4 The variance-covariance matrix of the baseline

The calculated values of the standard deviation of the observation of the unit weight (*rms*) is $s_0 = 0.3957$.

The co-factor matrix, $\mathbf{Q}_{xyz}$, denomination m^2 (*vektskoeffisient-matrisen*):

The matrix is symmetrical around the main diagonal.

The matrix is full (the upper triangle is shown in the table below).

The elements of the matrix show that *the three coordinate differences are correlated*, as the elements outside the main diagonal (the covariances) are different from zero.

a posteriori rms : 0.3957			
Co-factor matrix (upper triangle [m*m]) :			
	qx	qy	qz
qx	+5.9344000E-008	+5.1850000E-009	+5.4495001E-008
qy		+2.5848999E-008	+9.3720001E-009
qz			+2.2508399E-007

The standard deviation of ΔX , ΔY and ΔZ in the table can be calculated from the values of the standard deviation of the observation of unit weight and the co-factor matrix:

The variance-covariance-matrix: $\mathbf{C}_{xyz} = s_0^2 \cdot \mathbf{Q}_{xyz}$, denomination mm^2

$$\mathbf{C}_{xyz} = \begin{pmatrix} \sigma_{xx}^2 & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy}^2 & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz}^2 \end{pmatrix} = \begin{pmatrix} 0.00929 & 0.00081 & 0.00853 \\ 0.00081 & 0.00404 & 0.00147 \\ 0.00853 & 0.00147 & 0.03524 \end{pmatrix} \text{denomination: mm}^2$$

The standard deviations of the observations are obtained by taking the square roots of the elements along the main diagonal, and thus obtaining the values in chapter 9.4.3.

Observe the size of the variances and the co-variances that are computed in the baseline processing, they are far too low. See chapter 11.4.1 covering the standard deviations, where it is recommended to scale the values of the standard deviation by a factor of ca. 10, or the variance-covariance matrix with a factor of $10^2 = 100$.

In our example with an observation period of almost 5 hours (normally 10-60 minutes), we must assume that we obtain a lower values of the elements of the variance/covariance matrix, compared to a shorter observation period. This baseline is using far more observations and thus normally obtain lower values of the accuracy measures, compared to an estimation of this baseline by using a shorter observation period.

PART B: TRANSFERRING THE GNSS BASELINE TO THE MAP PROJECTION PLANE

This is one of the main problems/task concerning the processing of a GNSS baseline. This task is described in this part of the compendium:

- Transferring the original observations of a GNSS baseline from Cartesian 3D coordinates in the WGS84 datum (ΔX , ΔY , ΔZ) to decomposed observations in the map projection plane (and often in another datum). The decomposed observations are distance, bearing and zenith angle.

See a textbook (for instance “GNSS” by Hofmann-Wellenhof) or internet for more about the datum WGS84.

10. GNSS baselines and GNSS network

One of the main problems of the GNSS calculations is described in this part of the compendium:

The translation of measured GNSS baselines (ΔX , ΔY , ΔZ) from the datum WGS84 to the decomposed and derived observations in the map projection plane (distance and bearing) and a zenith angle or a height difference referred to the geoid (mean sea level, not the ellipsoid).

10.1 The GNSS baseline in WGS84

The first order network of control points of Norway and of a municipality (like Trondheim) usually is formed as a network of GNSS baselines. A similar network is also used when the aim is to obtain very good quality of the coordinates (high accuracy and precision), as illustrated in figure 10.1. These networks consist of polygons with three, four or five corners, dependent of the wanted accuracy of the new control points. This is described in the Norwegian standard “Satellitbasert posisjonsbestemmelse”; in it is found on the home page of the National Mapping Authority (“Kartverket”).

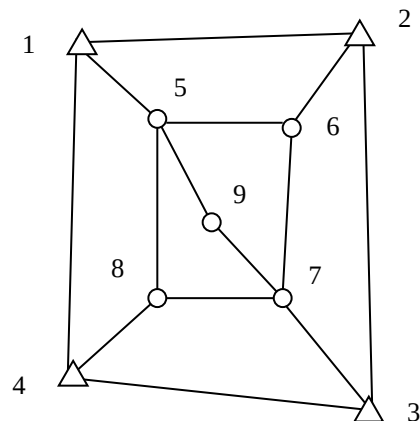


Figure 10.1 GNSS baselines in network of control points

From a baseline processing we obtain among other factors the following coordinate differences:

ΔX , ΔY , ΔZ	3D coordinate differences in a mean global geocentric coordinate system called "Conventional Terrestrial Reference System, CTRS", also named CTS.
--------------------------------------	---

After January 10th 1987 the Navstar GPS system is using the WGS84 datum, which is a version of a CTRS system. See chapter 9.2 in “GNSS” for a closer description of WGS84.

The ellipsoid in this datum is the WGS84 ellipsoid, which has the following parameters:

The semi major axis	$a = 63\,78137.0000\text{ m}$
The flattening	$f = 1/298.257223563$

From a baseline processing we also obtain the standard deviations s_{xx} , s_{yy} , s_{zz} of respectively ΔX , ΔY , ΔZ , the correlation matrix of ΔX , ΔY , ΔZ , geocentric coordinates and geographic coordinates (latitude and longitude) and ellipsoidal height of the end points of the baseline. See chapter 9. The absolute coordinates have a far lower accuracy than the coordinate differences.

10.1.1 Variance/covariance matrix of a GNSS baseline

See also example exercise no. 1 on itslearning.

The correlation coefficient between two variables, is calculated from the standard deviations of the two variables and the covariance between the two variables:

$$k_{ij} = \frac{\sigma_{ij}}{\sigma_i \cdot \sigma_j} \quad , \quad k_{ii} = 1 \quad , \quad -1 \leq k \leq 1$$

The correlation matrix of ΔX , ΔY , ΔZ (the elements in the matrix are the correlation coefficients):

$$\mathbf{K} = \begin{bmatrix} 1 & k_{xy} & k_{xz} \\ k_{yx} & 1 & k_{yz} \\ k_{zx} & k_{zy} & 1 \end{bmatrix} \quad (10.1)$$

The variance-covariance matrix (**C**) is obtained by the following matrix multiplication:

$$\mathbf{C} = \begin{bmatrix} \sigma_{xx} & 0 & 0 \\ 0 & \sigma_{yy} & 0 \\ 0 & 0 & \sigma_{zz} \end{bmatrix} \cdot \mathbf{K} \cdot \begin{bmatrix} \sigma_{xx} & 0 & 0 \\ 0 & \sigma_{yy} & 0 \\ 0 & 0 & \sigma_{zz} \end{bmatrix} =$$

$$\mathbf{C} = \begin{bmatrix} \sigma_{xx}^2 & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy}^2 & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz}^2 \end{bmatrix} \quad (10.2)$$

The weight matrix of a baseline from point i to point j , is the inverse of the co-factor matrix:

$$\mathbf{P}_{ij} = \mathbf{Q}_{ij}^{-1} \quad (10.3)$$

10.1.2 An example of a result of a baseline processing

The baseline from control point HOLT to the control point 9504 is approximately 839 m, and it is measured with 1 frequency receivers (Trimble 1995, measuring only on L1). Processing the baseline with newer software will give a similar documentation file.

Notice the very low values of the accuracy measures. They have to be scaled with a factor of at least 10 to be closer to the real accuracy measures, see also chapter 11.4 (about variance/covariance propagation).

Standard deviation of the coordinate difference:

```
Vector Standard Deviations (m):
      ΔX              ΔY              ΔZ
      .4858198D-03    .3378463D-03    .1041006D-02
```

Standard deviation of the baseline (the slope distance):

```
slope distance (m)      839.0755      sigma (m)      0.0004
```

The correlation matrix of the baseline: Notice that the elements outside the main diagonal are different from zero: There are correlations between the baseline components:

Vector correlation matrix:

	ΔX	ΔY	ΔZ
ΔX	1.0000000		
ΔY	.1544859	1.0000000	
ΔZ	.6166803	-.0393102	1.0000000

The components of the baseline in the geocentric coordinate system and their standard deviations:

	Solution	Sigma
ΔX (m)	740.883	.000
ΔY (m)	-240.202	.000
ΔZ (m)	-312.158	.001

Converted to a local topographic coordinate system (with the origin in one end of the baseline):

y east (m)	-370.060
x north (m)	-752.562
z up (m)	27.473

Converted to geodetic coordinates (differences in latitude, longitude and ellipsoidal height):

Delta latitude (d m s)	0	0	-24.30515
Delta longitude (d m s)	0	0	-26.66800
Delta height (m)	27.5283		

10.2 Calculation of a GNSS network

There are two main ways to obtain coordinates of a network of control points, when the final coordinates are in the map projection plan (for instance UTM coordinates) and orthometric or normal heights (for instance NN1654 or NN2000 heights in Norway). Often the coordinates are in a different datum than WGS84, which is the GPS datum.

The two main ways are:

1. The first step is a *3D adjustment* using ΔX , ΔY , ΔZ as observations. Then a *transformation* from 3D coordinates X , Y , Z to coordinates and heights in the map projection plane. The software used for baseline processing has also a module that computes estimated values of the unknowns.
2. The first step is a *transformation of the observations* ΔX , ΔY , ΔZ to observations in the map projection plane (distance, bearing, zenith angle (or orthometric/normal height difference)). Then an *adjustment in the map projection plane* is done. This method is used in the land surveying programs which are in use in Norway (like Gemini and GIS/Line).

Both the methods are described more in detail in the next chapters.

10.3 Adjustment of a 3D GNSS network

By estimating of a 3D network we consider the baseline components ΔX , ΔY and ΔZ (Cartesian coordinate differences) as derived correlated observations, ergo 3 observations for each measured baseline.

Building up the GNSS network

When total stations are used in the measurements, the network is built up by using triangles.

A GNSS network can consist of squares of independent baselines. A "square" normally provides a sufficient number of excess measurements. One of the reasons why the GNSS network is still of good quality, is that a baseline has a plane bearing as one of the measurements (the other two measurements are

the distance and zenith angle). The bearing, distance and zenith angle can be derived from the coordinate differences of the base line (ΔY , ΔX , ΔZ), see chapter 10.8.

Example of a GNSS network. See figure 10.1

In figure 10.1 the points 1, 2, 3 and 4 are known points, the other 5 points are unknown points whose coordinates and heights are to be determined. A total of 14 independent baselines are measured, where 4 is between the known points for a check of the coordinates and heights of the known points.

Number of measurements: 14 baselines, 3 measurements per baseline (ΔY , ΔX , ΔZ): $n = 42$

Number unknowns: 5 new points, 3 unknowns per point (Y , X , Z): $e = 15$

System unknowns: Scale, rotation, vertical deviation (east and north value): $e_s = 4$

Number of excess measurements, the degree of freedom: $f = n - e - e_s = 23$

In this example, we have approximately 100% excess measurements: 42 measurements to determine the 19 unknowns and for an inspection of the coordinates of the known points.

In Norwegian, as this topic is not a topic in the course TBA4245:

Vi skal se på hvordan formelapparatet for en estimering av de romlige koordinatene blir, ved bruk av elementmetoden. Teorien bak elementmetoden er beskrevet i kompendiet "Matematisk statistikk og estimering (MKM), elementmetoden" av Jon Holsen/Terje Skogseth.

Feilligninger:

Som beskrevet ovenfor, består hver GNSS-vektor av 3 måleverdier, ΔX , ΔY og ΔZ . Vi ser på måleverdien ΔX , det er tilsvarende formler for ΔY og ΔZ :

Den generelle feilligningen for målingen ΔX_{ij} mellom to punkter (i og j) som skal beregnes, og begge punktene er ukjente og skal beregnes i utjevningen:.

Matematisk sammenheng, forventet verdi: $E(\Delta X) = X_j - X_i$

Opprinnelig feilligning: $\Delta X_{ij} + \hat{v}_{\Delta X_{ij}} = X_j - X_i$

NB: Legg merke til at de opprinnelige feilligningene er lineære.

Normalt innføres likevel foreløpige verdier på elementene (her: de ukjente koordinatene), for å få små tall i normalligningene (beregner tillegg i stedet for koordinater).

Innfører foreløpige verdier på elementene (de ukjente): $\Delta X_{ij} + \hat{v}_{\Delta X_{ij}} = X_j^0 + \Delta X_j - X_i^0 - \Delta X_i$

Stokker om, og får feilligningen for ΔX , der punktene i og j er ukjente punkter:

$$\hat{v}_{\Delta X_{ij}} = \Delta X_j - \Delta X_i - (-X_j^0 + X_i^0 + \Delta X_{ij}) \quad (10.4a)$$

ΔX_{ij} : måleverdi, fra vektorberegningen

$\hat{v}_{\Delta X_{ij}}$: utjevningstillegget til måleverdien ΔX_{ij} .

X_j^0 og X_i^0 : foreløpige X -koordinater i henholdsvis punkt j og punkt i

ΔX_j og ΔX_i : ukjente koordinat tillegg til X^0 i henholdsvis punkt j og punkt i .

For vektoren fra ukjent punkt 5 til ukjent punkt 6 på figur 10.1, blir feilligningene for de 3 måleverdiene:

$$\hat{v}_{\Delta X_{56}} = \Delta X_6 - \Delta X_5 - (-X_6^0 + X_5^0 + \Delta X_{56})$$

$$\hat{v}_{\Delta Y_{56}} = \Delta Y_6 - \Delta Y_5 - (-Y_6^0 + Y_5^0 + \Delta Y_{56})$$

$$\hat{v}_{\Delta Z_{56}} = \Delta Z_6 - \Delta Z_5 - (-Z_6^0 + Z_5^0 + \Delta Z_{56})$$

For vektoren fra kjent punkt 1 til ukjent punkt 5 på figur 10.1, blir feilligningene for de 3 måleverdiene:

$$\hat{v}_{\Delta X_{15}} = \Delta X_5 - (-X_5^0 + X_1 + \Delta X_{15})$$

$$\hat{v}_{\Delta Y_{15}} = \Delta Y_5 - (-Y_5^0 + Y_1 + \Delta Y_{15})$$

$$\hat{v}_{\Delta Z_{15}} = \Delta Z_5 - (-Z_5^0 + Z_1 + \Delta Z_{15})$$

Koeffisientene til feilligningene på matriseform.

Vi ser at det er 3 feilligninger for hver vektor som er målt. Punkt 1 har kjente koordinater, de andre punktene er ukjente i det frie nettet som brukes i beregningene.

Feilligningene: $\mathbf{V} = \mathbf{AX} - \mathbf{F}$

$$\mathbf{V} = \begin{pmatrix} \hat{v}_{\Delta X_{12}} \\ \hat{v}_{\Delta Y_{12}} \\ \hat{v}_{\Delta Z_{12}} \\ \vdots \\ \vdots \\ \hat{v}_{\Delta X_{89}} \\ \hat{v}_{\Delta Y_{89}} \\ \hat{v}_{\Delta Z_{89}} \end{pmatrix}_{42 \times 1} \quad \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & \cdots & \cdots & \cdots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \cdots & \cdots & \cdots & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \cdots & \cdots & \cdots & 0 & 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & & & & & & & & \\ \vdots & \vdots & \vdots & & & & & & & & & \\ \vdots & \vdots & \vdots & & & & & & & & & \\ 0 & 0 & 0 & \cdots & \cdots & \cdots & -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & \cdots & \cdots & 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \cdots & \cdots & \cdots & 0 & 0 & -1 & 0 & 0 & 1 \end{pmatrix}_{42 \times 24} \quad \mathbf{X} = \begin{pmatrix} \Delta X_2 \\ \Delta Y_2 \\ \Delta Z_2 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \Delta X_9 \\ \Delta Y_9 \\ \Delta Z_9 \end{pmatrix}_{24 \times 1}$$

Vi ser i $\mathbf{A}$ -matrisa at for måleverdien $\Delta Z_{\Delta Z_{89}}$, er det bare er de involverte Z-verdiene for punkt 8 og 9 som gir verdier ulik 0 i $\mathbf{A}$ -matrisa.

Punkt 1 inngår ikke, da det er et kjent punkt og skal da ikke ha utjevningstillegg.

Vektsmatrisa

For hver vektor består vektsmatrisa av en 3x3 matrise (3 måleverdier per vektor).

Siden det generelt er korrelasjon mellom de tre måleverdiene for en vektor, er vektsmatrisa generelt også fullbesatt (elementene utenfor diagonalen er ulik null).

$$\mathbf{P}_{ij} = \begin{pmatrix} p_{\Delta X \Delta X} & p_{\Delta X \Delta Y} & p_{\Delta X \Delta Z} \\ p_{\Delta X \Delta Y} & p_{\Delta Y \Delta Y} & p_{\Delta Y \Delta Z} \\ p_{\Delta X \Delta Z} & p_{\Delta Y \Delta Z} & p_{\Delta Z \Delta Z} \end{pmatrix}_{3 \times 3} \quad (10.4b)$$

Den fullstendige vektsmatrisa settes sammen av vektsmatrisene for hver vektor. For eksemplet på figur 10.1 blir det en 42x42-matrise (42 måleverdier)

$$\mathbf{P} = \begin{pmatrix} \mathbf{P}_{12} & & & \mathbf{0} \\ & \ddots & & \\ & & \mathbf{P}_{ij} & \\ & & & \ddots \\ \mathbf{0} & & & & \mathbf{P}_{89} \end{pmatrix}_{42 \times 42} \quad (10.4c)$$

I $\mathbf{P}$ -matrisa i (10.4c), er det antatt uavhengige vektorer, vist ved verdiene 0 mellom vektorene 1-2 og 8-9.

Ved samtidige målinger med tre eller flere mottakere, kan også vektorene fra samme måleperiode være korrelerte.

Vektskoeffisientmatrisa:

Av en vektorberegning, fås normalt ikke vekstmatrisa. Det er vektskoeffisientmatrisa, **Q**, som beregnes ved element-beregningen. Vekstmatrisa **P** kan beregnes av den kjente sammenhengen at den er den inverse av **Q**-matrisa:

$$\mathbf{P} = \mathbf{Q}^{-1}$$

For hvert punkt kan man beregne feilellipsoiden i forhold til det gitte punkt. Normalt brukes 95% - feilellipsoiden.

Normalligningene

Når alle elementene i feilligningene er satt opp og vekstmatrisa er beregnet, fås de endelige verdiene på elementene (her: de ukjente koordinatene), ved å løse normalligningene.

$$\mathbf{V} = \mathbf{A}\mathbf{X} - \mathbf{F} \quad \text{og} \quad \mathbf{P} \quad \text{gir:} \quad \hat{\mathbf{X}} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \cdot (\mathbf{A}^T \mathbf{P} \mathbf{F})$$

Det vises til kompendiet som er brukt i temaet Minste kvadraters metode, elementmetoden, for videre gang i beregningene, blant annet beregning av nøyaktighetsmål.

10.4 From geocentric to geodetic coordinates. Converting

10.4.1 Definition of several geographic reference systems

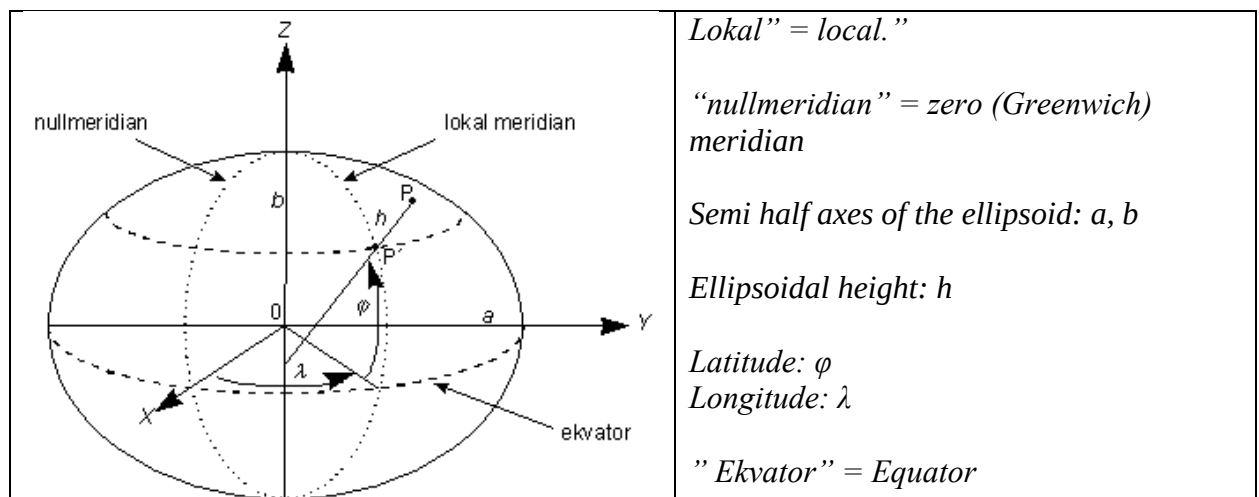


Figure 10.2 Coordinates

Starting point: Geocentric coordinates

Spatial coordinates such as latitude and longitude of a point, can be given in different reference systems:

- **Geographic coordinates:** Collective term for geodetic and astronomic coordinate systems
 - **Geodetic coordinates:** Latitude and longitude on the ellipsoid. Ellipsoidal height.
 - **Ellipsoidal coordinates:** Same as geodetic coordinates
 - **Astronomic coordinates:** Latitude and longitude with the vertical line (plumb line) as the reference line. Orthometric or normal height.

10.4.2 Conversion and transformation

Recommended terms of mathematical operations:

- **Conversion:** when doing calculations *within* the same datum
- **Transformation:** when doing calculations *between* different datums

Spatial (3D) GNSS networks of known and unknown control points will have a very accurate inner accuracy, while the absolute position of the network in the geodetic coordinate system will have a lower accuracy.

Converting to/from geodetic coordinates (latitude and longitude) and ellipsoidal height on the WGS84 ellipsoid, is done by using well known formulas from the compendium Geodesy part 2 (or from chapter 8.2.1 in the text book Hofmann-Wellenhof "GNSS" (2008)). The conversion gives unambiguous results, because the calculations are done within the same datum. Normally, in practice, we are using orthometric or normal heights, not ellipsoidal heights. Are the geoid undulation known for the actual ellipsoid, the conversion to orthometric/normal heights is easy to do. See figure 10.5.

h_o : orthometric height or normal height (height above the geoid)

h_e : ellipsoidal height (height above the ellipsoid)

N : geoid undulation (the difference between ellipsoidal and orthometric height in a point)

$$N = h_e - h_o$$

The calculated network on the ellipsoid can be mapped in different map projections.

Example, SkTrans (old name WSKTRANS)

The Norwegian Mapping Authorities: SkTrans is an official transformation program for Norwegian territory covering the datums NGO1948, ED50, EUREF89

From the manual of the SkTrans program:

The program transforms between the 3 concrete datums and the 3 types of coordinates and in (almost) all combinations. In SkTrans it is possible to transform coordinates of the Norwegian mainland.

When EUREF89 was well established in 1997 the North Sea countries discussed if the transformation formulae of the North Sea should be changed. But, they decided that the deformations in ED50 around the North Sea were so big in numerical values and so different in direction that it was not possible, even by using EUREF89, to obtain a mathematical formula which has residual smaller than 0.5 m. Therefore the North Sea countries decided to still use the formulas from 1990. However they recommended strongly that the users of coordinates in the North Sea to start using coordinates in EUREF89 also on the ocean between the North Sea countries.

The ocean north of 65° N to north of Svalbard: a 7 parameter transformation between ED50 and EUREF89 is valid. Up to 2000 a transition formulae was valid between 62° and 65° N.

☒ NGO1948-Gauss-K. N = 603286.859 E = -15548.034 Akse Akse 3	☐ ED50-UTM N = 7032805.632 E = 570298.285 Sone Sone 32	☐ EUREF89-UTM N = 7032598.412 E = 570217.469 Sone Sone 32
☐ ED50-Geodetisk Br = 063 24 55.04340 Ln = 010 24 28.45250		☐ EUREF89-Geodetisk Br = 063 24 53.96938 Ln = 010 24 22.82927
		☐ EUREF89-XYZ X = 2815010.096 Y = 516973.098 Z = 5680907.274
☒ Høyde i VREF1996 H = 48.061	☐ Ellipsoidisk høyde h = 87.618	☐ VREF1996 over ellipsoiden N = 39.557

Table 10.1

Enter the coordinates in one of the boxes (*from*), (in the figure: NGO1948 and VREF1996 = NN1954)), and the program calculates coordinates and heights in the other boxes (*to*). Also the geoid undulation is calculated.

Example, the North Sea Formula (1990)

- The block borders are given in the datum ED50. The quality of the coordinate values in the datum ED50 around the North Sea is non-homogeneous. Two reasons for this: ED50 is an old datum, where classical instruments like a theodolite (and no distance measurer) were used, and the realization of ED50 in each country was more or less done within each country and not as a joint project.
- The GNSS measurements are in the datum WGS84. WGS84 is a homogeneous datum.

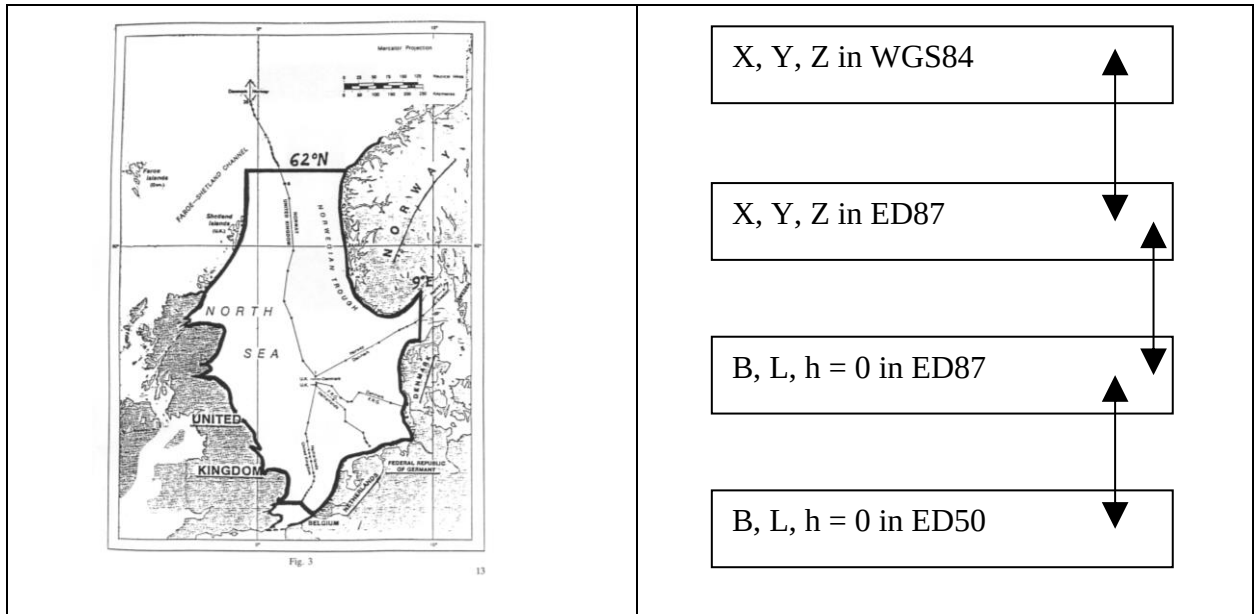


Figure 10.3 Area for the North Sea formulae. The transformation WGS84 – ED50

There is a common agreement between the countries for how to transform between WGS84 and ED50. The transformation methods are described in this report:

“The transformation between ED50 and WGS84 for exploration purposes in the North Sea, Geodetic publications 1990:1” issued by the Norwegian mapping authorities.

See the figure above giving the area in which the formulas are valid, the northern border is 62° North. ED87 is an improved and new version of ED50. ED87 is a homogeneous datum, as GPS measurements give the network a framework of accurate points and thereby a network with a homogeneous scale.

The way of transformation is given in the figure above. It is the transformation to and from ED50 that causes the greatest problems, as the coordinates given in ED50 is of lower accuracy (non-homogeneous). See the next chapters for more details and formulas of the transformation.

The North Sea, continued.

Guidance Note Number 10 from the European Petroleum Survey Group



<http://www.epsg.org/guides/docs/G10.pdf>

Overview: <http://www.epsg.org/guides/>

*Geodetic Transformations Offshore Norway***Summary of recommendations**

- (i) South of 62° North the 1991 North Sea formulae continues to be the recommended transformation between ED50 and WGS84 (paragraph 2). However for data management applications and real-time field operations where a two-step transformation and/or a polynomial transformation are not available, an approximation accurate to 1 metre may be used (paragraph 14).
- (ii) North of 62° North the recommended transformation between ED50 and WGS84 is as given in paragraph 10.
- (iii) No action is required from data managers following this clarification of geodetic transformation. Historically-derived ED50 coordinates may be assumed to be consistent with those derived through these 2001 transformations.
- (iv) Any operations which cross the parallel of 62° North should use only one transformation for the full survey area. The transformation adopted should be that which applies to the majority of the survey area.

10.5 Geocentric coordinates in different datums. Transformation

Most of the countries use another datum than WGS84. If the GNSS network (or a GNSS baseline) is to be transformed to another datum, the transformation parameters between the datum WGS84 and this datum must be known.

(1) Homogeneous networks of control points:

If the network has no constraints, meaning the coordinates of the common control points are of high quality, then a 7-parameter-transformation can be used. Common control points have coordinates in both datums involved in the transformation.

The 7 parameters are:

- Origin shift or translation: 3 (along each axes)
- Rotation: 3 (around each axes)
- Scale: 1 (the difference in scale in the two datums)

We can anticipate small values of the rotations (milligrad level), as the ellipsoids of the different datums are empirically oriented and placed almost in the same way in the global datums: The Z axis parallel to the rotation axis of the Earth, the origin of the ellipsoid in the gravitation centre of the Earth etc.

Example of 7-parameter-transformation: The North Sea, between ED87 and WGS84 (in reality WGS84*SEA)

Cartesian coordinates are used in the formulae below.

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{WGS84} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{ED87} + \begin{pmatrix} TX \\ TY \\ TZ \end{pmatrix} + \begin{pmatrix} D & -R_z & R_y \\ R_z & D & -R_x \\ -R_y & R_x & D \end{pmatrix} \cdot \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{ED87}$$

Translation ("datum shift"): $TX = -82,981 \text{ m}$ $TY = -99,719 \text{ m}$ $TZ = -110,709 \text{ m}$

Rotation in radians: $R_x = -0,5076 \cdot 10^{-6}$ $R_y = 0,1503 \cdot 10^{-6}$ $R_z = 0,3898 \cdot 10^{-6}$

Scale correction: $D = -0,3143 \cdot 10^{-6}$

(2) Non-homogeneous networks:

There are constraints in the network, for instance varying scale in parts of the network. Therefore a 7-parameter-transformation cannot be used.

Examples of non-homogeneous networks:

- ED50 (European datum) also used in The North Sea,
- The Norwegian datum NGO1948

Example of a non-homogenous transformation: Between ED50 and ED87 in the North Sea

Constants.	From ED50 to ED87		From ED87 to ED50	
	Latitude	Longitude	Latitude	Longitude
A ₀	-.556098E-05	.148944E-04	.556098E-05	-.148944E-04
A ₁	-.155391E-05	.268191E-05	.155391E-05	-.268191E-05
A ₂	-.402620E-06	.245290E-05	.402620E-06	-.245290E-05
A ₃	-.509693E-06	.294400E-06	.509693E-06	-.294400E-06
A ₄	-.819775E-06	.152260E-05	.819775E-06	-.152260E-05
A ₅	-.247592E-06	.910592E-06	.247592E-06	-.910592E-06
A ₆	.136682E-06	-.368241E-06	-.136682E-06	.368241E-06
A ₇	.186198E-06	-.851732E-06	-.186198E-06	.851732E-06
A ₈	.123350E-06	-.566713E-06	-.123350E-06	.566713E-06
A ₉	.568797E-07	-.185188E-06	-.568797E-07	.185188E-06
A ₁₀	-.232217E-08	.284312E-07	.232217E-08	-.284312E-07
A ₁₁	-.769931E-08	.684853E-07	.769931E-08	-.684853E-07
A ₁₂	-.786953E-08	.500828E-07	.786953E-08	-.500828E-07
A ₁₃	-.612216E-08	.415937E-07	.612216E-08	-.415937E-07
A ₁₄	-.401382E-08	.762236E-08	.401382E-08	-.762236E-08

Table 2. Polynomial constants for transforming ED50 – ED87.

As seen from table 2: constants (from ED87 to ED50) = – constants (from ED50 to ED87)
Computed values for Δ , which may be used for checking correct computing procedures, are shown in table 3A.

Table 10.2 Transformation between ED50 and ED87

A forth degree polynomial with 14 terms to obtain an optimal transformation when a non-homogeneous datum (ED50) is involved:

$$\Delta = A_0 + A_1 \cdot U + A_2 \cdot V + \dots + A_7 \cdot U^2 \cdot V + \dots + A_{14} \cdot V^4$$

where the latitude $U = B - 55$ degrees (B is the actual latitude of a point)
and the longitude $V = L$ (Grw) (L is the actual longitude of a point)
and A_i are tabulated values.

As we can see the procedure is much more complicated when one of the datums is non-homogeneous. For details, see the report (title is given on the previous page).

10.6 ΔX , ΔY , ΔZ transformed to a topocentric coordinate system

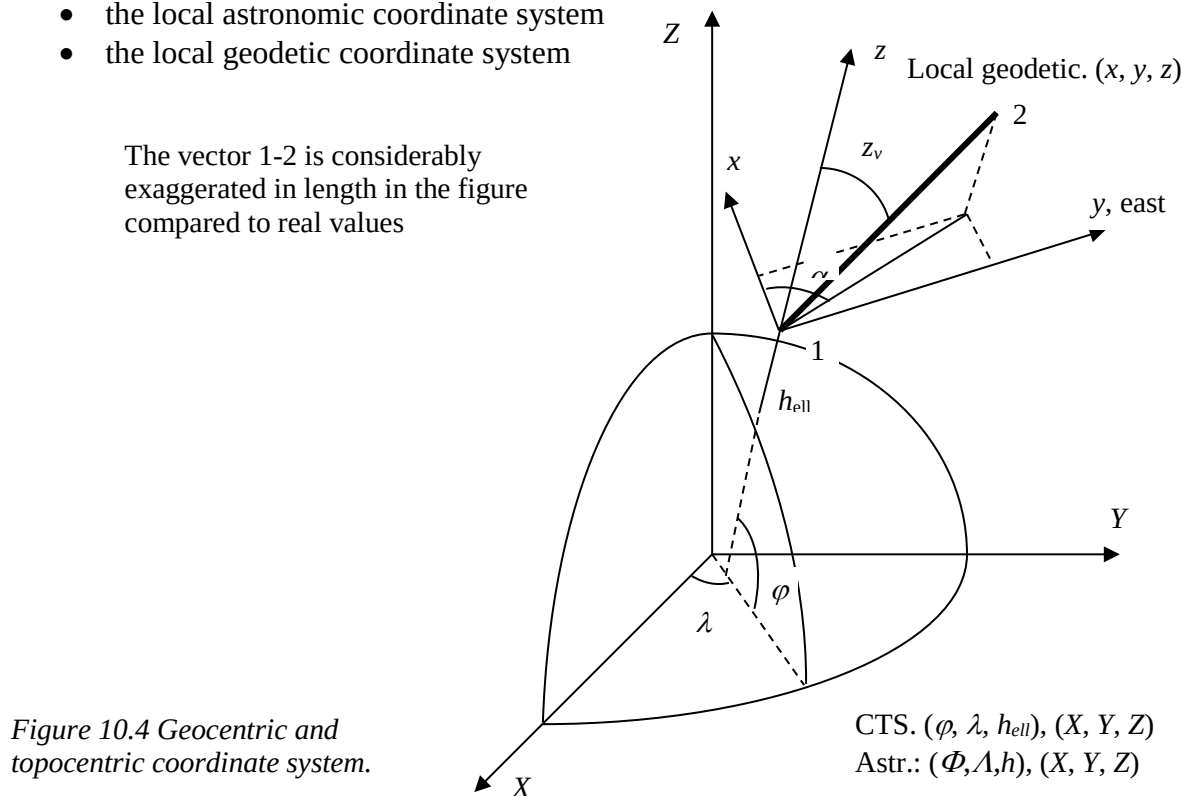
The components of the baseline, ΔX , ΔY , ΔZ , transformed to a topocentric system.

10.6.1 Local horizontal (toposentric) coordinate system

We differ between two local horizontal coordinate systems:

- the local astronomic coordinate system
- the local geodetic coordinate system

The vector 1-2 is considerably exaggerated in length in the figure compared to real values



The coordinate systems are called *topocentric* (they have the origin on the surface of the Earth) and they are defined below, see also figure 10.4:

- (1) The *local astronomic* (LA) coordinate system is linked to the geoid and the plumb line:
 - a) The origin is placed at the observation site, usually one of the ends of the baseline
 - b) The z axis coincides with the vertical line (the plumb line) going through the point.
 - c) The x axis is the intersecting line between the horizontal plane in the observation point and the vertical plane in this point containing the special North pole called IRP, international reference pole (earlier called the CIO pole). The position of IRP is decided by an international agreement, and thus also defining one point of the rotation axis of the Earth. The direction of the x axis gives *the astronomic north*.
 - d) The y axis is pointing towards east. The coordinate system is therefore a left hand system.
- (2) The *local geodetic* (LG) coordinate system is linked to the ellipsoid: The tangent plane and the ellipsoid normal:
 - a) The origin is placed on the ellipsoid normal going through the observation point, usually one of the ends of the baseline. In some cases the origin is where this ellipsoid normal is intercepting the ellipsoid or the geoid.
 - b) The z axis coincides with the ellipsoid normal going through the point. Also called the “up” axis.
 - c) The x axis is the intersecting line between a tangential plane in the point and a plane containing the rotation axis Z of the ellipsoid and the origin. The direction of the x axis gives *the geodetic (or ellipsoidal) north*.
 - d) The y axis is pointing towards east. The coordinate system is therefore a left hand system.

The mathematical differences between the two systems LA and LG are described in three formulas, where the *deflection of the vertical* (defined by the components ξ (north) and η (east) and

the difference between astronomic azimuth A and geodetic azimuth α are part of the formulas (the formulas are derived in chapter 12.1).

The difference in astronomic azimuth A and geodetic azimuth α is the 3rd formula, the **Laplace equation**:

$$(1) \quad A - \alpha = (\Lambda - \lambda) \cdot \sin \Phi = \eta \cdot \tan \Phi \quad (10.6)$$

The deflection of the vertical, also called the deflection of the plumb line. The deflection has 2 components, ξ and η :

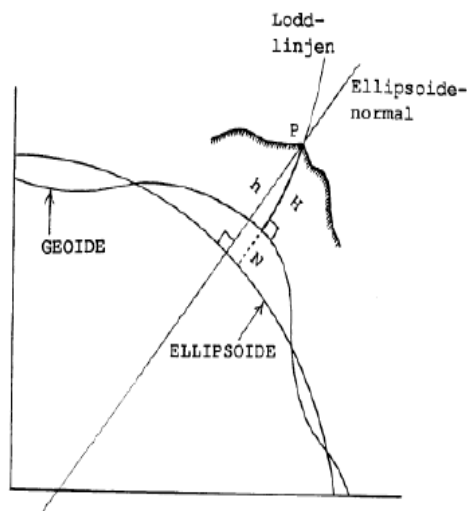
$$(2) \quad \text{The northern component: } \xi = \Phi - \varphi \quad (10.7a)$$

$$(3) \quad \text{The eastern component: } \eta = (\Lambda - \lambda) \cdot \cos \Phi \quad (10.7b)$$

The symbols used in the formulas are:

Λ : astronomic longitude and Φ : astronomic latitude

λ : geodetic longitude and φ : geodetic latitude



“Loddlinjen” = The plumb line or the vertical line

Figure 10.5 Geoid, ellipsoid, normal, plumb line

10.6.2 Matrix operators

In the derivation of the formulas for converting within the same datum, we use reflection and rotation matrices, which will give orthogonal transformations of the coordinate system. Here is an overview of the actual matrix operators:

1. Reflection

A reflection matrix is used to change the sign of a coordinate value, for instance when we go from a left hand system to a right hand coordinate system.

$$\mathbf{P}_1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} ; \quad \mathbf{P}_1 \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -x \\ y \\ z \end{bmatrix} \quad \mathbf{P}_1 \text{ changes the sign of } x \quad (10.8a)$$

$$\mathbf{P}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} ; \quad \mathbf{P}_2 \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ -y \\ z \end{bmatrix} \quad \mathbf{P}_2 \text{ changes the sign of } y \quad (10.8b)$$

$$\mathbf{P}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} ; \quad \mathbf{P}_3 \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ -z \end{bmatrix} \quad \mathbf{P}_3 \text{ changes the sign of } z \quad (10.8c)$$

2. Rotations

A rotation matrix is used to rotate the axes, for instance to obtain parallel axis.

The rotation matrices below describe *positive rotations*, which are right hand rotations for right hand coordinate systems and left hand rotations for left hand coordinate systems.

$$\text{The } x \text{ axis as the rotation axis: } \mathbf{R}_1(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \quad (10.9a)$$

$$\text{The } y \text{ axis as the rotation axis: } \mathbf{R}_2(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \quad (10.9b)$$

$$\text{The } z \text{ axis as the rotation axis: } \mathbf{R}_3(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (10.9c)$$

The specific transformations cannot change order in the product of the result matrix: The rotation matrices are not commutative.

A series of several rotations are performed from right to left:

$$\mathbf{R}_1(\alpha) \cdot \mathbf{R}_2(\beta) \cdot \mathbf{R}_3(\gamma) \quad (10.10)$$

Inverting and transforming the matrices:

$$\mathbf{P}_k^{-1} = \mathbf{P}_k \quad (10.11)$$

$$\mathbf{R}_k^{-1}(\theta) = \mathbf{R}_k(-\theta) \quad (10.12)$$

$$\mathbf{R}_k^T(\theta) = \mathbf{R}_k(-\theta) \quad (10.13)$$

10.6.3 Transformation from geocentric CTRS to a local geodetic system (LG)

The original coordinate system (datum) of the GNSS baseline observations is WGS84. The coordinates:

X, Y, Z : Cartesian coordinates, CTS, CTRS

B, L, H_{ell} ($\varphi, \lambda, H_{\text{ell}}$) : Geographic coordinates

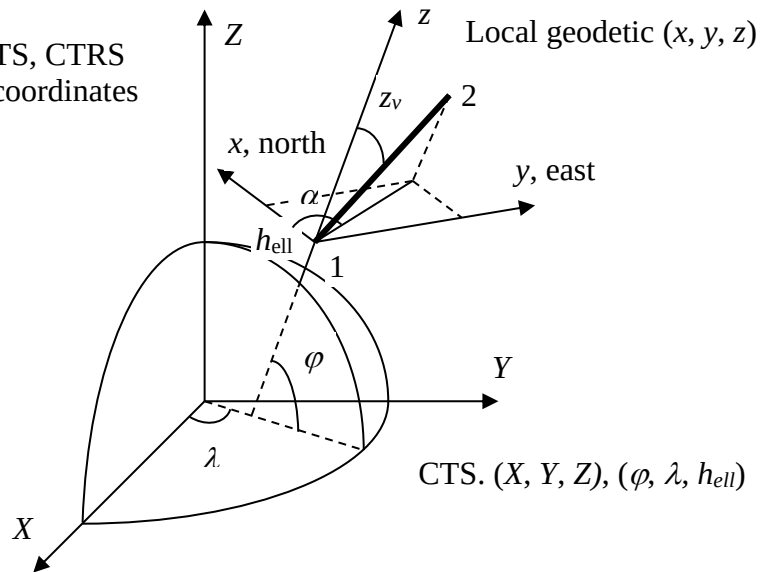


Figure 10.6

Local geodetic coordinates: x, y, z

We want the original GNSS observations transformed to the map projection plane.

First step is to convert from Cartesian coordinates (CTS) to local geodetic coordinates (LG):

$$\text{From } \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{CTS}} \text{ to } \begin{pmatrix} x \\ y \\ z \end{pmatrix}_{\text{LG}}$$

If the geographical coordinates for one of the end points of the GNSS baseline are given in the datum we want, the coordinate components can be converted to a local geodetic system.

In figure 10.5 the geographic coordinates (φ and λ) are assumed given for point 1, referred to the WGS84 ellipsoid. Are we working with a datum where the geometric centre of the ellipsoid do not coincide with or is not close to the gravity point of the Earth, we can shift the origin of the CTRS coordinate system to point 1.

Conversion from CTRS to LG of a baseline from point 1 to point 2, where the origin in the LG system is in point 1, see also figure 10.6, the solution:

$$(\bar{\mathbf{r}}_{12})_{\text{LG}} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\text{LG}} = \mathbf{P}_2 \cdot \mathbf{R}_2(\varphi - 90^\circ) \cdot \mathbf{R}_3(\lambda - 180^\circ) \cdot \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}_{\text{CTRS}} \quad (10.14)$$

Description, step by step:

1. Rotation (positive) around the Z axis ($\mathbf{R}_3$) by the angle $(\lambda - 180^\circ)$, making the X and x axes lying in the same plane. This is a rotation in the equatorial XY plane. See figure 10.7a.

The rotation matrix $\mathbf{R}_3$ transforms $(X, Y, Z)_{\text{CTRS}}$ to (X', Y', Z') :

$$\mathbf{R}_3 = \begin{bmatrix} \cos(\lambda - 180^\circ) & \sin(\lambda - 180^\circ) & 0 \\ -\sin(\lambda - 180^\circ) & \cos(\lambda - 180^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\cos \lambda & -\sin \lambda & 0 \\ \sin \lambda & -\cos \lambda & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

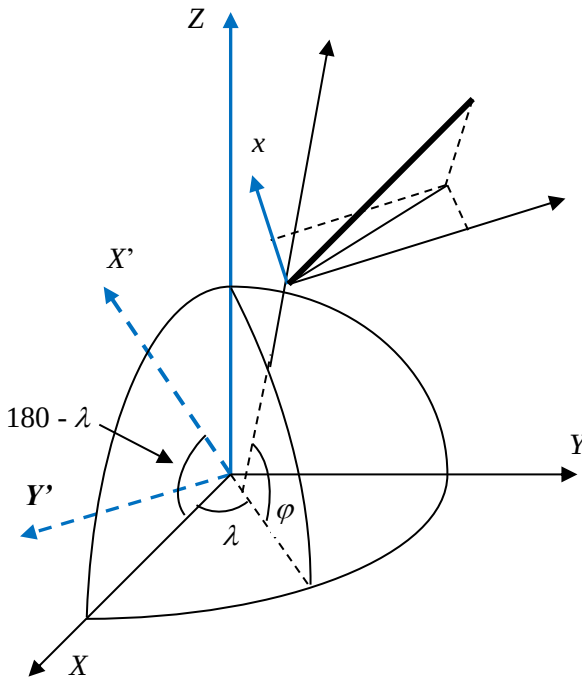


Figure 10.7a Rotation around the Z axis

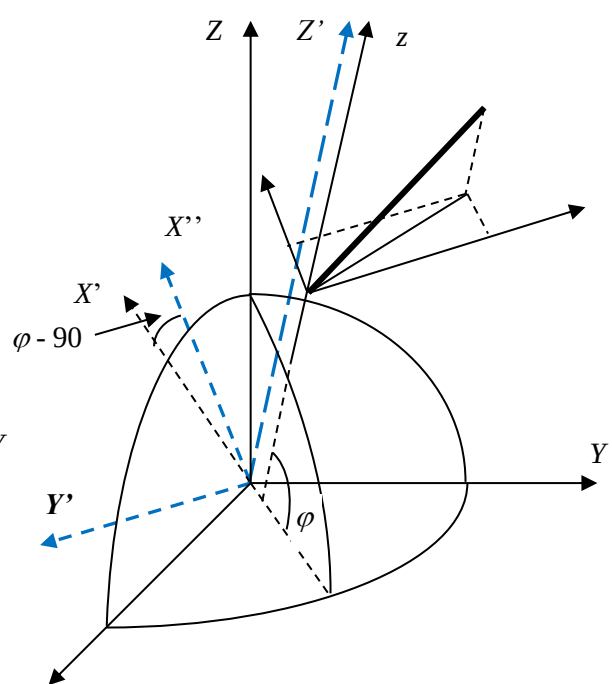


Figure 10.7b Rotation around the Y' axis

2. Rotation (positive) around the Y' axis on figure 10.7b ($\mathbf{R}_2$) by the angle $(\varphi - 90^\circ)$, making the X'' axis and the topocentric x axis parallel. This is a rotation in the X'Z plane.

$\mathbf{R}_2$ which is a rotation around the Y' axis, makes the $(X'Y')$ plane parallel with the $(xy)_{LG}$ plane.

$$\mathbf{R}_2 = \begin{bmatrix} \cos(\varphi - 90^\circ) & 0 & -\sin(\varphi - 90^\circ) \\ 0 & 1 & 0 \\ \sin(\varphi - 90^\circ) & 0 & \cos(\varphi - 90^\circ) \end{bmatrix} = \begin{bmatrix} \sin \varphi & 0 & \cos \varphi \\ 0 & 1 & 0 \\ -\cos \varphi & 0 & \sin \varphi \end{bmatrix}$$

3. Reflection. $\mathbf{P}_2$ changes the coordinate system from a right handed to a left handed system by rotating the Y' axis 180 degrees. See figure 10.7c.

$$\mathbf{P}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

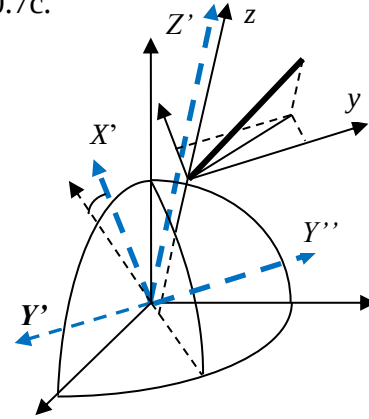


Figure 10.7c Reflection of the Y' axis

4. Multiplication from right to left in formula (10.14) gives the transformation matrix in formula (10.15):

$$\mathbf{T} = \begin{bmatrix} -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ -\sin \lambda & \cos \lambda & 0 \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix}$$

Translation or shift:

We must also compute values of the origin shift, a translation from the origin of the CTRS coordinate system to the origin of the LG coordinate system (where the origin is at one end of the baseline, point 1 on the figure).

The shift in point 1, which is the origin in the local geodetic coordinate system, is obtained by formula (10.15a):

$$\begin{bmatrix} X_0 \\ Y_0 \\ Z_0 \end{bmatrix}_{LG/CTRS} = \begin{bmatrix} X_1 \\ Y_1 \\ Z_1 \end{bmatrix}_{LG/CTRS} = \begin{bmatrix} -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ -\sin \lambda & \cos \lambda & 0 \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix} \cdot \begin{bmatrix} X_1 \\ Y_1 \\ Z_1 \end{bmatrix}_{CTRS} \quad (10.15a)$$

The coordinate values on the left hand side in (10.15a) are with the same origin as the CTRS coordinate system, but the axes are rotated and reflected (the Y axis) and thus parallel to the axes of the local geodetic coordinate system. The coordinate system is equal to the system with the axes X'' , Z' and Y'' (rotated 180 degrees), see figure 10.7c.

The formula (10.15a) for the other end of the baseline, for an arbitrary point i , here $i = 2$ is used, see the figures above:

$$\begin{bmatrix} X_2 \\ Y_2 \\ Z_2 \end{bmatrix}_{LG/CTRS} = \begin{bmatrix} -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ -\sin \lambda & \cos \lambda & 0 \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix} \cdot \begin{bmatrix} X_2 \\ Y_2 \\ Z_2 \end{bmatrix}_{CTRS} \quad (10.15b)$$

Subtracting the shift in (10.15a) from the coordinates of point $i = 2$ in (10.15b), and we obtain the local geodetic coordinates of point $i = 2$:

$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}_{LG} = \begin{bmatrix} X_2 \\ Y_2 \\ Z_2 \end{bmatrix}_{LG/CTRS} - \begin{bmatrix} X_0 \\ Y_0 \\ Z_0 \end{bmatrix}_{LG/CTRS} \quad (10.15c)$$

As we have corrected for the origin shift in (10.15a), we in (10.15c) obtain coordinates in the local geodetic system. The values in the matrices on the right hand side are in a rotated coordinate system where the axes are parallel to the axes in the LG coordinate system.

For a baseline:

The coordinate differences of the GNSS baseline from point 1 (origin) to point 2 is obtained by subtracting (10.15a) from (10.15b). The rotation matrix is the same in those two formulas, and $\Delta X = X_2 - X_1$ etc.:

$$\begin{bmatrix} \bar{x}_{12} \\ \bar{y}_{12} \\ \bar{z}_{12} \end{bmatrix}_{LG} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{LG} = \begin{bmatrix} -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ -\sin \lambda & \cos \lambda & 0 \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix} \cdot \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}_{CTRS} \quad (10.15d)$$

The equation (10.15d) using matrices: $\underline{(\bar{\mathbf{r}}_{12})_{LG}} = \underline{\mathbf{T}} \cdot \underline{(\bar{\mathbf{r}}_{12})_{CTRS}}$ (10.16)

The opposite way, from LG to CTRS, see also (10.12) and (10.13):

$$\underline{(\bar{\mathbf{r}}_{12})_{CTRS}} = \underline{\mathbf{T}^{-1}} \cdot \underline{(\bar{\mathbf{r}}_{12})_{LG}} \quad (10.17)$$

Observations in the Local geodetic (LG) coordinate system:

In the local geodetic coordinate system the following observations can be calculated in point 1 (the origin) and from point 1 to point 2 of the baseline:

NB: In the formulas is z a coordinate and z_V is a zenith angle.

Slope distance:	$S_{S12} = \left(x^2 + y^2 + z^2 \right)_{LG}^{\frac{1}{2}}$	(10.18a)
Horizontal distance:	$S_{H12} = \left(x^2 + y^2 \right)_{LG}^{\frac{1}{2}}$	(10.18b)
Azimuth from 1 to 2:	$\tan \alpha_{12} = \left(\frac{y}{x} \right)_{LG}$	(10.18c)
Zenith angle from 1 to 2, z_V : (Local Geodetic)	$\cos z_{V12} = \left(\frac{z_{12}}{S_{S12}} \right)_{LG}$ or $\tan z_{V12} = \left(\frac{S_{H12}}{z_{12}} \right)_{LG}$	(10.18d)
Height difference, local geodetic	$\Delta H_{LG} = z$	(10.18e)
Height difference, ellipsoidal	ΔH_{ell} (from the baseline processing, $\neq \Delta H_{LG}$)	(10.18f)

Table 10.3

From measured values in a Cartesian 3D coordinate system via local geodetic observations to observations in the map projection plane

The next and final step is to correct the distance, azimuth and the ellipsoidal zenith angle in (10.18) with the necessary corrections and obtaining observations in the map projection plane. See the next chapter.

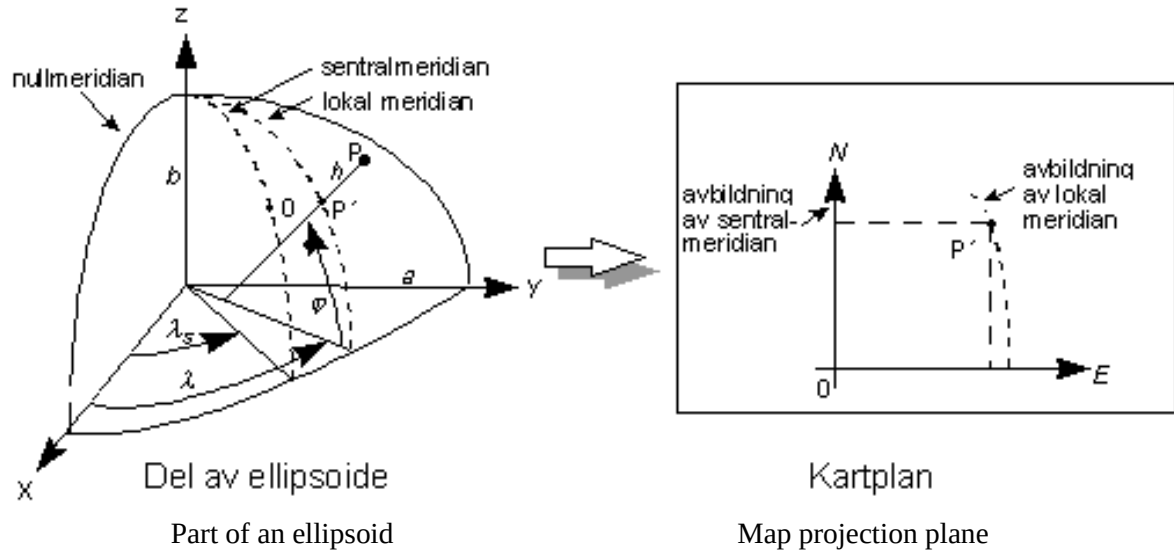


Figure 10.8

To be addressed: Transferring of measured GNSS baselines *from* the WGS84 datum ($\Delta X, \Delta Y, \Delta Z$) to the decomposed observations in the map projection plane (distance, bearing and zenith angle/height difference) and also probably in another datum.

Which values of the latitude and the longitude of a point should be used in the formula (10.15)?

The coordinate differences of the GNSS baseline are originally in the WGS84 datum. For the old national datums (for instance the Norwegian NGO1948) the axes of the ellipsoid chosen are oriented in such a way that these axes are approximate parallel to the axes in WGS84. This approximate parallelism makes it possible to use local values of the latitude and the longitude in the conversion in equation (10.15) (for instance NGO1948 values of latitude and longitude), without a critical loss of accuracy. The reason for this is:

The error or the loss of accuracy is propagated to the zenith angle and the plane bearing. If we therefore introduce rotation (in the horizontal plane) and the two components of the deflection of the vertical as unknown in the adjustments, a smaller difference in the orientation of the axes of the ellipsoids will be corrected.

Example: Latitude and longitude of a point in Trondheim in different datums:

Datum:	WGS84/EUREF89	ED50	NGO1948
Latitude(grads min sec):	63 24 53,9692	63 24 55,0432	63 24 52,4474
Longitude (grads min sec):	10 24 22,8294	10 24 28,4526	10 24 41,8604

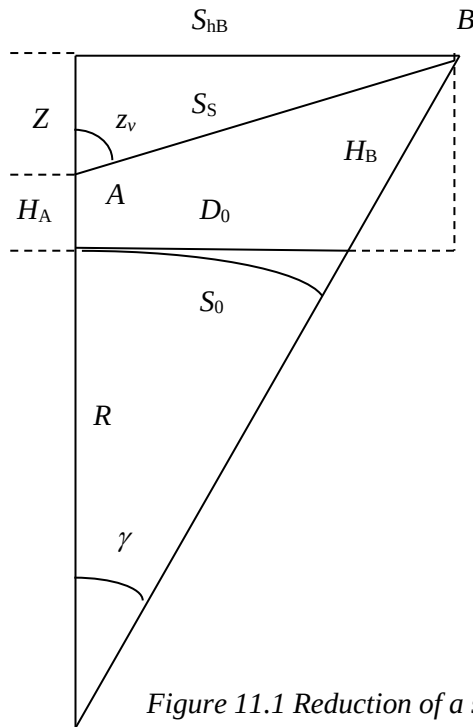
11. Reduction to the ellipsoid and to the projection plane

The correction formulas of the reduction of the distance and the azimuth from the ellipsoid to the map projection plane are described in the compendium of Holsen, "Geodesy part 2". See this compendium for more about these corrections/reductions.

See also the newest version of example exercise no. 1 (with the datums EUREF89 for coordinates and NN2000 for heights) and 5 at Blackboard (course Geodesy).

11.1 Reduction of distance

The formula (10.18) gives the slope distance between the points A and B. The slope distance must be corrected to a distance in the map projection plane.



- S_S : The slope distance from A to B
 S_{hB} : The horizontal distance in the height H_B
 D_0 : The tangential length in $H = 0$
 S_0 : The wanted ellipsoidal distance

From figure 11.1 we obtain:

$$S_{hB} = S_S \cdot \sin z_v \quad (11.1)$$

$$\tan \gamma = \frac{S_{hB}}{R + H_A + Z} \quad (11.2)$$

$$S_0 = R \cdot \gamma \quad (11.3)$$

NB: We do not need an accurate value of the radius of curvature, R , in the formulas (11.2) and (11.3), but it is important to use the *same value* of R in both the formulas.

Figure 11.1 Reduction of a slope distance

The following procedure can also be used:

$$D_0 = R \cdot \frac{S_{hB}}{R + H_A + Z} \quad \text{and then:} \quad S_0 = D_0 - \frac{1}{3} \frac{D_0^3}{R^2} \quad (11.4)$$

The distance on the ellipsoid S_0 is reduced to the actual map projection plane, for instance NGO1948-axis 3, by the following formula where y is an east coordinate in the map projection plane.

NB: (11.5) is an Earth-is-a-sphere-formula:

$$S_k = S_0 + \frac{S_0}{6R^2} (y_A^2 + y_B^2 + y_A \cdot y_B) \quad (11.5)$$

See Holsen "Geodesy part 2" and the example exercises on Blackboard for more about reduction of distances: The ellipsoidal correction formula and the formula of the reduction of the distances when a UTM map projection is used.

UTM:

Using an UTM projection we must take into consideration that the projection has a scale factor of 0.9996 (is "shrank") and that the east coordinate has the value of 500 000 m along the north axis. The y -values in formula (11.5) are the eastward distance from the north axis. Thus the UTM east value must be reduced by 500 000 m. And the theoretical correct coordinate values in (11.8) we obtain by increasing the UTM coordinates by the factor 1/0.9996 after the east value is reduced by 500 000 m. Then both the traps of the UTM are taken into consideration.

11.2 The plane bearing in the map projection plane, φ_{AB}

See in Holsen "Geodesy part 2" for more about the formulas (11.6), (11.7) and (11.8), among other things the complete formulas where the ellipsoidal terms are included in addition to the terms in (11.6) and (11.8).

NB: The meridian convergence can also be computed with latitude and longitude as arguments instead of x and y , in this case we do not need to calculate the latitude of the foot point (see Geodesy part 2).

Using an UTM projection we must take into consideration that the projection has a scale factor of 0.9996 (is “shrank”) and that the east coordinate has the value of 500 000 m at the north axis. The y-values in formula (11.5) are the eastward distance from the north axis. Thus the UTM east value must be reduced by 500 000 m. See below for more.

The formula (10.18) gives the azimuth from A to B, this azimuth is derived from the GNSS baseline which is measured between A and B. This azimuth must be corrected/reduced to a plane bearing.

$$\varphi_{AB} = \alpha - c_A - \delta_{AB} \quad (11.6)$$

c_A : The plane Gaussian meridian convergence

δ_{AB} : The reduction of the direction, defined as $\varphi_{sf} - \varphi_{pl}$

Formula, the meridian convergence c_A :

$$c_A = \frac{t_f y_A}{N_f} - \frac{t_f y_A^3}{3N_f^3} (1 + t_f^2 - \eta_f^2 - \eta_f^4) + \dots \text{ in radians} \quad (11.7)$$

$$t_f = \tan \varphi_f$$

$$\eta_f^2 = e'^2 \cos^2 \varphi_f$$

$$e'^2 = (a^2 - b^2)/b^2 \quad \text{is the square of the 2nd numerical eccentricity}$$

φ_f is the latitude of the foot point. The foot point is the point on the Central meridian that has the same x value (in the plane) as the actual point.

The reduction of directions, where x and y are coordinates in the map projection plane.

NB: (11.8) is an Earth-is-a-sphere-formula, see compendium Holsen “Geodesy part 2” for an ellipsoidal correction formula:

$$\delta_{AB} = \frac{\rho}{6R^2} (2y_A + y_B) \cdot (x_B - x_A) \quad (11.8)$$

$$\delta_{BA} = \frac{\rho}{6R^2} (2y_B + y_A) \cdot (x_A - x_B)$$

See Holsen “Geodesy part 2” and the example exercises on Blackboard for more about reduction of directions: The ellipsoidal correction formula and the formula of the reduction of the distances when a UTM map projection is used.

UTM:

Using an UTM projection we must take into consideration that the projection has a scale factor of 0.9996 (is “shrank”) and that the east coordinate has the value of 500 000 m at the north axis. The y-values in formula (11.8) are the eastward distance from the north axis. Thus the UTM east value must be reduced by 500 000 m. And the theoretical correct coordinate values in (11.8) we obtain by increasing the UTM coordinates by the factor 1/0.9996 after the east value is reduced by 500 000 m.

Then both the traps of the UTM are taken into consideration.

11.3 Numerical example. Datum NGO1948

See example exercise no. 1 (newest version) on Blackboard for an example where UTM coordinates are used.

Spatial baseline components in the datum WGS84 of the baseline from point A to B:

$$\Delta X = -20396.697 \quad \Delta Y = +20173.189 \quad \Delta Z = +8776.514$$

Geodetic coordinates of point A are given in the old Norwegian datum NGO1948, where the Modified Bessel ellipsoid is used. The coordinates in the map projection plane is:

$$\left. \begin{array}{l} \varphi_A = 63.25101216^\circ \\ \lambda_A = 8.50000000^\circ \text{ Ø. Gr. w} \end{array} \right\} \rightarrow x_A = 586954.175 \quad y_A = -111644.356$$

Ellipsoidal height of point A: $H_A = 500.000 \text{ m}$

The Cartesian coordinate differences ΔX , ΔY and ΔZ are first transformed to the local geodetic system, using (10.15). Coordinates in the local geodetic system of point B, with origin in the from-point A:

Point	x	y	z
A	0.000	0.000	0.000
B	19301.378 m	22966.428 m	100.000 m

The formulas (10.18) give the distance, the azimuth and the ellipsoidal zenith angle from A to B:

$$\begin{array}{ll} S_s = 30000.1667 & S_{hB} = 30000.000 \\ \alpha = 49.955733^\circ & z_v = 99.78779 \text{ gon} \end{array}$$

Axis III in the datum NGO1948 is 10.72291667° East of Greenwich ($10^\circ 43' 22.5''$)

The difference in longitude from the north axis to point A:

$$\Delta\lambda = \lambda - 10.72291667^\circ = -2.22291667^\circ$$

The radius of the curvature in the direction from A to B, see Holsen "Geodesy part 1" for details about the formula:

$$R_\alpha = \frac{MN}{N \cos^2 \alpha + M \sin^2 \alpha} = 6390930.76 \text{ m}$$

See figure 11.1:

$$\tan \gamma = \frac{S_{hB}}{R + H_A + Z} = \frac{30000}{6391531} \Rightarrow \gamma = 0.268927861^\circ$$

$$S_\circ = \gamma \cdot R = 29996.9634 \text{ m}$$

$$D_\circ = \frac{R \cdot S_{hB}}{R + H_A + Z} = \frac{6390931 \cdot 30000}{6391531} = 29997.184 \text{ m}$$

$$S_\circ = D_\circ - \frac{1}{3} \frac{D_\circ^3}{R^2} = 29996.9636 \text{ m}$$

After the preliminary coordinates of point B are calculated (or are known), the numerical value of the distance between A and B in the map projection plane, S_k (preliminary value), can be calculated by formula (11.5), see also figure 11.3. See below for the calculation of the preliminary coordinates of B.

$$S_\circ = 29996.9636 \text{ m}$$

$$\text{See below for details: } S_k - S_\circ = +3.6775 \text{ m} \quad S_k = 30000.6411 \text{ m}$$

Calculations of the corrections on the azimuth from (10.18), c_A and δ_{AB} :

To use the formulas below for the meridian convergence, c_A , the latitude of the foot point F must be calculated, φ_f , and also the principle radius of curvature, N_f , in the point F. The foot point is the point on the Central meridian that has the same x value (in the plane) as the actual point.

NB: The meridian convergence can also be computed with latitude and longitude as arguments instead of x and y , in this case we do not need to calculate the latitude of the foot point (see Geodesy part 2).

With $n = (a - b)/(a + b)$, where a and b are the semi axis of the ellipsoid, we use the following formula for the length of the meridian arc along the Central meridian (the north axis):

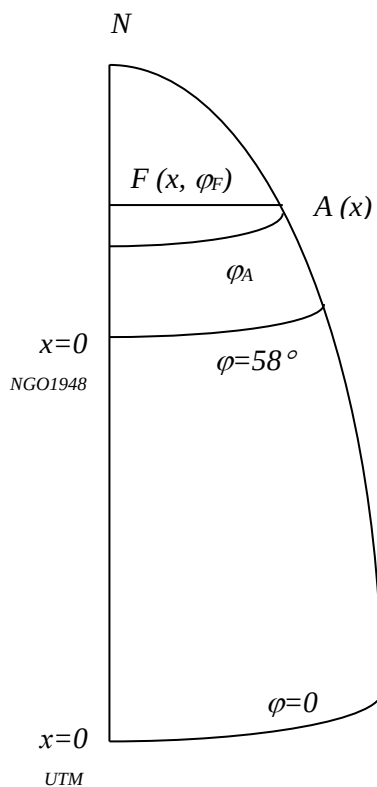
$$x = \frac{a}{n+1} \left[\left(1 + \frac{1}{4}n^2 + \frac{1}{64}n^4 \right) \frac{\Delta\varphi^\circ}{\rho^\circ} - 3 \left(n - \frac{1}{8}n^3 \right) \cos 2\varphi \cdot \sin \Delta\varphi + \frac{15}{8} \left(n^2 - \frac{1}{4}n^4 \right) \cos 4\varphi \cdot \sin 2 \cdot \Delta\varphi + \dots \right]$$

x is the length of the meridian arc along the Central meridian (the north axis) and x is mapped in true scale ($m = 1$) in the Gaussian map projection, see chapter 4 in “Geodesy part 2”. See also formula (8.22) in “GNSS” by Hofman-Wellenhof (2008), for the computation of the latitude of the foot point.

In our case, datum NGO1948 and the Gauss-Krüger-projection, the length of x is the length of the meridian arc from 58° (where the origins are) and to point F. This formula has the desired accuracy when doing the calculations within a meridian quadrant (0-90 degrees). $\Delta\varphi$ is the difference in latitude between 58° and the point we shall calculate the x value of, in our case point F.

But here the x value is used in the formulas, we have to find the solution by doing an iteration (stepwise approach). In the first iteration $\Delta\varphi = \varphi_A - 58^\circ$, while the last two terms in the formula above is neglected. In the next iterations all the terms are included in the calculations and the $\Delta\varphi$ on the right hand side is the mean value of the latitude of 58° and F:

$$\varphi = (58^\circ + 58^\circ + \Delta\varphi) / 2 = 58^\circ + \Delta\varphi / 2$$



After some rounds of iterations we obtain:

$$\varphi_f = 58^\circ + \Delta \varphi$$

$$N_f = \frac{a}{(1 - e^2 \sin^2 \varphi_f)^{1/2}} = a \frac{\cos \beta_f}{\cos \varphi_f}$$

where $e^2 = \frac{a^2 - b^2}{a^2}$

$$\text{and } \tan \beta_f = \frac{b}{a} \cdot \tan \varphi_f$$

and β is the reduced latitude, see Holsen "Geodesy part 1"

Figure 11.2 Foot point and origins

The meridian convergence c_A can now be calculated by using (11.7).

$$\varphi_f = 632683675^\circ \quad N_f = 6394536.797 \quad c_A = -2.205819 \text{gon}$$

For calculation of preliminary values of the coordinates of point B, a preliminary value of the plane bearing in the map projection plane is calculated:

$$\alpha - c_A = 57.713297 \text{ gon}$$

$$x_B^\circ = x_A + S_\circ \cdot \cos (\alpha - c_A) \approx 605446 \text{ m} \quad y_B^\circ = y_A + S_\circ \cdot \sin (\alpha - c_A) \approx -88025 \text{ m}$$

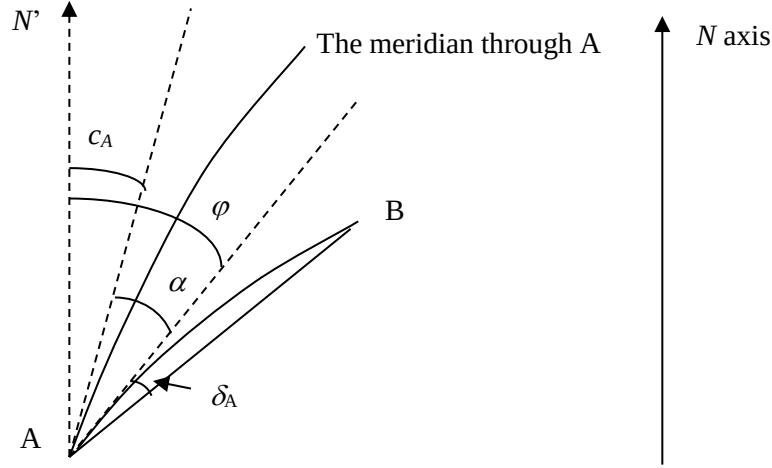


Figure 11.3. A and B in the map projection plane. Meridian convergence, reduction of directions

Calculation of the reduction of the directions

Below the values of x , y and R are entered in km:

$$\delta_{AB} = \frac{\rho^{\text{mgon}}}{6 \cdot 6390.193^2} (-111.64 \cdot 2 - 88.09) \cdot 18.49 = -1.496 \text{ mgon}$$

$$\delta_{BA} = \frac{\rho^{\text{mgon}}}{6 \cdot 6390.193^2} (-88.03 \cdot 2 - 111.64) \cdot (-18.49) = 1.382 \text{ mgon}$$

The plane bearing from A to B, see also figure 11.3:

$$\begin{aligned} \varphi_{AB} &= \alpha - c_A - \delta_{AB} = 55.50637 + 2.20582 + 0.001496 = 57.713686 \text{ gon} \\ \varphi_{BA} &= \varphi_{AB} + 200 = 257.713686 \text{ gon} \end{aligned}$$

The distance in the map projection plane

To obtain the distance in the map projection plane, S_k , the formula (11.5) is used:

$$S_k = 29996.9636 \cdot \left[1 + \frac{1}{6 \cdot 6390.93^2} \left((-111.64)^2 + (-88.03)^2 + 111.64 \cdot 88.03 \right) \right]$$

$$S_k = 30000.6411 \text{ m}$$

For the 3rd observation, the orthometric (normal) zenith angle or the orthometric (normal) height difference, see chapter 12.2, or the example exercise no. 5 on itslearning.

We have calculated the plane distance and the plane bearing: Final values of the coordinates of point B in the map projection plane can be computed, NGO1948 values:

$$x_B = x_a + S_K \cdot \cos \varphi_{AB} = 605448.205 \text{ m}$$

$$y_B = y_a + S_K \cdot \sin \varphi_{AB} = -88022.135 \text{ m}$$

NB! The computation of point B is done without any control and the coordinates are therefore not trustworthy. But, in this numerical example the reductions/corrections were the main issue to demonstrate.

A good check of the computation is to compute the geographic coordinates of point B, by using the First geodetic main task and then transform (φ_B, λ_B) to (x, y) in the datum NGO1948 and axis III.

$$\left. \begin{array}{l} \varphi_B = 63,4234178^\circ \\ \lambda_B = 8,95991057^\circ \end{array} \right\} \rightarrow \begin{array}{l} x_B = 605448,203\text{m} \\ y_B = -88022,135\text{m} \end{array}$$

Which coincide with the calculations in this numerical example, we are within the accuracy of the manual arithmetical calculation.

11.4 Variance-covariance propagation

See textbooks and examples in another course(s) on Blackboard covering the basic theory of the variance-covariance propagation law. The textbook “Skogseth/Holsen: Mathematical statistics” is copied to Blackboard in the courses Theoretical and Applied geomatics, and Geodesy

We have shown how measured GNSS baselines are transferred from coordinate differences in a geocentric coordinate system to derived observations in the map projection plane.

Doing an adjustment of a GNSS network of control points where there are measured redundant observations (measured with control, more than “necessary”), we also want to compute the accuracy measures of the observations, the variance/covariance matrix of the measurements. From this variance/covariance matrix of the measurements and the geometry of the network, the variance/covariance matrix of the adjusted coordinates can be computed.

Generally, when $\mathbf{y}$ is a linear or linearized function of $\mathbf{x}$, then the variance-covariance propagation from the variance/covariance matrix of $\mathbf{x}$ to the variance/covariance matrix of $\mathbf{y}$ is:

$$\text{The function (mathematical connection/relation): } \mathbf{y} = \mathbf{A}\mathbf{x} \quad (11.9)$$

$$\text{The variance/covariance propagation law: } \mathbf{C}_{yy} = \mathbf{A}\mathbf{C}_{xx}\mathbf{A}^T \quad (11.10)$$

$$\text{or: } \mathbf{Q}_{yy} = \mathbf{A}\mathbf{Q}_{xx}\mathbf{A}^T \quad (11.11)$$

$\mathbf{C}_{xx}, \mathbf{C}_{yy}$: The variance-covariance matrices

$\mathbf{Q}_{xx}, \mathbf{Q}_{yy}$: The co-factor matrices

The relations between the variance-covariance matrices and the co-factor matrices:

$$\mathbf{C}_{xx} = \sigma_0^2 \mathbf{Q}_{xx}$$

$$\mathbf{C}_{yy} = \sigma_0^2 \mathbf{Q}_{yy}$$

where σ_0^2 is the variance of the observation with weight 1, the unit weight.

11.4.1 Variances-covariances computed from $\mathbf{C}_{CTRS}$

Here the Local Geodetic zenith angle is the third derived observation in addition to the horizontal distance and the azimuth in the formulas (10.18a to e) of the local geodetic system. See chapter 11.4.2 where the Local Geodetic height difference, ΔH_{LG} , is used instead of the Local Geodetic zenith angle.

In the GNSS baselines processing, also the variance-covariance matrix of the baseline components ΔX , ΔY and ΔZ , $\mathbf{C}_{CTRS}$, are computed. See an example in chapter 10.1, where the calculation of $\mathbf{C}_{CTRS}$ is shown.

Equation (10.16) shows the mathematical connections between the geocentric and local horizontal coordinates:

$$(\bar{\mathbf{r}}_{12})_{LG} = \mathbf{T} \cdot (\bar{\mathbf{r}}_{12})_{CTRS}$$

Thus the variance-covariance propagation from the geocentric to the local geodetic (topocentric) system can be derived directly from (10.16):

$$\mathbf{C}_{LG} = \mathbf{T} \cdot \mathbf{C}_{CTRS} \cdot \mathbf{T}^T \quad (11.12)$$

The final step is to derive the variance-covariance matrix for the horizontal distance, the azimuth and the zenith angle z_v .

The mathematical connections in the formulas (10.18b), (10.18c) and (10.18.d) are linearized by finding the total differential, NB: Here the azimuth, the *horizontal* distance and the *zenith angle* is used:

$$\begin{bmatrix} d\alpha_{12} \\ dS_{H12} \\ dz_{v12} \end{bmatrix} = \begin{bmatrix} -\sin \alpha_{12} \cdot \rho / S_{H12} & \cos \alpha_{12} \cdot \rho / S_{H12} & 0 \\ \cos \alpha_{12} & \sin \alpha_{12} & 0 \\ \cos z_v \cdot \cos \alpha_{12} \cdot \rho / S_{S12} & \cos z_v \cdot \sin \alpha_{12} \cdot \rho / S_{S12} & -\sin z_v \cdot \rho / S_{S12} \end{bmatrix} \cdot \begin{bmatrix} dx_{12} \\ dy_{12} \\ dz_{12} \end{bmatrix} \quad (11.13a)$$

Introducing the coefficient matrix, $\mathbf{F}$, NB: With different elements in the third row compared to (11.14.b):

$$\begin{bmatrix} d\alpha_{12} \\ dS_{12} \\ dz_{v12} \end{bmatrix} = \mathbf{F} \cdot \begin{bmatrix} dx_{12} \\ dy_{12} \\ dz_{12} \end{bmatrix} \quad (\rho = 200 \text{ gon} / \pi) \quad (11.13b)$$

In formula (11.13) two of the elements in the coefficient matrix $\mathbf{F}$ are equal to zero, and these two can normally be set to 0.

The element f_{31} in the 3rd column is equal to 0, as the azimuth is not dependent on the variation of the zenith angle (the z coordinate is not a part of the formula (10.18b)).

The element f_{32} in the 3rd column is equal to 0, as the horizontal distance is independent of the value of the zenith angle in formula (10.18d). NB: Here the *horizontal* distance is used. If the slope distance is used, then the element f_{23} is not equal 0 as the zenith angle is an argument in (10.18a)

Below the derivation of some of the elements in equation (11.13) are shown

(1) Element f_{21} , from the mathematical connection in formula (10.18d):

$$\frac{dS}{dx} = \frac{d \left[(x^2 + y^2)^{0.5} \right]}{dx} = \frac{1}{2} \cdot S_H^{-1} \cdot 2 \cdot x = \cos \alpha_{12}$$

The element f_{31} , from the mathematical connection in formula (10.18c):

$$(i): \quad \frac{d \cos z_v}{dx} = \frac{d \cos z_v}{dz_v} \cdot \frac{dz_v}{dx} = -\sin z_v \cdot \frac{dz_v}{dx}$$

$$(ii): \quad \frac{d \cos z_v}{dx} = \frac{d \left[z \cdot (x^2 + y^2 + z^2)^{-0.5} \right]}{dx} = z \cdot (-0.5) \cdot S_S^{-3} \cdot 2 \cdot x = -\frac{z \cdot x}{S_S^3}$$

and getting from (i) and (ii):

$$\frac{dz_v}{dx} = \frac{z \cdot x}{\sin z_v \cdot S_S^3} = \frac{z \cdot x}{S_H \cdot S_S^3} = \frac{z}{S_S} \cdot \frac{x}{S_H} \cdot \frac{1}{S_S} = \frac{\cos z_v \cdot \cos \alpha}{S_S}$$

(2) The element f_{32} , from the mathematical connection in formula (10.18c), identical derivations as for f_{31} :

$$\frac{dz_v}{dy} = \frac{\cos z_v \cdot \sin \alpha}{S_s}$$

(3) *Approximation.* The zenith angle, z_v , differs with a few gons/grads from a right angle (100 gon/grads) for an average GNSS baseline. A distance of 5 km and a height difference of 500 m gives a zenith angle of $z_v = 94$ gon, if the distance is 15 km (an the same height difference) the zenith angle is $z_v = 98$ gon, The factor $\cos z_v$ will therefore without considerably loss of accuracy be set to zero. Then the approximate value for the elements f_{31} and f_{32} in the 3rd row of the coefficient matrix $\mathbf{F}$ are 0..

Using the exact formulas and inserting numerical values into this element, show that the numerical values of this element normally is so small that it without loss of accuracy can be set to zero.

11.4.2 Variances-covariances computed from $\mathbf{C}_{CTRS}$. Using ΔH_{LG} instead of z_v

Here the Local Geodetic height difference is the third derived observation in addition to the horizontal distance and the azimuth in the formulas (10.18a to e) of the local geodetic system. See chapter 11.4.1 where the Local Geodetic zenith angle, z_v , is used instead of the Local Geodetic height difference, ΔH_{LG} .

This method is normally used in the commercial land surveying programs, for instance GIS/Line. We can also use more simplified formulas to calculate the corrections from ellipsoidal heights to heights referred to the geoid (mean sea level).

The mathematical connection between the local geodetic coordinates xyz and the height difference is only dependent on the z value:

$$\Delta H_{LG} = z$$

We have to derive new formulas for the elements in the third row of (11.13a).

NB: Here the azimuth, the *horizontal* distance and the *Local Geodetic height difference* is used:

$$\begin{bmatrix} d\alpha_{12} \\ dS_{H12} \\ dH_{12} \end{bmatrix} = \begin{bmatrix} -\sin \alpha_{12} \cdot \rho / S_{H12} & \cos \alpha_{12} \cdot \rho / S_{H12} & 0 \\ \cos \alpha_{12} & \sin \alpha_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} dx_{12} \\ dy_{12} \\ dz_{12} \end{bmatrix} \quad (11.14a)$$

Introducing the coefficient matrix, $\mathbf{F}$, NB: With different elements in the third row compared to (11.13.b):

$$\begin{bmatrix} d\alpha_{12} \\ dS_{12} \\ dH_{12} \end{bmatrix} = \mathbf{F} \cdot \begin{bmatrix} dx_{12} \\ dy_{12} \\ dz_{12} \end{bmatrix} \quad (\rho = 200 \text{ gon} / \pi) \quad (11.14b)$$

11.4.3 Final variance-covariance matrix in the map projection plane

The derived coefficient matrix $\mathbf{F}$ gives the variance-covariance matrix of the azimuth, horizontal distance and the zenith angle z_v (in chapter 11.4.1), or the variance-covariance matrix of the azimuth, horizontal distance and the height difference (in chapter 11.4.2):

$$\mathbf{C}_{(\alpha, s, z_v)} = \mathbf{F} \cdot \mathbf{C}_{LG} \cdot \mathbf{F}^T = \mathbf{F} \cdot \mathbf{T} \cdot \mathbf{C}_{CTRS} \cdot \mathbf{T}^T \cdot \mathbf{F}^T \quad (11.15)$$

The derived observations, α , S and z_v are correlated, as the original observations of the baseline, ΔX , ΔY and ΔZ , are correlated.

These derived observations can be included in an adjustment where also classical observations like directions, distances and zenith angles measured by a total station or a theodolite are used. In this case with mixed observations we have to be careful to handle the corresponding variance-covariance matrices correct.

Remarks:

The computation of the variance-covariance matrix $\mathbf{C}_{(\alpha, S, Z_V)}$ described in this chapter is complicated.

Moreover can the computations of $\mathbf{C}_{\text{CTRS}}$ be somewhat uncertain, see the example in the chapters 9.4 and 10.1.2, showing unrealistic low numerical values of the standard deviations of the measured coordinate differences of the GNSS baseline.

Before an adjustment of a network of control points are performed, the values of the elements in the variance-covariance matrix must be evaluated and eventually scaled by a factor of ca 10, if the numerical values are unrealistically low. The commercial adjustment programs have an option for up scaling all the variance-covariance matrices by a factor, in our case 10^2 . The advantage of such a common up scaling of all the baselines is that the accuracies between the baselines are kept.

An alternative method of obtaining the variance-covariance matrix in the map projection plane is to use empirical values, as shown in the next chapter.

11.4.4 The variances-covariances computed empirically

Not covered in details this year, as the method has some large weaknesses, and therefore normally is not used in the adjustments.

The starting point is the empirical accuracy of the base lines, for instance:

Horizontal distance: $5 + 1 \cdot S_{km}$ (in mm)

Height difference: $8 + 1 \cdot S_{km}$ (in mm)

Remarks, weaknesses:

We get realistic values of the standard deviations using this method, by using empirical values. On the other hand the information of different accuracies of different baselines are not known empirically, compared to the other method in the previous chapter.

The correlations between the three derived observations bearing, distance and zenith angle cannot be determined on a simple way empirically. The correlation matrix in formula (10.1) can be given as some average values which normally will be the same for all the baselines.

In Norwegian:

Dette er en alternativ metode sammenlignet med beregning av varians/kovariansmatriser ved bruk av feilforplantning, som vist i forrige kapittel.

Erfaringsmessig (empirisk) har man funnet at nøyaktigheten man oppnår med en en-frekvent satellitmottaker for landmålingsformål, kan karakteriseres ved følgende nøyaktigheter (konstantleddene er for høyde, spesielt for tofrekvent måleutstyr, men her vises prinsippet for beregningene):

Horisontalavstand: $5 + 1 \cdot S_{km}$ (i mm)

Høydeforskjell: $8 + 1,5 \cdot S_{km}$ (i mm)

Tilhørende standardavvik blir da:

$$\text{Horisontalavstand: } \sigma_s = \left(5^2 + 1 \cdot S_{km}^2\right)^{1/2} \text{ mm} \quad (11.16)$$

$$\text{Høydeforskjell: } \sigma_h = \left(8^2 + 1,5 \cdot S_{km}^2\right)^{1/2} \text{ mm} \quad (11.17)$$

Med standardavvikene i (11.16) og (11.17) kan andre standardavvik beregnes ved bruk av feilforplantningsloven. Eksempelvis velges en teodolittmålt retning som middel av flere satser, som vektsenhet med tilhørende standardavvik σ_0 .

Den generelle sammenhengen mellom varians og vekt gir da vektene på avstand, asimut og senitvinkel:

$$\begin{aligned} p_r &= 1 \quad (\text{valgt, vekten p\aa vektsevenheten}) \\ \text{og} \quad p_s &= \frac{\sigma_0^2}{\sigma_s^2} \end{aligned} \quad (11.18)$$

$$\text{og videre} \quad p_\alpha = \frac{\sigma_0^2}{\sigma_\alpha^2} \quad (11.19)$$

$$p_{zv} = \frac{\sigma_0^2}{\sigma_{zv}^2} \quad (11.20)$$

Variansene som ogs\aa er brukt til beregningene av vektene, beregnes slik:

σ_s regnes direkte av (11.16).

Ved beregning av σ_α antas at standardavviket tvers p\aa en retning er den samme som langs en retning, alts\aa:

$$\sigma_\alpha^2 = \frac{\sigma_s^2}{S^2} \cdot \rho^2 = \frac{1}{S_{km}^2} \cdot (5^2 + 1 \cdot S_{km}^2) \cdot 10^{-12} \cdot \rho^{mgon^2} \quad (11.21)$$

Tilsvarende beregnes variansen for senitvinkelen:

$$\sigma_{zv}^2 = \frac{1}{S_{km}^2} \cdot (8^2 + 1,5 \cdot S_{km}^2) \cdot 10^{-12} \cdot \rho^{mgon^2} \quad (11.22)$$

In the formulas (11.21) and (11.22) the transformation from radians to milligon is done with the factor:

$$\rho^{mgon} = \frac{200000}{\pi}$$

PART C: GNSS AND HEIGHTS. GEOID MODELLING. CORRECTIONS TO/FROM THE MAP PROJ. PLANE.

- **Chapters 12 and 13:** The transformation from measured ellipsoidal heights/zenith angles to orthometric/normal heights/zenith angles. Key words: The deflection of the vertical (“loddavvik”), geoid heights (ellipsoid-geoid separation)
- **Chapters 14:** Corrections on the measured values when going to or from the ellipsoid/geoid and the map projection plane

12. Heights. From ellipsoidal to orthometric/normal heights

In practice we are using heights referred to the surface of the sea, for instance the mean sea level: Orthometric or normal heights, not ellipsoidal heights.

If the ellipsoid-geoid separation (geoid height «geoidehøyde») known for the actual ellipsoid in the chosen datum, the transformation from ellipsoidal to orthometric height easy to do.

h_o : orthometric height (height above the geoid). Normal height is height above the quasi geoid.

h_e : ellipsoidal height (height above the ellipsoid).

N : ellipsoid-geoid separation (geoid height, «geoidehøyde»). The difference between ellipsoidal and orthometric/normal height in a point. NB: geoid height is not the height above the geoid!

$$N = h_e - h_o$$

Figure 12.1 shows some important issues of the connection between the ellipsoid and the geoid as reference surfaces

- Plumb line (“loddlinje”), ellipsoid normal (“ellipsoidenormal”), deflection of the vertical (“loddavvik”)
- Geoid, ellipsoid
- Ellipsoid-geoid separation (geoidal height, “geoidehøyde”), do not use geoid undulasjon, old name

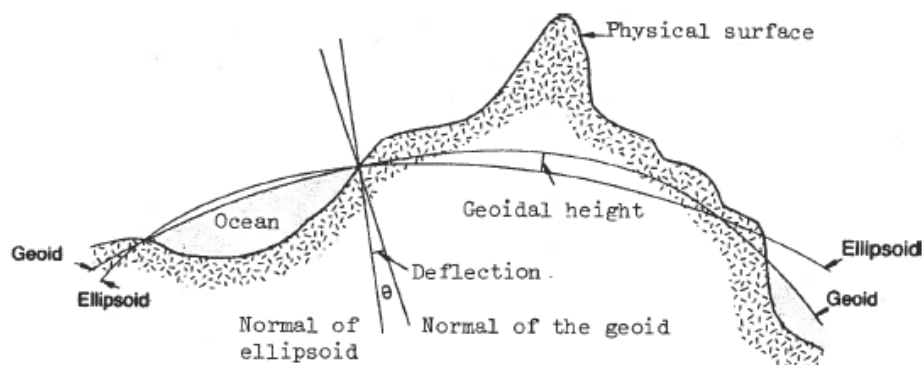


Figure 12.1 Geoidal height, deflection of the vertical. Shown in a point where geoid and ellipsoid intersect: the plumb line (“loddlinjen”) is the geoid normal.

12.1 Formulas: Deflection of the vertical. Laplace equation.

The zenith angles, z_v , calculated from measured satellite baselines (GNSS/GPS) are free of refraction «refraksjon/lysbøying»), as we are not aiming between the ends of the baseline. The zenith angle is an *ellipsoidal zenith angle*. This means that the reference direction ($z_v = 0$) is the *ellipsoid normal* in the point where z_v is calculated in.

When we measure a zenith angle with a total station or theodolite, the reference direction ($z = 0$) is the plumb line (“loddlinjen”) in the point where z is calculated in.

The difference between these two values of the zenith angle is the deflection of the vertical («loddavviket»), if we disregard the uncertainties of measurements. See figure 12.1.

The deflection θ (theta) is the angle between the normal of the ellipsoid and the vertical (plumb line).

We can express it by, see also figures 12.4a, b and c:

- (1) The magnitude θ and the azimuth A_v from the north axis to the plumb line
- (2) More used, the two components of the deflection θ (theta)
 - ξ (xi, ksi): The component in the north-south direction (positive value on figure 12.4a and b)
 - η (eta): The component in the east-west direction (positive value on figure 12.4a and b)

The total deflection of the vertical («loddavvik») θ , in a point is the angle between the ellipsoid normal and the plumb line in the point. The total deflection θ is split into two components, a component along the meridian, called ξ (xi, ksi), and one component perpendicular to the meridian, called η (eta). Figure 12.2 shows the north component of the deflection of the vertical.

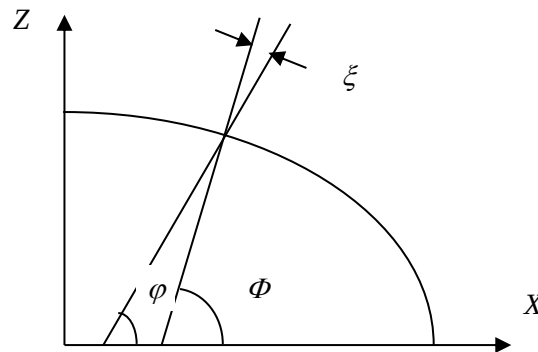


Figure 12.2 The north component of the deflection of the vertical

Names of the coordinates, see also the figures 12.2 and 12.3:

Φ : astronomic latitude («bredde»), the reference line is the plumb line, «loddlinjen»

φ : ellipsoidal (geodetic) latitude, the reference line is the ellipsoid normal

Λ : astronomic longitude («lengde»), the reference line is the plumb line

λ : ellipsoidal (geodetic) longitude, the reference line is the ellipsoid normal

The north component of the deflection, ξ , has a positive value on figure 12.2, where the plumb line is rotated more towards the north pole (and the Z axis, the rotation axis of the ellipsoid).

=> ξ is positive on figure 12.2.

The equation of ξ :

$$\xi = \Phi - \varphi$$

The east component of the deflection, η , see figure 12.3.

$N = R_N$: The radius of the curvature perpendicular to the meridian (see Holsens "Geodesy part 1")

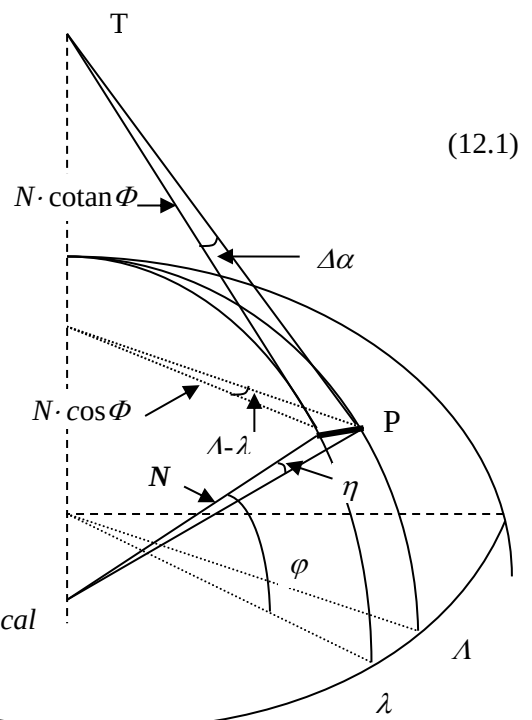


Figure 12.3 The east component of the deflection of the vertical

On figure 12.3 both the astronomic values (latitude, longitude, plumb line) and the ellipsoidal values (latitude, longitude, ellipsoid normal) in point P are found. The astronomic values are transferred to, or plotted on, the ellipsoid. Therefore there are two points P on the figure 12.3: One point with ellipsoidal values and the second point P with astronomic values.

On figure 12.3 there are 3 angles with a common chord, “korde”, k :

- $A - \lambda$: The difference between astronomic and geodetic longitude in point P
- η : The difference between the east component of the ellipsoid normal and the east component of the plumb line in point P; is the east component of the deflection of the vertical
- $\Delta\alpha$: The difference between the ellipsoidal and astronomic azimuth in point P; is the angle in the top point T

As the 3 angles above are very small (the value of the deviation of the vertical is normally a few milligon), the chord is set equal to the bow of the acute (“spisse”) triangles on the figure 12.3. This is an approximation, as the triangles have unlike bows. The length of a bow b is equal the angle times the radius of the curve/bow. Setting the two bows to be equal in length, and we obtain the following connection:

$$b = \eta \cdot N = (A - \lambda) \cdot (N \cdot \cos \Phi)$$

and we obtain the equation of the east component of the deflection of the vertical, η :

$$\eta = (A - \lambda) \cdot \cos \Phi \quad (12.2)$$

Alternative derivation of equation (12.2):

By using the chord, k , instead of the bow(s) in the same two acute triangles:

$$\sin \frac{1}{2}(A - \lambda) = \frac{k}{2N \cdot \cos \Phi} \quad (12.3a)$$

$$\text{and} \quad \sin \eta / 2 = \frac{k}{2N} \quad (12.3b)$$

$$(12.3a) \text{ divided by } (12.3b): \quad \sin \frac{1}{2}(A - \lambda) = \frac{\sin \frac{1}{2}\eta}{\cos \Phi} \quad (12.3c)$$

Comparing left and right hand side in (12.3c), and setting $\sin x = x$ as $A - \lambda$ and η are very small angles:

$$(A - \lambda) \approx \eta / \cos \Phi$$

And we obtain equation (12.2): $\eta = (A - \lambda) \cdot \cos \Phi$

The connection between astronomic and ellipsoidal azimuth, the Laplace equation:

Figure 12.3 gives also the connection between astronomic and ellipsoidal azimuth:

A : astronomic azimuth
 α : ellipsoidal azimuth

The difference in latitude, $(A - \lambda)$ gives a difference between the two azimuth values:

$$\Delta\alpha = A - \alpha$$

The bow b in point P: $b = N \cdot \eta$

In the triangle with vertex in T: $\Delta\alpha = b / (N \cotan \Phi)$

Substituting b : $\Delta\alpha = N \cdot \eta / (N \cotan \Phi) = A - \alpha$

The two equations of $\Delta\alpha$ gives, when also equation (12.2) is substituting η :

$$A - \alpha = (\Lambda - \lambda) \cdot \cos \Phi \cdot N / (N \cotan \Phi)$$

The equation of the differences in azimuths, the *Laplace equation*:

$$A - \alpha = (\Lambda - \lambda) \cdot \sin \Phi \quad (12.4)$$

Equation (12.4) is called the *Laplace equation*. It can be used for placing the ellipsoid optimal to the body of the earth (to the geoid). In the classical datum ED50 107 Laplace stations were used in the calculations. In a Laplace point the astronomic azimuth is measured (by observations to the stars).

12.2 The ellipsoidal zenith angle corrected for deflection of the vertical

From a GNSS baseline an *ellipsoidal zenith angle* can be derived. Orthometric/normal heights (with the geoid/quasi geoid as the zero surface) are used in Norway, for instance in the height datum NN2000. The ellipsoidal zenith angle must therefore be corrected to obtain the *astronomic (orthometric/normal) zenith angle* for use in the height calculations.

Figure 12.4a shows a 3D sketch where the plumb line («loddlinja») is placed in such way that the components of the deflection of the vertical, ξ and η , are positive: The plumb line is rotated towards north and east compared to the ellipsoid normal.

Figure 12.4b shows a plane which is perpendicular on the ellipsoid normal. The plane is also shown on figure 12.4a.

The deflection of the vertical («loddavviket»), θ , and with that the components ξ and η , are small angles, normally ca. a few milligon. The arc bows corresponding to these angles can therefore without loss of accuracy be considered as straight lines in this plane, see figure 12.4b.

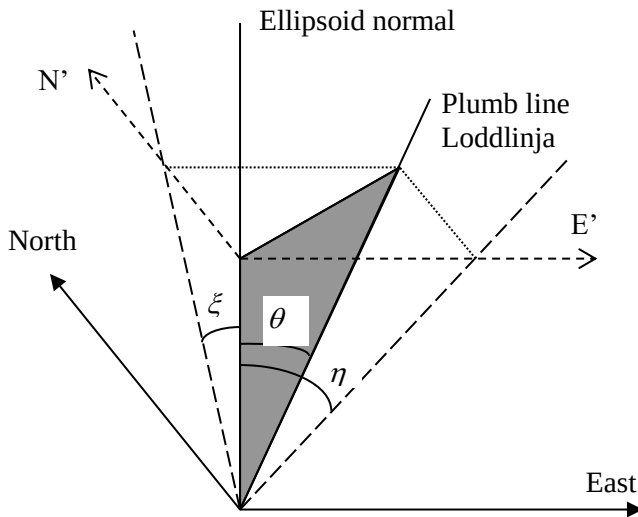


Figure 12.4a Deviation of the vertical («loddavvik»)

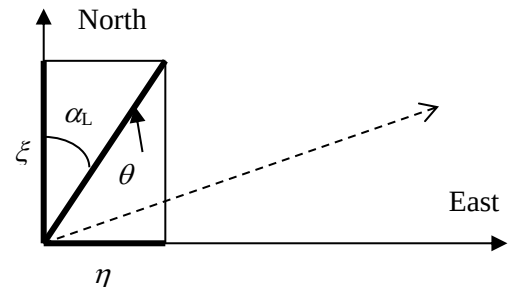


Figure 12.4b Components of the derivation, ξ , η

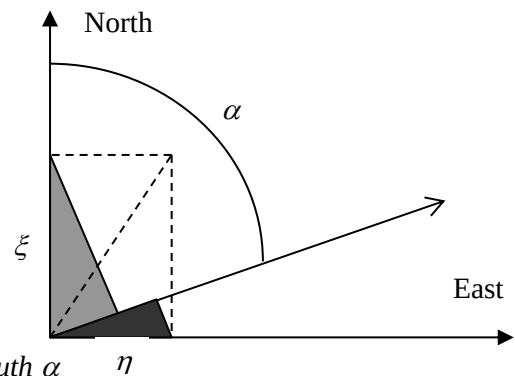


Figure 12.4c Deviation of an arbitrary direction with azimuth α

Figure 12.4c shows the impact of the deflection of the vertical on the zenith angle on a horizontal direction with an azimuth α . The two components of the plumb line are decomposed to a component along the direction with azimuth α , and a component crosswise (perpendicular to) this direction. Summarizing these two components along the actual direction (the legs (“katetene”) in the actual direction of the black (from η) and grey triangle (from ξ) on figure 12.4c, we obtain the impact of the deflection of the plumb line in the direction with azimuth α :

$$\Delta z_v = \xi \cdot \cos \alpha + \eta \cdot \sin \alpha \quad (12.5)$$

Correction on the ellipsoidal zenith angle

Figure 12.5a shows $-\Delta z_v$ as the correction on the ellipsoidal zenith angle to obtain the astronomic (orthometric/normal) zenith angle.

$$z_{\text{ort}} = z_{\text{ell}} - \Delta z_v = z_{\text{ell}} - \cos \alpha \cdot \xi - \sin \alpha \cdot \eta \quad (12.6)$$

z_{ort} : Can be calculated from the orthometric heights, see an example in chapter 12.3.

z_{ell} : Can be calculated from the GNSS/GPS measurements, see equation 10.18.

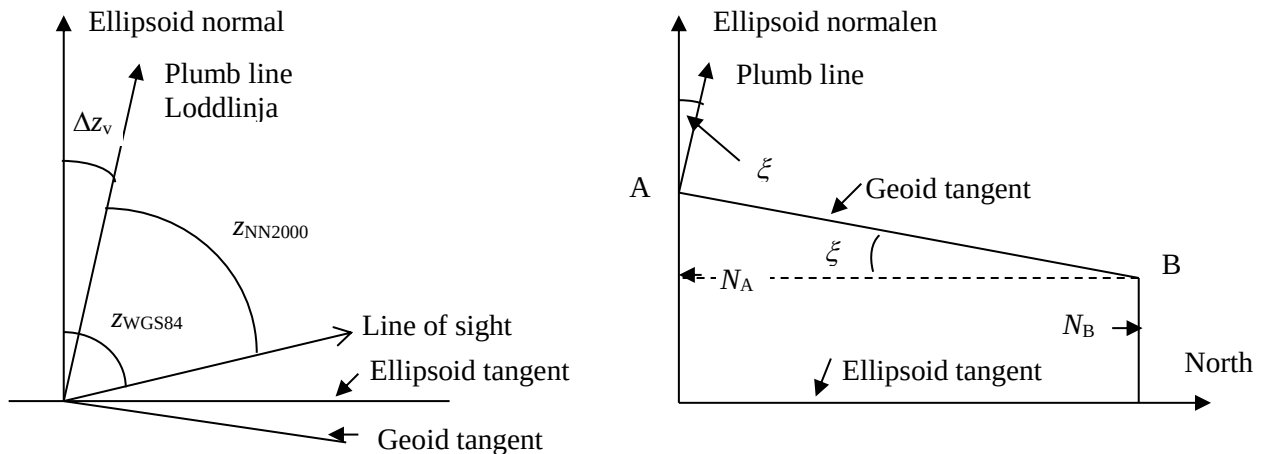


Figure 12.5a Zenith angle, deflection of the vertical. Figure 12.5b Geoid heights, deflection of the v.

Figure 12.5b shows that a positive north component of the deflection of the vertical, ξ , gives a decreasing («avtagende») geoid height, N , in the north direction, as the geoid height in A, N_A , is larger than N_B in B. Quite analogous will a positive η give a decreasing («avtagende») geoid height, N , in the east direction.

Estimation of the two components of the deflection of the vertical.

The observation equation (“feilligningen”) of the zenith angle

The two components of the deflection of the vertical, ξ and η are normally not known.

But, if we have measured GPS baselines from a control point («fastmerke») with a known NN2000 height to several (minimum two) other points with known NN2000 heights, can ξ and η be estimated. The known heights must be of high quality: The heights must preferably be precise levelled (“nivellerte”) heights, and the points should be distributed well in the horizontal plane.

With more than three known points with both ellipsoidal and NN2000 heights, the components ξ and η can be estimated by using the least squares method (LSM), “minste kvadraters metode, MKM”.

An ellipsoidal zenith angle is a derived («avledet») observation, and a correction, v , is applied to this observation in the estimation (correction, «utjevningstillegg»). From equation (12.6) we obtain:

$$z_{\text{ell}} + v = z_{\text{ort}} + \cos \alpha \cdot \xi + \sin \alpha \cdot \eta$$

The observation equation of an ellipsoidal zenith angle is obtained by isolating the correction on the left hand side:

$$v = \cos \alpha \cdot \xi + \sin \alpha \cdot \eta + (z_{\text{ort}} - z_{\text{ell}}) \quad (12.7)$$

An estimation by using LSM will give the unknowns components ξ and η .

See also chapter 13.1 how the two components of the deflection of the vertical can be calculated from a known geoid model.

The ellipsoid-geoid separation (geoid height), N , is defined as the difference between ellipsoidal and orthometric heights in a point (ellipsoidal minus orthometric):

$$N = H_e - H_o$$

Figure 13.1 shows that the geoid heights in Norway in the datum EUREF89 is varying between ca. +17 meters in north and +48 meters in south. In Norway the ellipsoid is lower than the geoid, => always positive N 's.

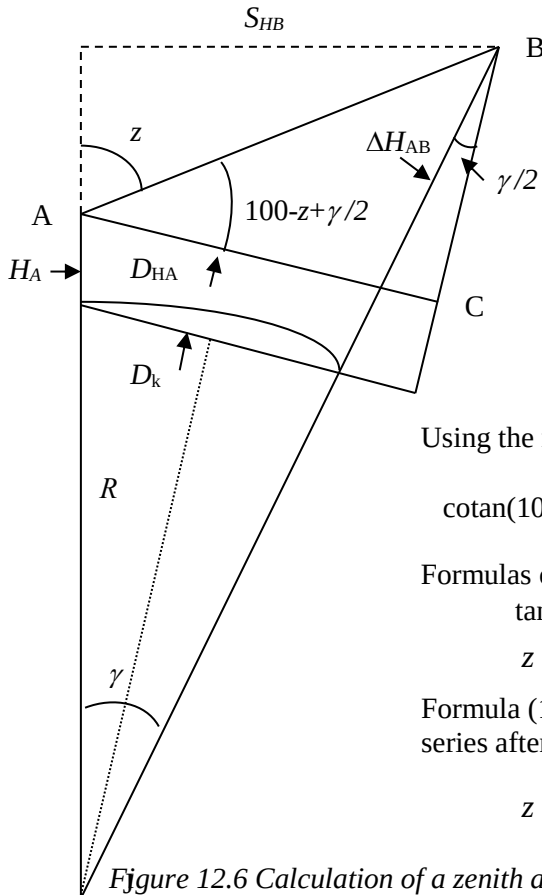
12.3 Calculation of a zenith angle from known orthometric height difference

Numerical example:

This chapter shows how the orthometric zenith angle can be calculated from a known orthometric height difference.

Use: For use in the equations (12.6) and (12.7) in the last chapter.

NB: The distances in figure 12.6 below are very exaggerated compared to measured distances in practice. The distance D_{HA} is ca. 3300 km.



Height difference: $\Delta H_{AB} = H_B - H_A$

D_k is horizontal distance (chord) in $H = 0$.

D_{HA} is the chord distance in height H_A :

$$D_{HA} = \frac{R + H_A}{R} \cdot D_k$$

The angle γ on figure 12.6 can be calculated, see also the example in chapter 11.1:

$$\sin \frac{\gamma}{2} = \frac{D_{HA}}{2 \cdot (R + H_A)}$$

Using the right-angled triangle ABC:

$$\cotan(100 - (z - \gamma/2)) = (D_{HA} + \Delta H_{AB} \cdot \sin(\gamma/2)) / (\Delta H_{AB} \cdot \cos(\gamma/2))$$

Formulas of the zenith angle:

$$\tan(z - \gamma/2) = D_{HA} / (\Delta H_{AB} \cdot \cos(\gamma/2)) + \tan(\gamma/2) \quad (12.8)$$

$$z = \arctan\left(D_{HA} / (\Delta H_{AB} \cdot \cos(\gamma/2))\right) + \gamma/2 \quad (12.9)$$

Formula (12.10) is a series expansion of formula (12.9), aborting the series after two terms (an approximate formula, but Ok for this use):

$$z = \arctan\left(\frac{\Delta H_{AB}}{D_{HA}}\right) + \frac{D_{HA} \cdot \rho^{\text{gon}}}{2R \cdot \sin^2 z} + \dots \quad (12.10)$$

Figure 12.6 Calculation of a zenith angle

13. Geoid modelling

To be able to transfer between ellipsoidal heights and orthometric/normal heights (NN2000 heights), we must know how the ellipsoid-geoid separation (geoid height), N , is varying in the actual area. The geoid height is defined as the difference between the reference ellipsoid and the geoid. See figure 12.1.

Modelling the geoid covering Norway consists of two main parts:

- **Carpet (“teppe”)**: The *shape of the geoid* is calculated from gravity measurements of the mainland and the ocean areas close to the shores. This is normally done by using airplanes and helicopters. From these measurements we obtain the shape of the geoid, which is a physical surface. This shape can be seen on figure 13.1, a bulky surface.
The geoid model made by only gravity measurements do not give sufficient absolute accuracy of the geoid model, but the relative shape has sufficient accuracy. See chapters 13.2 and 13.3.
In this compendium we do not go into details about the gravity measurements, see text books in *physical geodesy*.
- **Pegs (“knagger”)**: Measuring and calculating the *ellipsoid-geoid separation* (geoid height) in a limited number of points. From the information in these points, we can land the carpet on these pegs or points, and obtain a good geoid model. In 2017 the accuracy of this landed model is ca. 1-2 cm. See figure 13.3. in chapter 13.2.
In this compendium we compute the ellipsoid-geoid separation (geoid height) and find geoid models by using only measured points with ellipsoidal heights (from GNSS/GPS) and measured heights in the height datum NN2000 (or the old datum NN1954). These models are suitable for smaller areas.

Comment 2017: The height reference models are improved yearly by using more points with known ellipsoidal and NN2000 heights, and by using newer and better gravimetrical measurements. The latest HREF for obtaining NN2000 heights from ellipsoidal heights is per October 2017 the HREF model [Href2015B_NN2000_EUREF89](#).

13.1 Geoid height models, calculation of deflection of the vertical, «loddavvik»

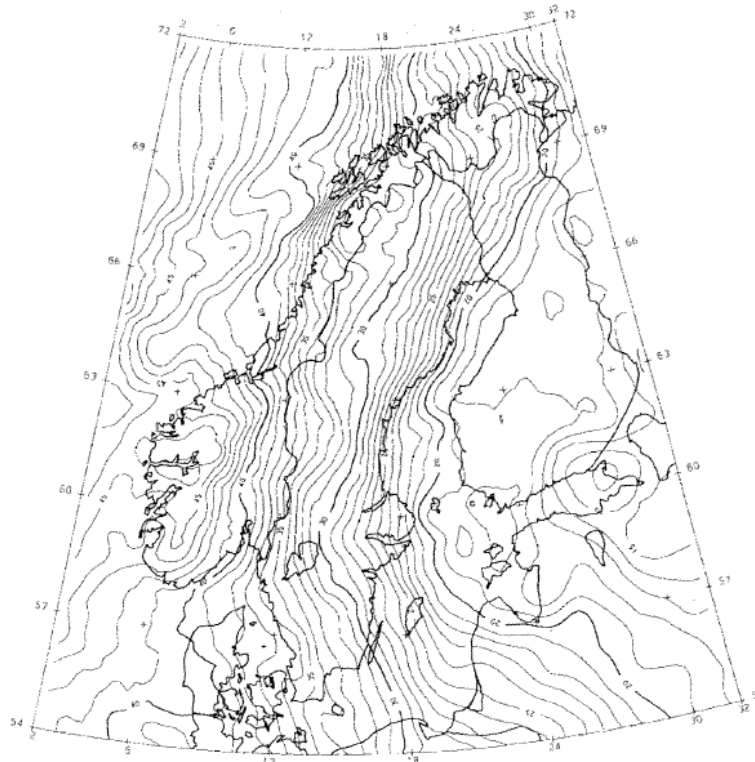


Figure 13.1 Geoid heights. The Nordic geoid (NKG89) equidistance 1m

Within smaller areas, ca. 15 km x 15 km, can the two components of the deflection of the vertical, «loddavviket», ξ and η , be considered as approximately constants. But, for larger areas will the values of ξ and η be significantly dependent of the coordinates. See figure 13.1, where the geoid heights from the reference ellipsoid GRS80 are shown. This ellipsoid is very close to the WGS84 ellipsoid in shape and how they are linked to the geoid.. The figure shows the Nordic geoid from 1989 (NKG89), a result of a Nordic cooperation in the «Nordiske Geodetiske Kommisjonen».

See also in chapter 13.2 about a newer version called NKG96, and about geoid heights of the height reference surface VREF96 from 1996, which the Norwegian mapping authority, «Kartverket» was using in their transformation program SkTrans. Geoid heights of VREF1996 in figure 13.2.

Geoid heights in an area are described by using a geoid height model.

The geoid height N will be a function of where we are: $N = F(\text{coordinates})$

In this chapter a polynomial model of 1st or 2nd order is used. Such a model is well suitable for use in smaller areas.

The deflection of the vertical («loddavviket») express how the geoid height, N , is changing in different directions from a point. See figure 12.5b in chapter 12.2. Mathematically this is expressed with the following connections between N , ξ and η , where the *negative* sign in the formulas is due to a *positive* value of ξ and η will give a decreasing value of the geoid height when going respectively northwards and eastwards from a point:

North component of the deflection of the vertical («loddavviket»):

$$\xi = -\frac{\partial N}{\partial x} \quad (13.1)$$

East component of the deflection of the vertical («loddavviket»):

$$\eta = -\frac{\partial N}{\partial y} \quad (13.2)$$

13.1.1 Geoid model with a constant deflection of the vertical “loddavvik”

With constant components of the deflection of the vertical in an area, a polynomial model of 1st order can be used to describe the geoid height (the ellipsoid-geoid separation), N :

$$N = Dx + Ey + F \quad (13.3)$$

D and E are coefficients. They are either known, or are to be estimated.

The coordinates x (north) and y (east) are in the map projection plane.

Differentiating of (13.3) according to the definitions in equations (13.1) and (13.2) gives the two components of the deflection of the vertical (“loddavvikskomponentene”):

$$\xi = -\frac{\partial N}{\partial x} = -D \quad \eta = -\frac{\partial N}{\partial y} = -E \quad (13.4)$$

(13.4) is inserted into (13.3), and we obtain the geoid height in point i :

$$N_i = -\xi \cdot x_i - \eta \cdot y_i + F \quad (13.5)$$

The calculations are made easier by choosing a local origin in the project area, and use local coordinates (coordinate differences) instead of the absolute EUREF89-UTM coordinates (or old NGO1948 coordinates). The constant F will in this case with local origin, be the geoid height, N , in the point that is chosen as the local origin, N_{LO} .

$$\text{The geoid height in point } i: \quad N_i = -\xi \cdot \Delta x_i - \eta \cdot \Delta y_i + N_{LO} \quad (13.6)$$

We need to have/determine ellipsoidal heights in minimum 3 points with known orthometric heights to be able to calculate the 3 unknowns in formula (13.6), ξ , η and F , or D , E and F (see (13.4)).

Are there more than 3 points with known heights, the unknowns are estimated by using the least squares method (LSM), “minste kvadraters metode (MKM)”.

See equation (13.11b and c) for how the observation equations («feilligningen») are established.

Are the components of the deflection of the vertical known, equation (13.6) can be used to calculate relative geoid heights (ΔN), or absolute values, N , if the geoid height in for instance the local origin is known.

Numerical example of a geoid height model:

Deflection of the vertical (“loddavvik»), constant values: $\xi = +2,259$ mgon $\eta = +1,897$ mgon
Geoid height model, with a local origin with a geoid height of $N = 39,844$ m:

$$N = -\frac{2,259}{\rho^{\text{mgon}}} \cdot \Delta x - \frac{1,897}{\rho^{\text{mgon}}} \cdot \Delta y + 39,844$$

The components ξ and η must be given with denomination radians in equation (13.6), therefore the conversion factor ρ is implemented, here. $\rho = 200\,000$ mgon / π .
See also a numerical example in chapter 13.1.3.

13.1.2 Geoid model, where the deflection of the vertical is not constant

With varying values of the components of the deflection of the vertical in an area, a polynomial model of 2nd or even higher order can be used to describe or model the geoid height, N .

A polynomial of 2nd order can be written like this:

$$N = Ax^2 + Bxy + Cy^2 + Dx + Ey + F \quad (\text{second order polynomial}) \quad (13.7)$$

A, B, C, D, E and F are coefficients that either are known, or are to be estimated.
The coordinates x (north) and y (east) are in the map projection plane.

Differentiating of (13.7) according to the definitions in equations (13.1) and (13.2) gives the two components of the deflection of the vertical (“loddavvikskomponentene”):

$$\xi = -\frac{\partial N}{\partial x} = -2Ax - By - D \quad \eta = -\frac{\partial N}{\partial y} = -Bx - 2Cy - E \quad (13.8)$$

Mathematical descriptions of the variations of the components ξ and η :

$$\xi = \xi_0 + a_1\Delta x + a_2\Delta y \quad \eta = \eta_0 + a_2\Delta x + a_3\Delta y \quad (13.9)$$

ξ_0 and η_0 are the components in a local origin in the project area.
 Δx and Δy are the coordinate differences from this local origin to an arbitrary point.
The coefficients a_i are either known or shall be estimated.

A comparison of (13.8) and (13.9) gives:

$$\begin{aligned} A &= -a_1 / 2 & B &= -a_2 & C &= -a_3 / 2 \\ D &= -\xi_0 & E &= -\eta_0 & F &= H_{\text{ell}} - H_{\text{ort}} \text{ in the local origin} \end{aligned} \quad (13.10)$$

We need to have/determine ellipsoidal heights (with GPS/GNSS) in minimum 6 points with known orthometric heights to be able to calculate the 6 unknowns in formula (13.10), A, B, C, D, E and F , or $\xi_0, \eta_0, a_1, a_2, a_3$ and F , (see (13.8)).

Are there more than 6 points with known heights, the 6 unknowns are estimated by using the least squares method (LSM), “minste kvadraters metode (MKM)”.

13.1.3 Geoid model. The observation equations, LSM (“feilligninger, MKM”)

The observation equations when a polynomial model of max 2nd order is used, up to six parameters, for instance A, B, C, D, E and F . The 1st order polynomial in chapter 13.1.1 has constant components of the deflection of the vertical, ξ and η . In this case $A = B = C = 0$.

Three types of observation equations below:

- 1: Zenith angles as the observations: Ellipsoidal zenith angle must be calculated from the GNSS observations. Probably easier to use 2a or 2b.
- 2a: Geoid heights as the observations: Easy to calculate as this observation is the difference between ellipsoidal height and the NN2000 height, $N = H_e - H_o$
- 2b: ΔF is the unknown element instead of F : Same as 2a, but using a preliminary/approximate value of the parameter F .

1: Zenith angles as the observations:

Equation (13.9) and (13.10) inserted into the observation equation («feilligningen») (12.7):

Gives an observation equation with the *zenith angles as the observations*:

$$v = \cos \alpha \cdot (\xi_0 + a_1 \cdot \Delta x + a_2 \cdot \Delta y) + \sin \alpha \cdot (\eta_0 + a_2 \cdot \Delta x + a_3 \cdot \Delta y) + (z_{ort} - z_{ell}) \quad (13.11a)$$

2a: Geoid heights as the observations:

The observation equation with the *geoid heights as the observations*, see the mathematical connections between the observation and the coordinates and the unknowns in equation (13.7):

$$N = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$$

$$N_i + v_i = A \cdot x_i^2 + B \cdot x_i \cdot y_i + C \cdot y_i^2 + D \cdot x_i + E \cdot y_i + F$$

$$\Leftrightarrow v_i = x_i^2 \cdot A + x_i \cdot y_i \cdot B + y_i^2 \cdot C + x_i \cdot D + y_i \cdot E + F - N_i \quad (13.11b)$$

The sixth unknown, F , is the geoid height in the local origin. This can easily be seen when we insert the local coordinates $x = y = 0$ of the local origin into (13.7): $N_{LO} = F$.

The observation equation (“feilligningen”) of the local origin point: $v_{LO} = F - N_{LO}$.

2b: ΔF is the unknown element instead of F :

We can use a preliminary or approximate value, F_0 , of the unknown element F , where $F = F_0 + \Delta F$.

Choosing $F_0 = N_{measured} = N_{LO}$ in the point chosen as the local origin.

The observation equation with $F_0 = N_{measured} = N_{LO}$, and here ΔF is the unknown element instead of F :

$$v_i = x_i^2 \cdot A + x_i \cdot y_i \cdot B + y_i^2 \cdot C + x_i \cdot D + y_i \cdot E + \Delta F - (N_i - N_{LO}) \quad (13.11c)$$

After ΔF is estimated we find the estimated value of F : $F = F_0 + \Delta F$

From the observation equations to the estimated values of the elements, using (13.11b or 13.11c):

Using matrices, the observation equation matrix $\mathbf{V}$ is: $\mathbf{V} = \mathbf{Ax} - \mathbf{f}$

- The unknowns A, B, C, D, E and F are elements in the $\mathbf{x}$ matrix.
- The coordinates gives the elements in the coefficient matrix $\mathbf{A}$.
- In (13.11b) the constant matrix $\mathbf{f}$ consists of the measured values of the geoid heights N_i .
- In (13.11c) the constant matrix $\mathbf{f}$ consists of the difference between N_i and the preliminary value of N in the point chosen as the local origin.

See the further work from the observation equations to the solution of the normal equations in another compendium.

Comments:

Using a polynomial of 2nd order, we have six unknown parameters, A, B, C, D, E and F , or $\xi_0, \eta_0, a_1, a_2, a_3$ and F . We must be critical of the estimated values of the unknown elements, and

check by statistical testing if the estimated values of the elements are significantly different from zero. See the numerical example in the next chapter, 13.1.3, for the procedure.

Are the two components of the deflection of the vertical «loddavvikskomponentene» and the coefficients A to E known, equation (13.7) can be used to calculate either relative geoid heights (ΔN).

We can calculate absolute values, N , if the geoid height in for instance the local origin is known (the coefficient F is known).

In a larger area (for instance a municipality, «kommune») normally the components ξ and η are not constant.

An alternative to a polynomial model of higher order, is to divide the area in smaller areas, and calculate and use constant ξ and η within each part of the total area. The advantage is simpler formulas, the disadvantage is no continuity in the border areas between the smaller areas. This method was in use at the start of modelling the geoid height,

Now (2017) the transformation between ellipsoidal height to heights in the reference system NN2000, is done by using height reference models developed by Norwegian mapping authority “Kartverket”.

An example from 2000:

From the height reference surface VREF1996 we (or the processing program) can calculate the deflection of the vertical, ξ and η , in each point of a network of new and known control points («fastmerker»). Thus the correction from ellipsoidal to orthometric/normal zenith angle is done by using unique and optimal ξ and η values derived in each point.

The advantage is that we are using a continuous geoid height surface with no border problems as in the method by dividing an area in smaller parts with constant pairs of ξ and η in each area.

This method is implemented into the commercial land surveying programs, and is used in 2017.

13.1.3 Geoid model, numerical example

A local GPS network of control points is established in Orkanger municipality, «kommune», west of Trondheim. One of the purposes was to estimate a geoid height model in that area. The local network consists of 9 control points, see the figure.

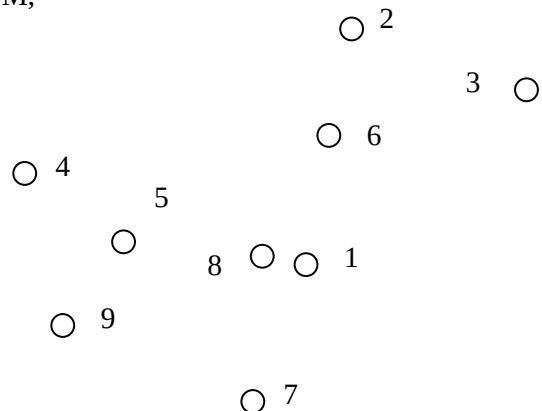
The heights of these 9 control points were also measured-in by precise levelling «presisjonsnivelllement» with the precise bench marks “presisjons-høydefastmerker” established by the Norwegian mapping authority as the starting points, the known points.

The ellipsoid-geoid separation, the geoid height, can be calculated in each of the 9 points as: $N = H_e - H_o$

NB: The values of this geoid height model differ with ca. 13 cm from the geoid height model VREF1996 (using WSkTrans 3.1), see Geir W. Simensens dr.ing. report, on page 139, about normal heights.

In the table below the coordinates are in EUREF89-UTM, and the heights are ellipsoidal heights.

Point	N	E	H_{ELL}
1	7014965,167	540789,507	74,508
2	7026565,573	546422,244	50,278
3	7023037,921	553522,734	63,214
4	7020314,990	528232,098	351,269
5	7016965,877	532989,173	255,747
6	7020964,316	543579,953	59,187
7	7008381,424	539364,262	71,810
8	7015200,583	539920,443	69,128
9	7012840,637	530064,371	217,804



From the measurements, various models of the geoid in the area can be estimated. Normally we have to test several models to find the most optimal model.

The area is ca. 18,2 km (south-north) times 25,3 km (west-east). The terrain is relatively smooth; not cliffy, no large heights differences, and no steep slopes. Together with the size of the area, this indicate that a 2nd order polynomial model can be used, where control point 1 is used as a local origin, and x, y are local coordinates:

$$N = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$$

The local coordinates, with point 1 as origin, the measured geoidal heights, N , are not shown here:

Point	x	y	N	Point	x	y	N
1	0,000	0,000		6	5999,149	2790,446	
2	11600,406	5632,737		7	-6583,743	-1425,245	
3	8072,754	12733,227		8	235,416	-869,064	
4	5349,823	-12557,409		9	-2124,530	-10725,136	
5	2000,710	-7800,334					

In this project we have 9 observations of N , 1 in each point. The number of unknown parameters are maximum 6 (A to F). Therefore the least squares method (LSM, "MKM") must be used to estimate the parameters.

First a 6 parameter model was estimated. A statistical test of the values of the parameters showed that parameter A was not significantly different from 0. Therefore this model was rejected, or simplified, and a 5 parameter model without A was estimated and tested. All 5 parameters were significantly different from 0. In addition a 3 parameter model was estimated, just to have a comparison between a simple 3 parameter model and an optimal 5 parameter model.

The results of the estimations of the coefficients / parameters for the two most interesting models, with their standard deviations, s , are shown below (results in radians and meter):

(1): 3 parameter model. A, B and C is set to zero. $N = Dx + Ey + F$

$$\begin{aligned} D &= -3,08 \cdot 10^{-5} & \text{and} & & s_D &= 3,21 \cdot 10^{-6} \\ E &= -4,13 \cdot 10^{-5} & \text{and} & & s_E &= 2,22 \cdot 10^{-6} \\ F &= 40,92 \text{ m} & \text{and} & & s_F &= 0,018 \text{ m} \end{aligned}$$

(2): 5 parameter model. A is set to zero. $N = Bxy + Cy^2 + Dx + Ey + F$

$$\begin{aligned} B &= -1,44 \cdot 10^{-9} & \text{and} & & s_B &= 1,55 \cdot 10^{-10} \\ C &= +5,85 \cdot 10^{-10} & \text{and} & & s_C &= 6,65 \cdot 10^{-11} \\ D &= -3,45 \cdot 10^{-5} & \text{and} & & s_D &= 8,33 \cdot 10^{-7} \\ E &= -3,21 \cdot 10^{-5} & \text{and} & & s_E &= 1,03 \cdot 10^{-6} \\ F &= 40,93 \text{ m} & \text{and} & & s_F &= 0,005 \text{ m} \end{aligned}$$

The degree of freedom, f , in the estimation of the coefficients (parameters) in model (1) and (2)?

(1) $f = n - e = 9 \text{ points} - 3 \text{ unknowns } (D, E, F) = 6$

(2) $f = n - e = 9 \text{ points} - 5 \text{ unknowns } (B, C, D, E, F. \text{ Not } A) = 4$

Statistical test if the coefficients in model (2) are significant different from zero:

A student-T test (we have estimated standard deviations) with significance level $\alpha = 5 \%$ (is normally used). Using a two-tailed test, we use $\alpha / 2 = 2,5 \%$ in the student-T table.

Tabulated student-T value: $T_{f, \alpha/2} = T_{4, 2,5} = 2,78$

Test value obtained from the LSM estimations: $t = \left| \frac{x}{s_x} \right|$

$$\text{Calculated test values: } t_B = \left| \frac{-1,44 \cdot 10^{-9}}{1,55 \cdot 10^{-10}} \right| = 9,3, \text{ and } t_C = 8,8, t_D = 41,4 \quad t_E = 31,17 \quad t_F = 8190$$

Conclusion: All the 5 parameters are significantly different from 0, as the calculated test values are larger than the tabulated value of 2,78.

The components of the deflection of the vertical (“loddavvikskomponentene”) in point 1, when the 3 parameter model (1) is used:

We know that the derivative of the surface gives the components of the deflection of the vertical, see also formula (13.4). We obtain the two components that *are valid for the whole area*:

$$-\frac{\partial N}{\partial x} = -D = +\xi = +3,08 \cdot 10^{-5} \text{ rad} = +1,96 \text{ mgon}$$

$$-\frac{\partial N}{\partial y} = -E = +\eta = +2,63 \cdot 10^{-5} \text{ rad} = +2,63 \text{ mgon}$$

The components of the deflection of the vertical (“loddavvikskomponentene”) in point 1, when the 5 parameter model (2) is used:

Same procedure as above but more terms, see also formula (13.8).

The components of the deflection of the vertical are *varying from point to point*. Entering the local coordinates $x = y = 0,00$ for point 1 (the local origin), and we obtain the components ξ and η in point 1:

$$-\frac{\partial N}{\partial x} = -By - D = +\xi \Rightarrow \xi_1 = +3,45 \cdot 10^{-5} \text{ rad} = +2,20 \text{ mgon}$$

$$-\frac{\partial N}{\partial y} = -Bx - 2Cy - E = +\eta \Rightarrow \eta_1 = +3,21 \cdot 10^{-5} \text{ rad} = +2,04 \text{ mgon}$$

The ellipsoid-geoid separation (geoid height) in the points 1, 2 and 4, by using both model (1) and (2):

Model (1):	$N_1 = 40,92 \text{ m}$	$N_2 = 40,330 \text{ m}$	$N_4 = 41,274 \text{ m}$
Model (2):	$N_1 = 40,93 \text{ m}$	$N_2 = 40,273 \text{ m}$	$N_4 = 41,338 \text{ m}$

Conclusion:

Use model (2). All the 5 coefficients are significant, and the check in points 1, 2 and 4 showed that the difference in height above the sea for a point is 6 cm in points 2 and 4, compared to a simpler 3 parameter model.

The size of the area is ca. 18 km x 25 km and has a height difference of ca. 300 m. It is as expected that the components of the deflection of the vertical will vary in the area.

13.2 VREF1996. Reference surface for heights

The chapters 13.2 and 13.3 are background stuff, and are not part of the curriculum (required reading) where deeper understanding is demanded. The chapters were originally written in 2000, and need to be updated with the current situation.

The methods to obtain a reference surface for heights or a geoid model are ca. the same in 2017. Thus, the description of the used methods is still valid.

VREF1996 is a geoid model which is (2017: was) used in the official transformation program WSkTrans developed by the Norwegian mapping authorities («Kartverket»), to generate corrections between ellipsoidal heights and NN1954 heights (2017: NN2000 heights). VREF1996 was introduced in 1996.

The reasons for introducing the reference model, or the surface model, VREF1996 is stated in the preliminary consultation document («høringsutkast») from August 2000, for the Norwegian standard ”Koordinatbasert referansesystem. Datum, koordinatsystem, transformasjon, konvertering og avbildning” from the Norwegian mapping authorities («Kartverket»). English title: «Coordinate based reference system. Datum, coordinate system, transformation, converting and mapping”.

In Norway the official height datum is NN1954 (2017: NN2000). To be able to correct the ellipsoidal heights to precise height above the geoid, gravimetric measurements (measurements of gravity, force of gravity) at a selection of the levelled bench mark (“nivellamentsfastmerkene”) in the official precise network of bench marks. Such measurements was not available in sufficient degree in 1954. Instead it was in a limited extent applied a *spheroidic-orthometric correction* based on theoretical values of the gravity. See more about this in the Norwegian compendium from B. Langseth about height references. Calculations has shown that the NN1954 heights are approximately heights above the *quasi geoid*.

A *quasi geoid* is a geoid like surface, but it is not a equipotential surface as the geoid is. The quasi geoid is, as stated above, based on theoretical values of the gravity, and not on the real variation of the density of the earth.

Further, in the calculation of the datum NN1954, the land uplift after the ice age 10 000 years ago, was not corrected for. It was not well enough known to be able to quantify the uplift and its variation around in Norway. This introduced errors, as the observation period was 1916-1954. The height datum NN1954 is referenced to 1954 as the year when the height calculations were completed / finished.

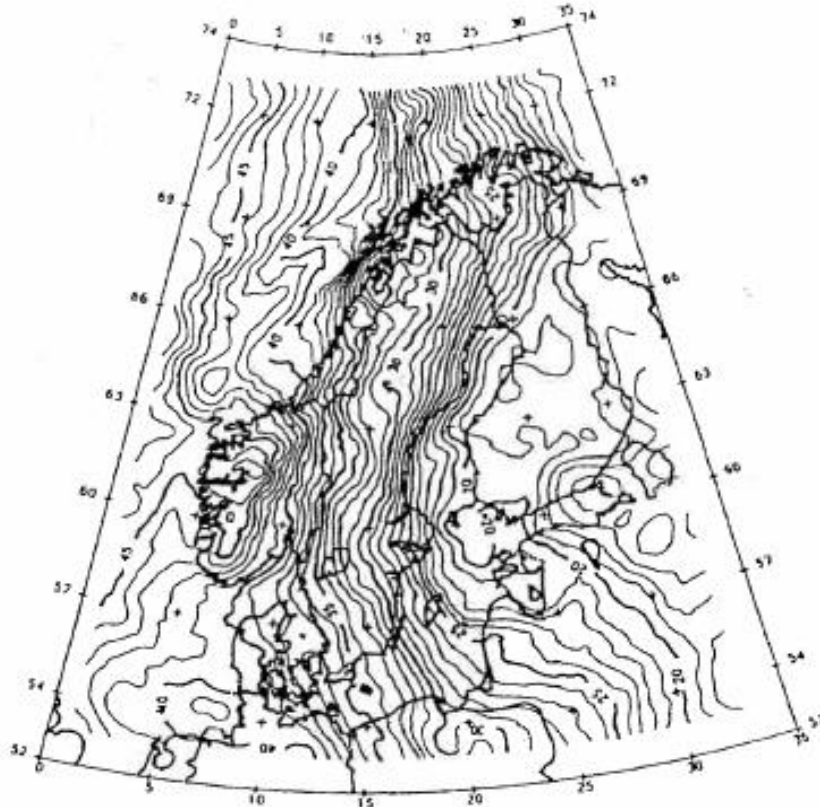


Figure 13.2 VREF1996

Ellipsoidal heights measured with satellite based systems (as GPS) were in the 1990-ties transformed to heights in NN1954 by using the Nordic NKG89 geoid as the reference surface. See figure 13.1 in chapter 13.1, and below in chapter 13.2.1.

The NKG89 geoid model gave a difference of up to 0,9 m on a height measured by GPS, compared with a levelled («nivellert høyde») to the same point, when the levelling started in a control point with known height in NN1954. Therefore the NKG89 geoid model was too inaccurate to be used to calculate “good enough” orthometric heights from the GPS measurements.

13.2.1 The reference surface VREF1996

In 1996 there were ca. 30 bench marks («nivellamentsfastmerker») that also had ellipsoidal heights calculated from GPS measurements. The geoid surface was adjusted optimally to these 30 bench marks, and thus the geoid surface was fitted optimally to the zero surface of the height datum NN1954. See figure 13.3 showing the distribution of the 30 bench marks the NKG geoid model was adjusted to. This surface was named VREF1996, as this surface could not be considered as a genuine geoid.



Figure 13.3 The 30 common points

In 1997 more bench marks with ellipsoidal heights were added, and the total number of such common points increased from 30 to ca. 50. The surface model kept its name, VREF1996.

With VREF1996 we obtained NN1954 heights from the GPS measured ellipsoidal heights. A test in a number of check points showed that there were a difference of less than 0,05 m on a GPS measured ellipsoidal height corrected with VREF1996, compared to a measured value in the height datum NN1954. It was assumed that the height model VREF1996 is less than 0,1 m from a correct value.

New versions of VREF, VREF2000

In 1999 a new version of VREF was tested. It was based on 190 bench marks (compared to 30 in the original 1996 version) and ca. 1000 check points. 81% of the check points had a deviation of less than 0,03 m between ellipsoidal heights corrected with VREF2000 and the NN954 height. With GPS as the dominating method for measuring coordinates, a increasing number of bench marks with heights in two different systems contribute to an improved geoid height model. There is also a continuous work going on with collecting new and checking old sets of gravity data, to improve the gravimetric geoid model.

Comment 2017:

The height reference models are improved yearly by using more points with known ellipsoidal and NN2000 heights, and by using newer and better gravimetric measurements. The latest HREF for obtaining NN2000 heights from ellipsoidal heights is per October 2017 the HREF model [Href2015B_NN2000_EUREF89](#).

The accuracy is assumed to be ca. 0,01-0,02 m. The Norwegian mapping authority (“Kartverket”) has a goal to have an accuracy of less than 0,01 m of the geoid surface.

13.3 The gravimetric geoid models NKG89 and NKG96

NKG: The Nordic commission of geodesy, «Nordiske kommisjonen for geodesi».

The models are based of gravity measurements of the Nordic countries and the ocean areas close to the shores. Digital terrain models are used to remove effects from the terrain in the gravity data, and to improve the representation of the geoid on shorter wavelengths. These geoid models are calculated as quasi geoids, as the calculated geoid heights are referred to the surface of the earth, see a definition of a quasi geoid.

The NKG models are therefore pure *gravimetric* geoid models: There are no GPS measurements in levelled (“nivellerte”) points included or used in the NKG models.

NKG96n (n for Norwegian version of the Nordic NKG96) is a development and an improvement of NKG89. This model is supplemented with gravity data from NGU («Norges geologiske undersøkelser», Geological survey of Norway), and a digital terrain model with a grid of 100 x 100 m is used, compared to a grid of 1 x 1 km for NKG89.

14. Corrections of the observations

When we consider which corrections to apply to the observations, the following questions are central:

- In which reference system is the measurements carried out?
- In which reference system shall the results be given in?

Are there different reference systems used in the measurements and the calculations, corrections must be applied. For instance corrections for the deflection of the vertical must be applied on the ellipsoidal zenith angle (from the GNSS measurements) to obtain orthometric heights as the geoid is the zero reference plane.

14.1 Corrections on the GNSS measurements

14.1.1 The zenith angle from the GNSS measurements

The measured zenith angle is with the ellipsoid as the reference plane.

The heights shall be computed with the geoid as the reference plane.

- We have to correct for the deflection of the vertical, as shown in chapter 12, equation (12.6):

$$z_{\text{ort}} = z_{\text{ell}} - \Delta z_v = z_{\text{ell}} - \cos \alpha \cdot \zeta - \sin \alpha \cdot \eta \quad (14.1)$$

Must we also correct for the refraction of the light and for the earth curvature?

- The sight of line between the two points of the baseline is influenced by the refraction. But there is no use of the sight of line when GNSS measurements are performed, therefore we shall not correct for the refraction.
- The correction for the earth curvature must be applied to obtain the correct height difference between the points. This can be done by correcting the zenith angle, or by calculating how much the curvature of the earth is in meters (see a textbook in basic land surveying).

14.1.2 Plane bearing and (horizontal) distance from the GNSS measurements

The reference plane for the measurements: The ellipsoid

The reference plane for the computations: The ellipsoid

- The same reference plane for the measurements and the computations: No corrections because of the deflection of the vertical.

See chapter 11.1 and 11.2 for other corrections which must be applied to the plane bearing and the plane distance.

14.2 Correction on the total station measurements

14.2.1 The zenith angle measured by a total station

The total station is set up with the vertical axis of the total station in the plumb line. The geoid is therefore the reference plane.

Orthometric or normal heights are computed, and therefore the geoid is the reference plane.

- The same reference plane for the measurements and the computations: No corrections because of the deflection of the vertical.

Must we also correct for the refraction of the light and for the earth curvature?

- It is sighted from the station point to the other point, therefore the corrections for the refraction of the light must be applied.
- The correction for the earth curvature must be applied to obtain the correct height difference between the points. This can be done by correcting the zenith angle, or by calculating how much the curvature of the earth is in meters (see a textbook in basic land surveying).

14.2.2 Horizontal direction from the total station measurements

The reference plane for the measurements: The geoid, the deflection of the vertical

The reference plane for the computations: The ellipsoid

- Different reference planes for the measurements and the computations: Corrections because of the deflection of the vertical must be applied.

In chapter 11.2 the component of the deflection of the vertical in the direction between two points derived:

$$\Delta z_v = \xi \cdot \cos \alpha + \eta \cdot \sin \alpha \quad (14.2)$$

The horizontal direction: It is the component of the deviation of the vertical *orthogonal* to the sight of line that is of interest.

Figure 14.1 shows the effect of the deviation of the vertical on the horizontal direction with the azimuth α . Compare with the figure 12.4.c (the component of the deflection of the vertical in the direction of the sight of line).

The two components of the deflection of the vertical are decomposed to a component along the line with azimuth α and a component orthogonal to the line.

Summarizing the two parts of the components orthogonal on the line of sight with azimuth α , and we obtain the component of the deflection of the vertical which is affected on the horizontal direction (see the actual cathetus (small leg) in the black and grey triangle on figure 14.1):

$$\delta_L = \xi \cdot \sin \alpha - \eta \cdot \cos \alpha \quad (14.3)$$

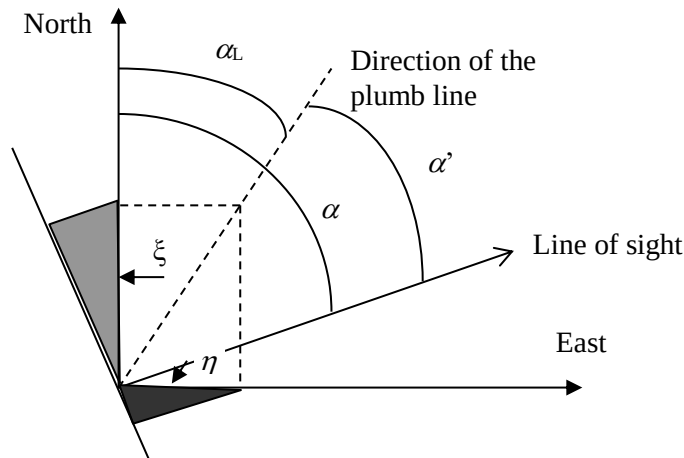


Figure 14.1 The vertical deflection of the vertical. Correction on the horizontal direction

The component of the deflection of the vertical orthogonal to the line of sight, is affecting the horizontal direction in the same way as a vertical axis error of a theodolite.

The vertical axis of the theodolite (the rotation axis) is in the plumb line, but the vertical axis should have been in the normal of the ellipsoid, as the ellipsoid is the plane of the computations.

A general equation for the correction on a horizontal direction caused by a vertical axis error, see also a textbook in basic land surveying:

$$f_v = \theta \cdot \sin \alpha' \cdot \frac{1}{\tan z} \quad (14.4)$$

θ : The vertical axis error is equal to the deviation of the vertical

α' : The horizontal angle between the plane through the plumb line and the ellipsoid normal, and the plane containing the line of sight (see figure 14.1)

z : The zenith angle

The connection between (14.3) and (14.4) is to be derived, see also figure 14.1:

$$\theta \cdot \sin \alpha' = \theta \cdot \sin(\alpha - \alpha_L) = \theta \cdot [\sin \alpha \cdot \cos \alpha_L - \cos \alpha \cdot \sin \alpha_L]$$

Inserting the components of the deviation of the vertical, and formula (14.3) is obtained:

$$\theta \cdot \sin \alpha' = \sin \alpha \cdot (\theta \cdot \cos \alpha_L) - \cos \alpha \cdot (\theta \cdot \sin \alpha_L) = \sin \alpha \cdot \xi - \cos \alpha \cdot \eta$$

As the correction is *from* the plumb line *to* the ellipsoid normal, the starting point is the component of the deflection of the vertical, δ_L in equation (14.3), but the sign must be changed.

And the correction of the measured horizontal direction is:

$$\delta = -f_v \cdot \frac{1}{\tan z} = (\eta \cdot \cos \alpha - \xi \cdot \sin \alpha) \cdot \frac{1}{\tan z} \quad (14.5)$$

See chapter 11.2 for other corrections that must be applied to obtain a direction in the map projection plane (about the reduction of the direction and the meridian convergence). See also page 1.9 and chapter 4 in Holsen "Geodesy part 2", also for the meridian convergence.

14.2.3 (Horizontal) distance from the total station measurements

The reference plane for the measurements: The geoid, the deflection of the vertical

The reference plane for the computations: The ellipsoid

- Different reference planes for the measurements and the computations: In theory the corrections because of the deflection of the vertical must be applied. The deviation of the vertical is an angle of a few milligrads (milligons). The effect of deviation of the vertical is therefore small and can normally be set to zero.

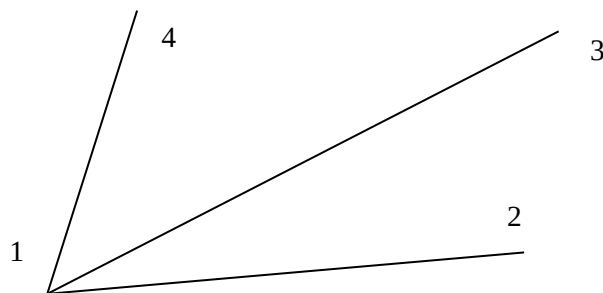
See chapter 11.1 for other corrections that must be applied to obtain a distance in the map projection plane.

14.3 Rotation and scale differences between WGS84 and another datum

See the remarks in chapter 10.6.3.

In land surveying programs like the Norwegian Gemini and GIS-Line, the rotation and the scale can be considered as additional unknowns.

In this way small corrections in rotations and scale between two different types of measurements and two different datums are calculated.



Figur 14.2 Tre vektorer

Skal GNSS-bestemte størrelser brukes sammen med konvensjonelle målinger i et gitt geodetisk grunnlag, er det aktuelt å beregne eventuelle rotasjons- og målestokksavvik mellom de to systemene. De samme

beregningene er også nødvendige, når GNSS-vektorer i datumet WGS84 skal innpasses i et annet datum, for eksempel NGO1948, se merknaden sist i kapittel 10.6.3.

Det kan gjøres når vektoren bestemmes mellom punkter som er gitt i det eksisterende geodetiske grunnlag. Koordinatdifferansen mellom gitte punkt transformeres til CTRS via lokalgeodetiske system. Det er her som ellers viktig at man kjenner geoidehøyden for WGS-ellipsoiden slik at man kan regne om til ellipsoidiske høyder.

Man kan forvente at rotasjonene er av differensiell størrelse. La oss anta at punktene i figur 14.2 er gitt i det eksisterende grunnlag og at de tre vektorer også er bestemt med GNSS.

Vi har da følgende ukjente parametere:

$\omega_x, \omega_y, \omega_z$: rotasjon om henholdsvis x, y, z –aksen.

m : målestokksavvik, avvik fra 1.

Hver vektor gir tre feilligninger. Da rotasjonene er små vinkler, kan vi benytte en forenklet rotasjonsmatrise i beregningen:

$$\begin{bmatrix} \hat{\epsilon}_x \\ \hat{\epsilon}_y \\ \hat{\epsilon}_z \end{bmatrix} + \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}_{\text{Gitt}} = \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix}_{\text{Målt}} + \begin{bmatrix} m & \omega_z & -\omega_y \\ -\omega_z & m & \omega_x \\ \omega_y & -\omega_x & m \end{bmatrix} \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \end{bmatrix} \quad (14.6)$$

Når transformasjonsparameterne er beregnet, kan alle ΔX , ΔY og ΔZ korrigeres for rotasjon og målestokk før de transformeres til det lokalgeodetiske system med ligningen (10.15).

I prinsippet kan man kombinere målinger av ulike slag, GNSS-målinger, konvensjonelle geodetiske målinger, astronomiske målinger, treghetsmålinger og fotogrammetriske målinger.

Ulike mottakere og ulike beregningsprogram kan gi noe forskjellig resultat ved måling av de samme vektorer. Det vil derfor være av interesse å undersøke om denne forskjell i resultat kan tolkes som rotasjoner og målestokkavvik. Det kan gjøres på samme måte som ovenfor.

Vi får feilligningssystem av samme type som ligning (10.4). Men man bør undersøke om de beregnede rotasjoner og målestokkavviket er signifikante.

I landmålingsprogrammer som Gemini og VG-Land, kan rotasjon og målestokk inngå som tilleggsukjente.

Dermed blir det korrigert for små forskjeller i rotasjon og målestokk mellom ulike typer målinger og ulike datum.

14.4 Corrections because of an eccentric station

14.4.1 Correction because of the height of the antenna

Den vektor som på første hånd beregnes, er avstanden mellom antennesentrene i de to stasjoner. Selv om antennen står sentrisk, vil antennesenteret være mer eller mindre eksentrisk i høyderetningen i det geosentriske koordinatsystemet.

De primært beregnede vektorkomponenter ΔX , ΔY og ΔZ , må derfor korrigeres for eksentrisitet i høyde for de to antennesentrene.

Korreksjonene er som følger:

$$\Delta\Delta X = \Delta H \cdot \cos\varphi \cdot \cos\lambda \quad (14.7)$$

$$\Delta\Delta Y = \Delta H \cdot \cos\varphi \cdot \sin\lambda \quad (14.8)$$

$$\Delta\Delta Z = \Delta H \cdot \sin\varphi \quad (14.9)$$

ΔH er antennesenterets høyde over punktet, og φ og λ er punktets geografiske koordinater. Disse beregnes under vektorberegningen.

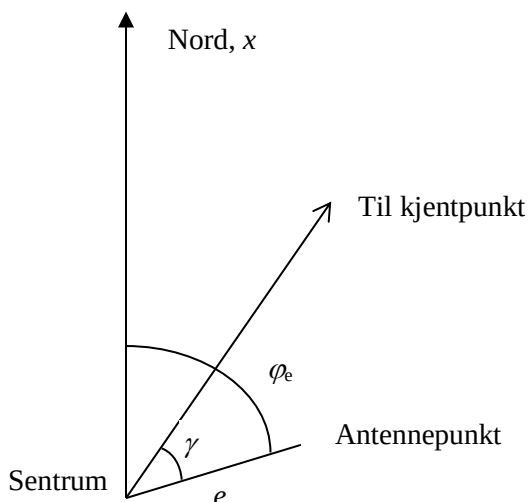
Korreksjonsformlene ligger i beregningsprogrammene for vektorberegninger, slik at korreksjonene påføres den beregnede vektor automatisk.

14.4.2 Correction because of an eccentric antenna

I enkelte tilfeller vil det være aktuelt å sette antennen eksentrisk i forhold til det punkt man ønsker som vektor-endepunkt. Årsaken til det kan være trær som stenger for sikt mot satellittene, eller at vedkommende punkt ligger for nær en husvegg, slik at signalrefleksjoner fra veggen vil gi mye støy på målingen (flerveisinterferens, multipath). De primært beregnede vektorkomponenter må derfor sentreres. Dette er i regelen ikke lagt inn i programmene for beregning av vektorer og må eventuelt beregnes i et separat program.

Definisjoner, se også figur 14.3:

- γ : målt horisontalvinkel mellom kjent retning og retningen fra sentrum til antennesenter.
- e : målt skråavstand fra sentrum til antennepunkt.
- φ_e : retningsvinkel i for eksempel NGO1948 for eksenterlinje e .
- Δh : beregnet høydeforskjell mellom sentrum og antennepunkt.
- e_h : horisontal avstand fra sentrum til antennepunkt.



Figur 14.3 Eksentrisk oppstilling

Fremgangsmåten blir som følger, går fra kartplanet til kartesiske koordinater (se kapittel 10.4):

1. Beregning av asimut α for eksenterlinje e

$$\alpha_e = \varphi_e + c_e + \delta_e, \text{ se formel (11.6) i kapittel 11 og } c_e \text{ beregnes med (11.7)}$$

2. Koordinater i det lokalgeodetiske system

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{LG} = \begin{bmatrix} e_h \cdot \cos \alpha \\ e_h \cdot \sin \alpha \\ \Delta h \end{bmatrix}$$

3. og videre til CTRS

$$\begin{bmatrix} \Delta\Delta X \\ \Delta\Delta Y \\ \Delta\Delta Z \end{bmatrix}_{CTRS} = \begin{bmatrix} -\sin \Phi \cdot \cos \lambda & -\sin \lambda & \cos \Phi \cdot \cos \lambda \\ -\sin \Phi \cdot \sin \lambda & \cos \lambda & \cos \Phi \cdot \sin \lambda \\ \cos \Phi & 0 & \sin \Phi \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{LG} \quad (14.10)$$

END

APPENDIX A. LITERATURE. INTERNET ADRESSES

In Norwegian:

Se spesielt på standardene fra Kartverket, de siste utgavene finnes på Kartverkets hjemmesider, <http://www.kartverket.no/>, under "standardisering".

The Web pages are Ok per January 2016.

The Norwegian standard "Satellittbasert posisjonsbestemmelse" from Kartverket	Kartverket
The Norw. standard "Koordinatbasert referansesystem. Datum, koordinatsystem, transformasjon, konvertering og avbilding" from Kartverket	Kartverket
The Norw. standard "Norges offisielle høydesystem og referansenivåer" from Kartverket	Kartverket
Terje Skogseth og Dag Norberg: "Grunnleggende landmåling". Gyldendal forlag 2014	
Hofmann-Wellenhof, Lichtenegger and Walse: "GNSS", SpringerWienNewYork, 2008	At Blackboard
Geir W. Simensen: Dr.ing.avhandling "Navstar GPS. Planlegging, kvalitetskontroll og høydebestemmelse (1992).. In Norwegian.	Paper version
Jon Holsen: "Geodesy part 1"	At Blackboard
Jon Holsen: "Geodesi part 2"	At Blackboard
Bjarne Langseth "Høyder og presisjonsnivellement" (in Norw., paper version)	Paper version
https://igscb.jpl.nasa.gov/ Orbital data etc. from International GNSS Service	Int. GNSS serv.
http://www.aiub.unibe.ch/content/index_eng.html The Astronomic institute at the University in Bern, Switzerland. Central in the GNSS world, many links. CODE Analysis Center. Ionosphere models. Bernese GNSS Software, etc..	Univ. Bern
http://earth-info.nga.mil/GandG/publications/tr8350.2/tr8350_2.html NIMA. Report with an official description and definition of WGS84 (the GPS datum) and the connections between WGS84 and other datums.	WGS84
http://itrf.ensg.ign.fr/ ITRF: International Terrestrial Reference Frame: The coordinate reference system and frame (ITRS/ITRF) given by IERS (see below), the most accurate global coordinate reference frame. Used by International GNSS Service (see above) for orbital data.	ITRF
http://www.iers.org/iers/EN/Home/home_node.html IERS: International Earth Rotation and Reference Systems Service. Calculate polar motion, time etc..	IERS
http://gauss.gge.unb.ca/GPS.INTERNET.SERVICES.HTML (overview of lot of web pages from the University of New Brunswick, Canada)	Overview. Links
http://omniweb.gsfc.nasa.gov/vitmo/iri_vitmo.html Real time Ionospheric maps. Ionosfærisk modell som kan kjøres på www The International Reference Ionosphere (IRI) is an international project	NASA Ionosph. model
http://iono.jpl.nasa.gov/ gim_dailymovie.html JPL-NASA, central lab for GPS. Real time Ionospheric global map	NASA JPL Ionosph. map
http://sidc.oma.be/ Royal Observatory of Belgium harbours the Sunspot Index Data center, the World data center for the Sunspot Index and space weather forecast.	Ionosphere center

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Key words in Norwegian

Old, not referred to pages.

A posteriori	geosentriske koordinater	potensialflate
A priori	GLONASS	projeksjon
absoluttbestemmelse	GPS	pseudoavstand, pseudorange
anti-spoofing	GPS-banedata	pseudokinematisk måling
A-S	GPS-tid	referanseellipsoide
asimut	gravimetrisk geoidmodell	referanseramme
astronomisk bestemte koordinater	GRS80	referansesatellitt
atomtid	heltallsukjente (på en satellittavstand)	referansestasjon, referansepunkt
avbildning	heltallsbestemmelse	referansesystem, koordinatbasert
avstandsreduksjon	horisonten	relativ måling og posisjonsbestemmelse
banedata	hovedmeridian	retningsreduksjon
baseline	høydevinkel	Rms
basislinje	IAT	romligg utjevning
C/A-kode	IERS	Rover
choke ring antenne	IGS	RTK
CODE	ikke-triviell vektor	SA
CTS, CTRS	initialisering	Sanntid
cut-off-angle	ionosfæren	seksagesimal sirkeldeling
cycle	ITRF	Selective Availability
cycle ambiguity	kartetisk koordinatsystem	sentralmeridian
cycle slip	kartprojeksjon	singel-differanse
datum	kartprojeksjonsplan, kartplan	solfleksyklus
differensiell beregning	kinematisk posisjonsbestemmelse	statisk måling
dobbeltdifferanse	klassisk statisk måling	stopp og gå, "stop and go"
eferemider	klokkefeil (ved GPS-målinger)	TAI
ekvipotensialflate	kodemåling	toposentrisk koordinatsystem
elevation, elevasjon	kodesignal	transformasjon
ellipsoide	konvertering	transformasjonsparametre
ellipsoideparametre	korttidsstatisk måling	trippeldifferanse
ellipsoidisk høyde	L1, L2	troposfæren
enkelttdifferanse	Laplace-ligning	tyngdekraft
enkeltpunktbestemmelse	loddavvik	uavhengige vektorer
epoke	loddlinje	UTC
EUREF89	loddretning	UTM
fasebrudd	lokalhorisontert koordinatsystem	UTM-projeksjon
fasedifferanse, faseforskjell	meridian	UTM-soner og -belter
fasemåling	meridiankonvergens	vektor, GPS-vektor
fix-løsning (på vektoren)	meridianplan	vektorberegning
flattrykning (ellipsoidens)	mobil mottaker	vertikalt datum
flerveisinterferens	multipath	VREF1996, VREF2000
float-løsning (på vektoren)	NGO1948	WGS84
fotpunkt	NGO-striper	
GDOP	nivåflate	
geodesi	NKG	
geodetisk bredde	NN1954	
geodetisk lengde	normalhøyde	
geodetiske geografiske koordinater	nullmeridianplan	
geografisk bredde	on the fly	
geografisk lengde	ortometrisk høyde	
geografiske koordinater	PDOP	
geoide	periode (for et signal)	
geoideavstand	P-kode	
geoidehøyde	posisjonering	
geoidmodeller	posisjonsbestemmelse	

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